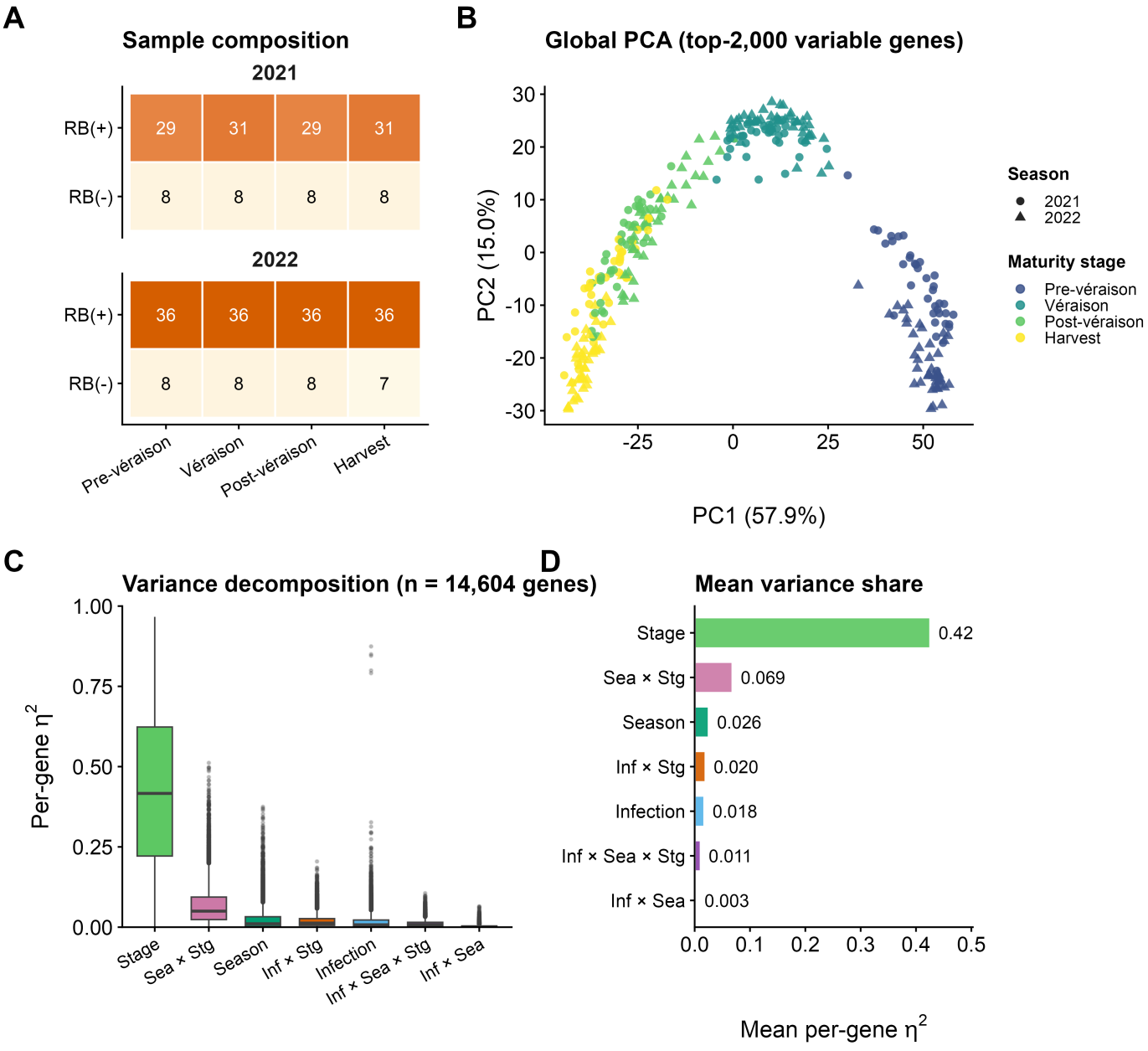
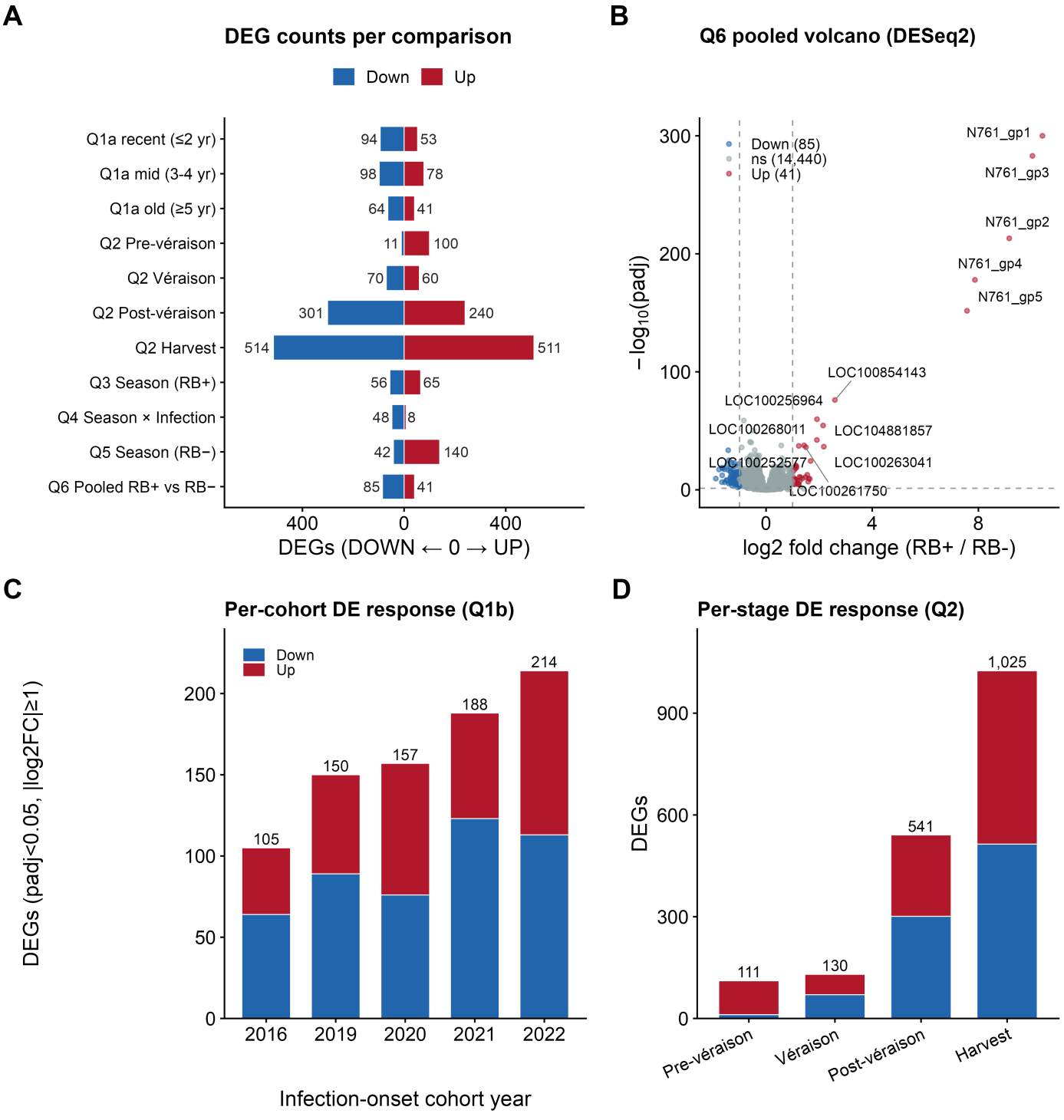
Study overview and global variance structure

327 samples (264 RB+, 63 RB−), 14,604 expressed genes; stage explains the most transcriptome variance



Differential expression landscape (DESeq2 reanalysis)

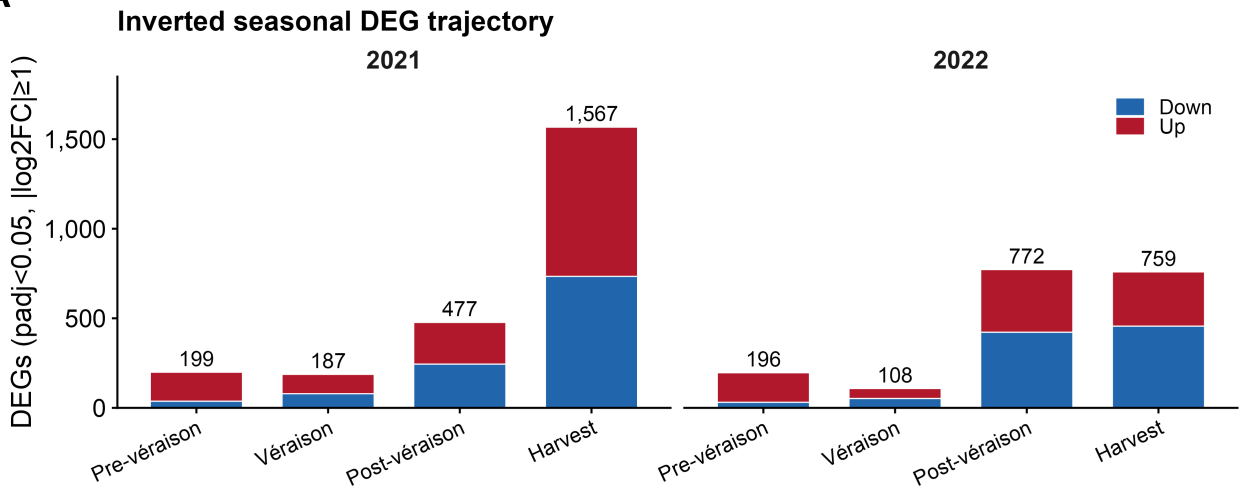
All contrasts refit with covariate-adjusted DESeq2; significance: padj < 0.05 and |log2FC| ≥ 1



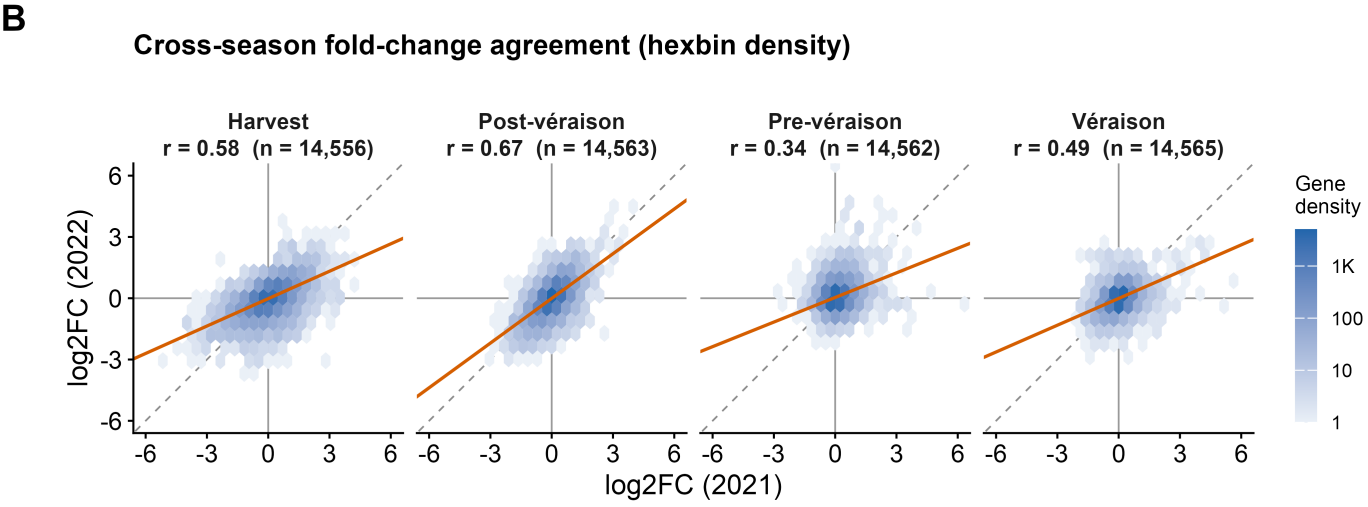
Inverted seasonal trajectory: cool 2021 peaks late, hot 2022 peaks early

Per-stage × per-season DESeq2 contrasts (RB+ vs RB−); raw titer averaged across infection-onset cohorts

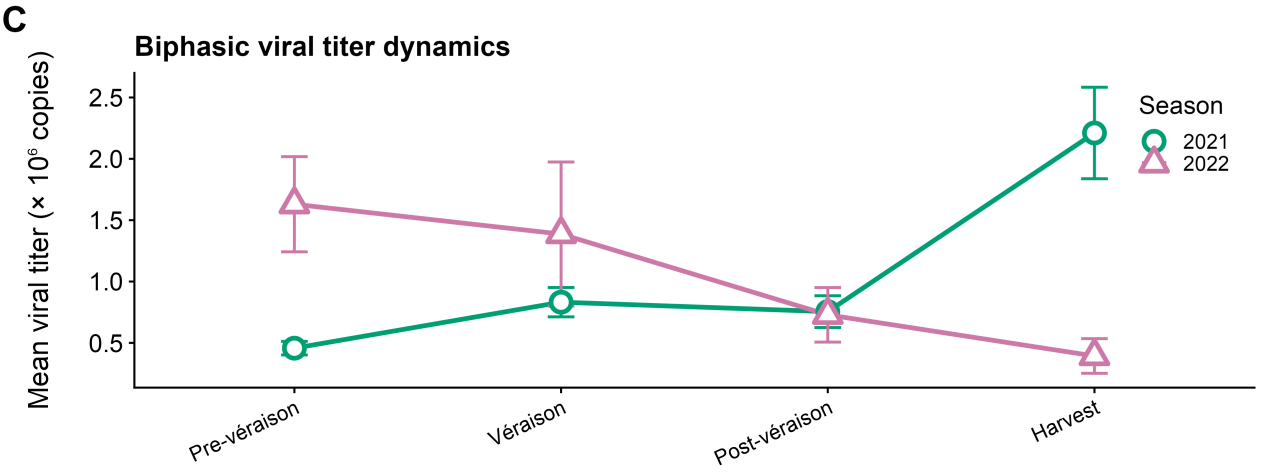
A



B



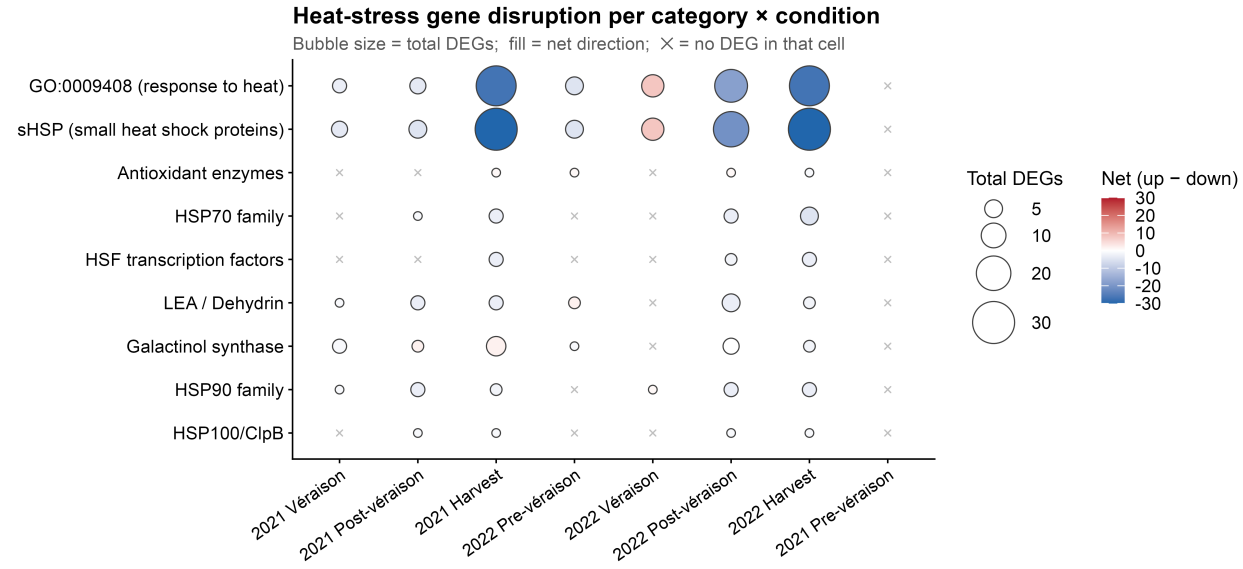
C



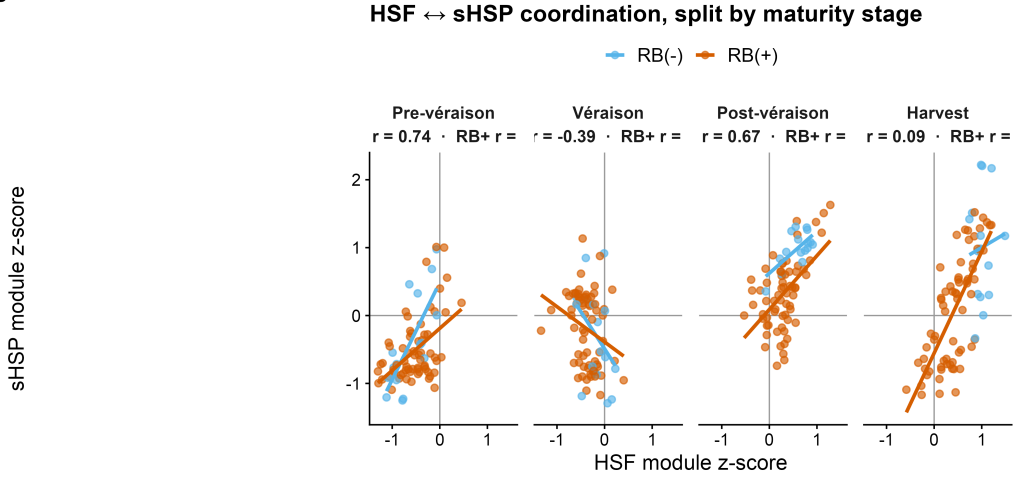
Heat-stress disruption: sHSP suppression and HSF coordination

115-gene heat panel (9 categories); HSF + sHSP module z-scores per sample; rows in C clustered

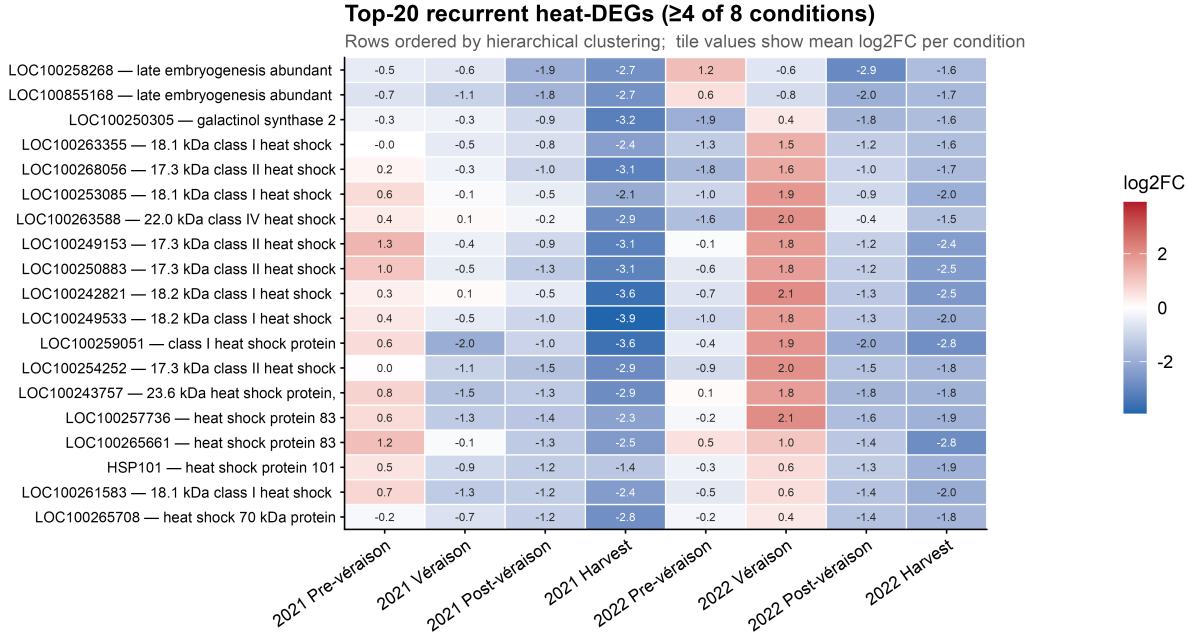
A



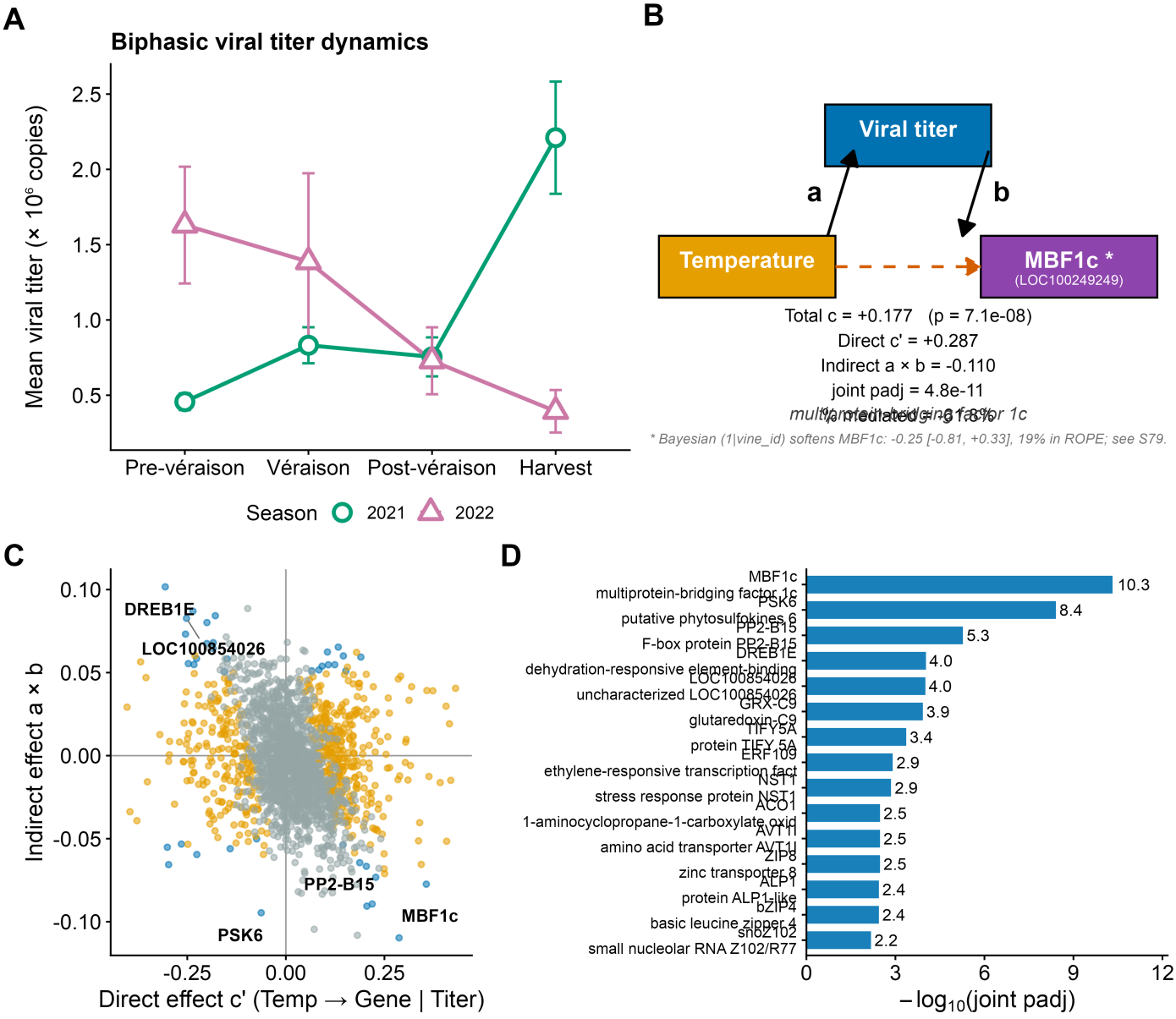
B



C



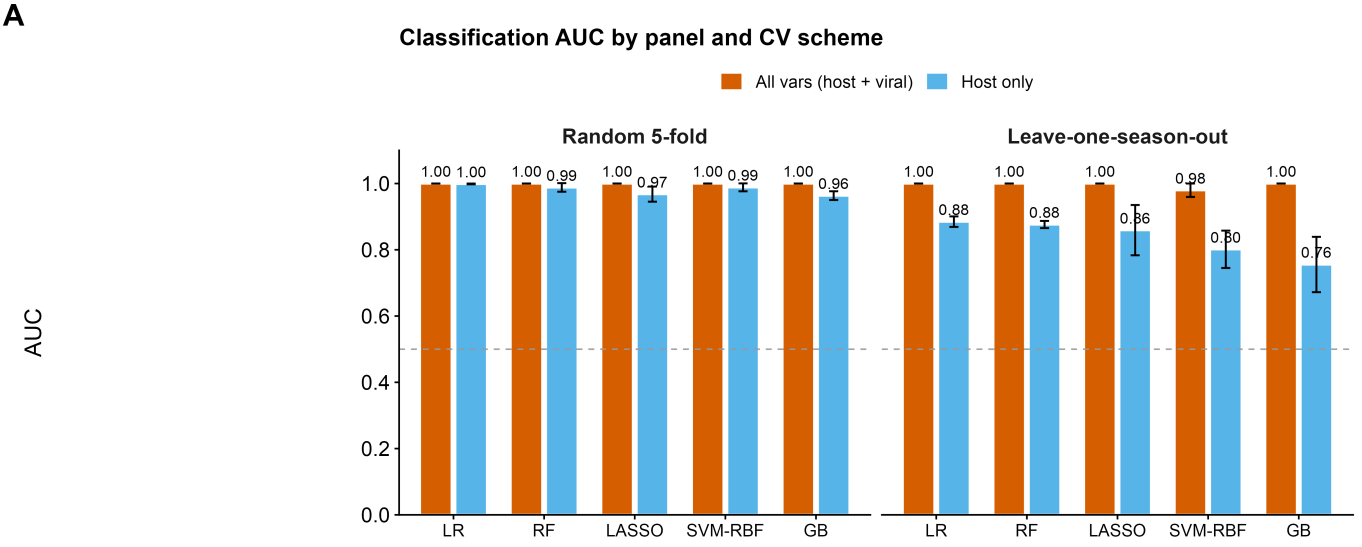
Causal mediation: temperature → viral titer → gene expression



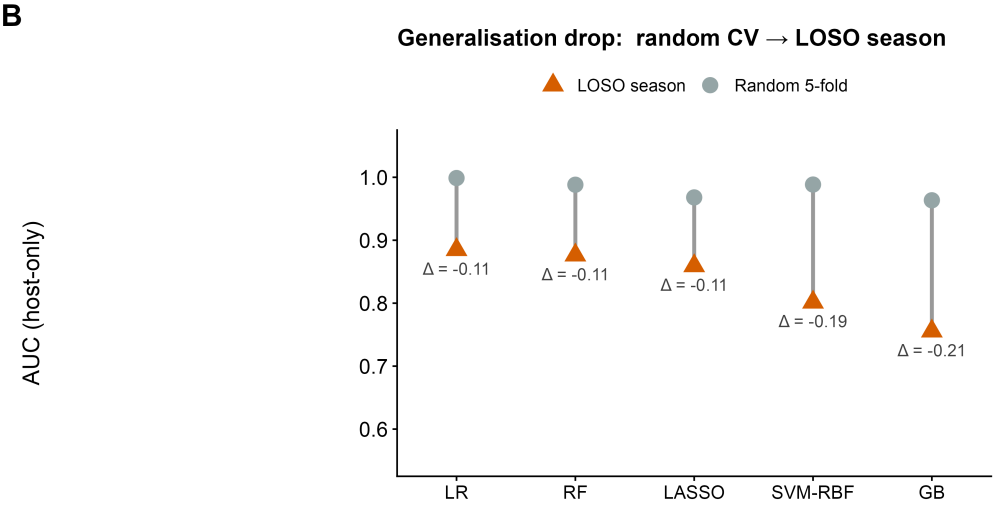
Reframed ML biomarker analysis: AUC=1.000 was an artifact of viral reads

Random 5-fold AUC stays ≈ 1.000 with all variables, drops to 0.76-0.88 under leave-one-season-out CV with host genes only

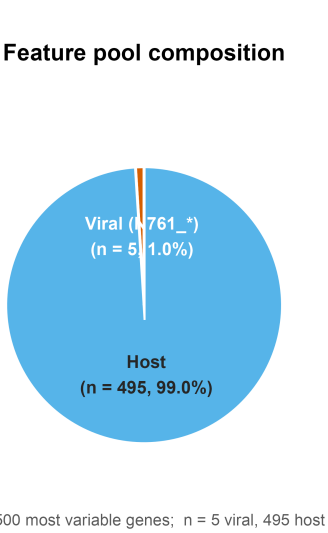
A



B

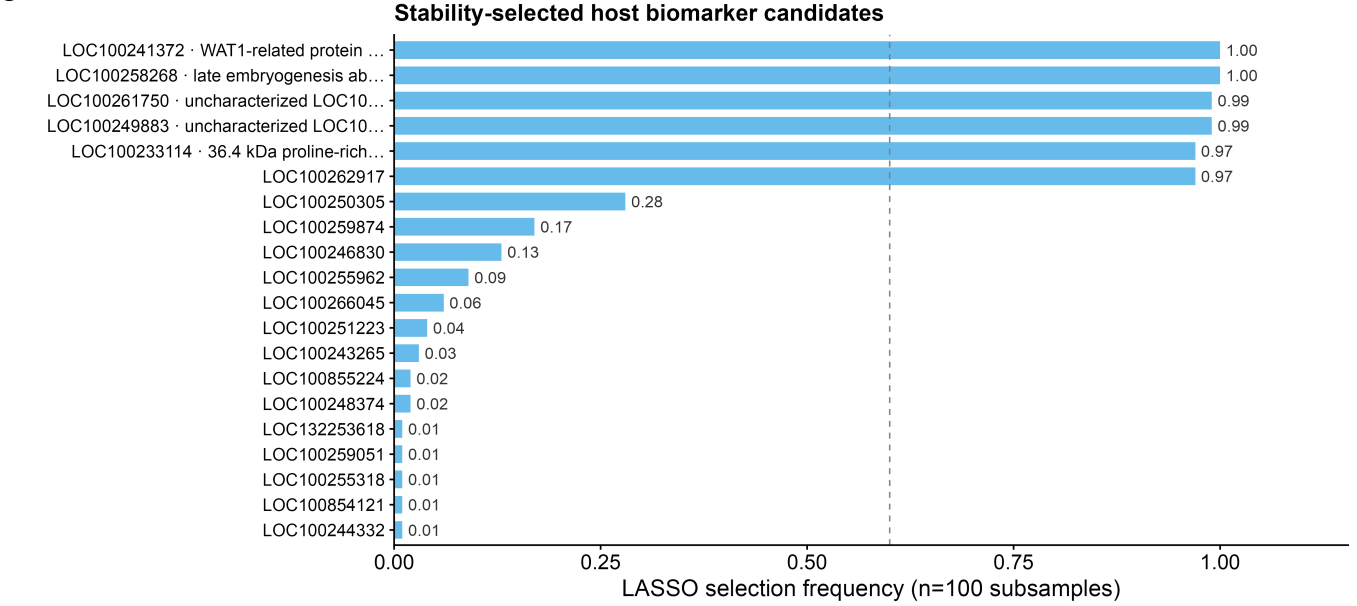


D



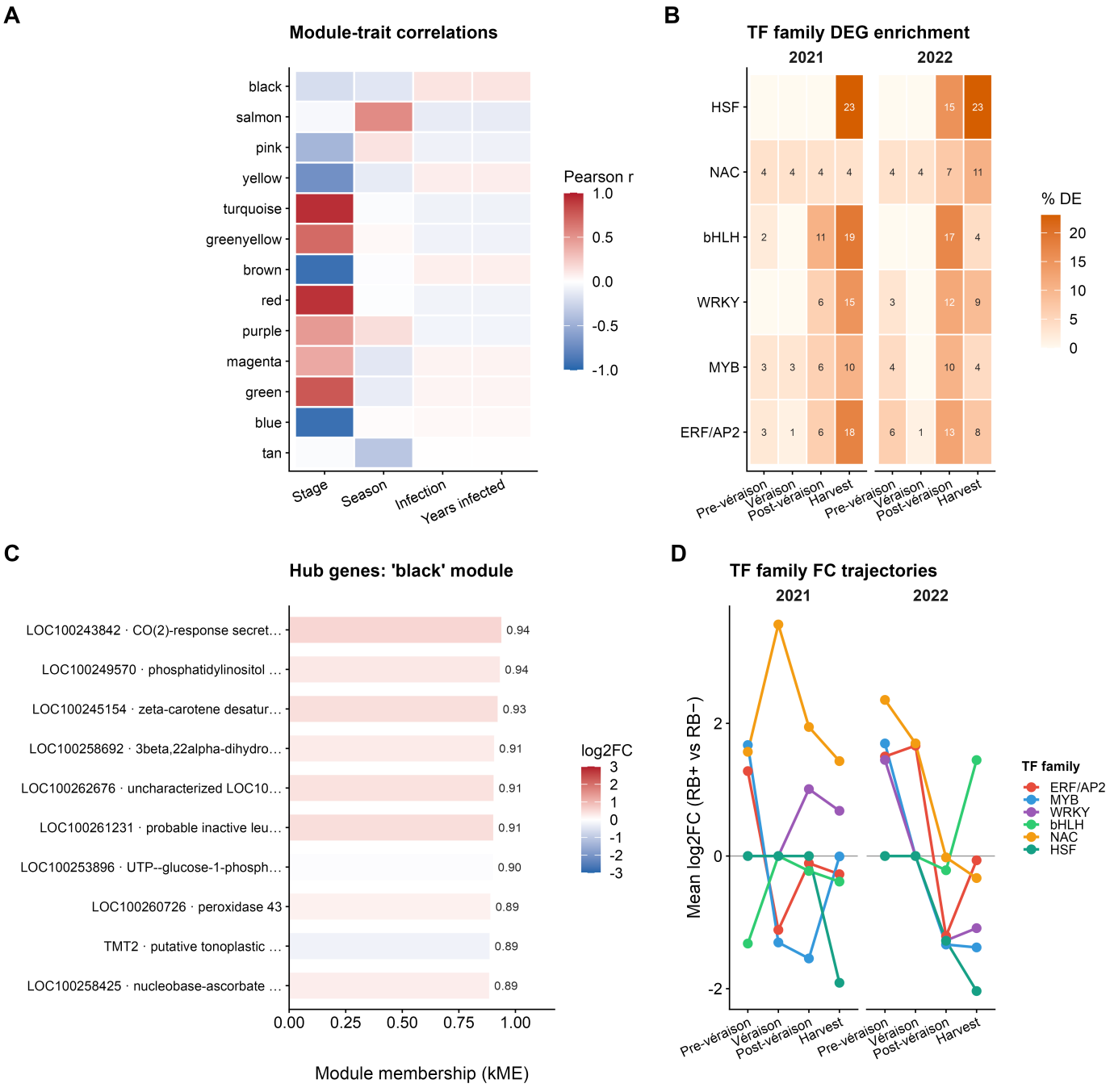
Top 500 most variable genes; n = 5 viral, 495 host

C



Co-expression modules and transcription-factor regulators

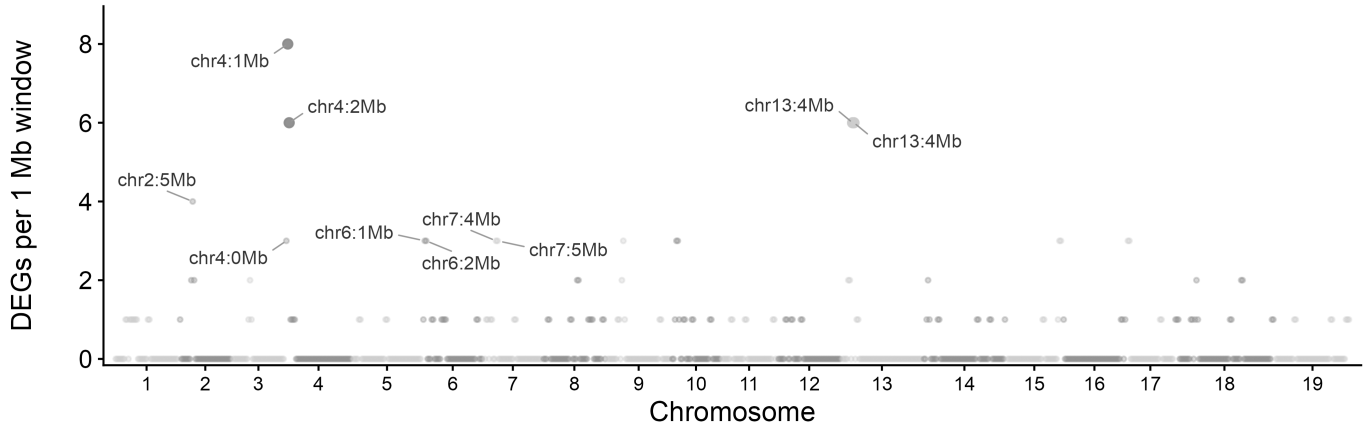
13 WGCNA modules across 4 traits; 6 TF families (267 genes) profiled across 8 stage × season conditions



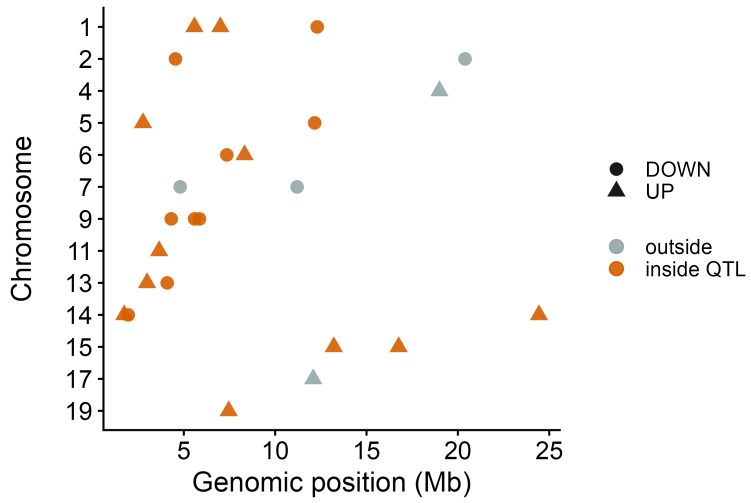
Genomic localisation and biomarker – QTL overlap

DEG hotspots in 1 Mb windows; 25/25 biomarkers (100.0%) within 1 Mb of a known QTL
2.29x enrichment vs 5000-permutation null (p < 0.0002)

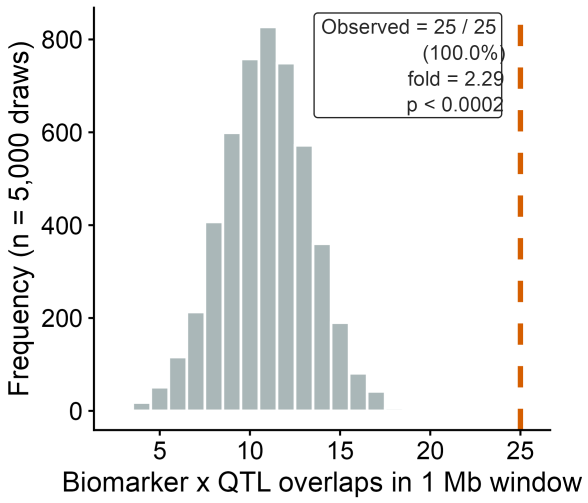
A



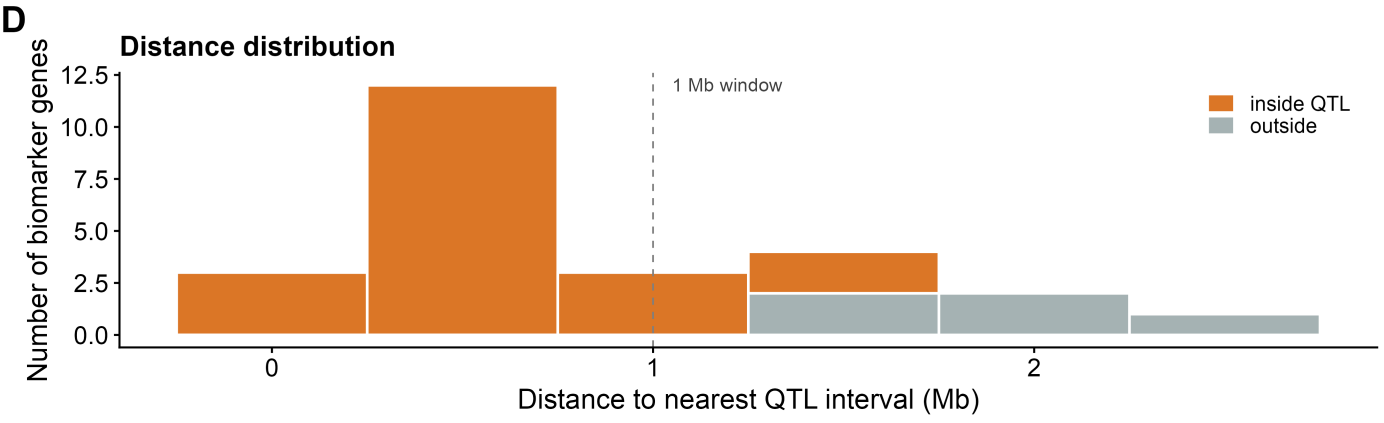
B



C



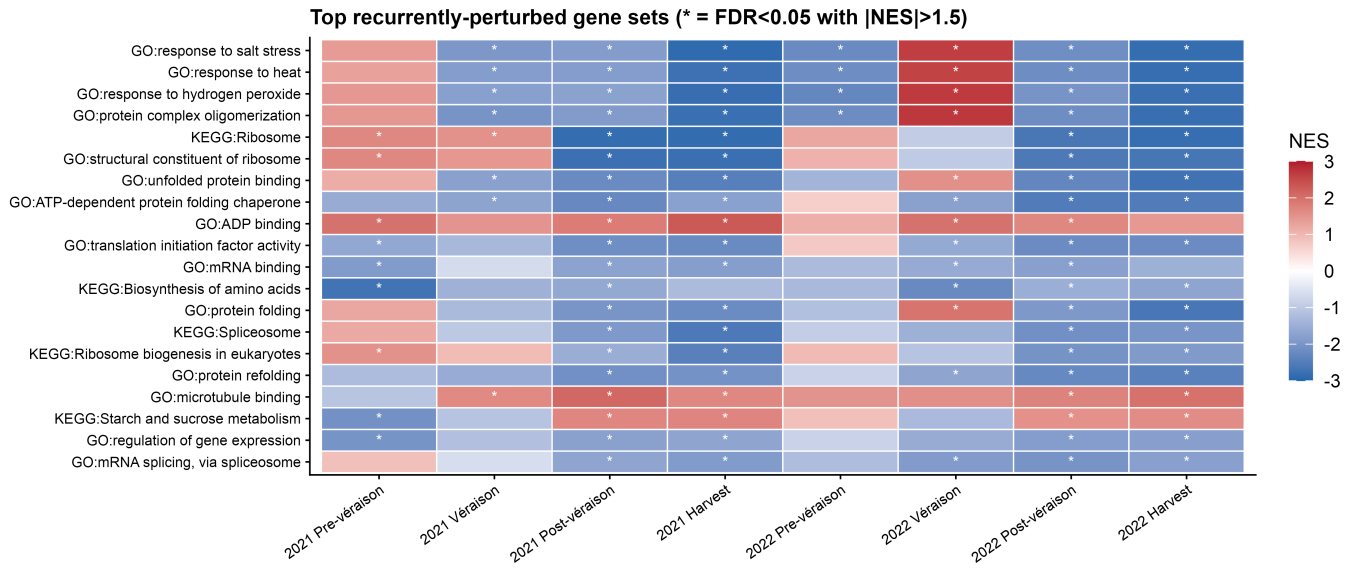
D



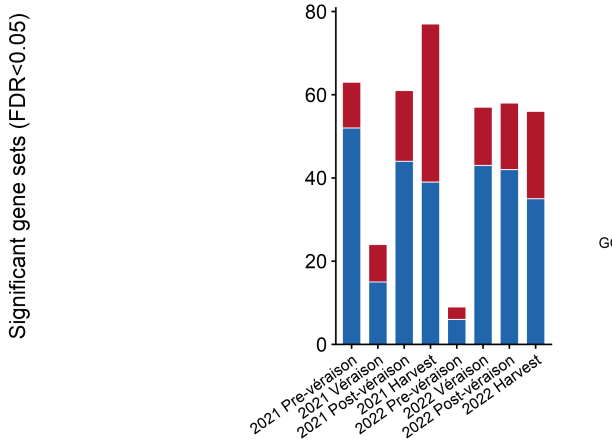
Gene-set enrichment (GSEA) across stage × season conditions

375 gene sets × 8 conditions (500 permutations); 3,000 tests, 270 significant
77% of significant calls are suppression (NES<0)

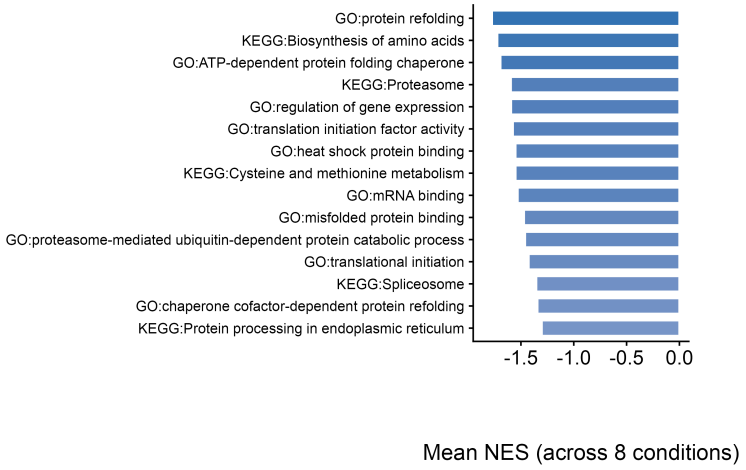
A



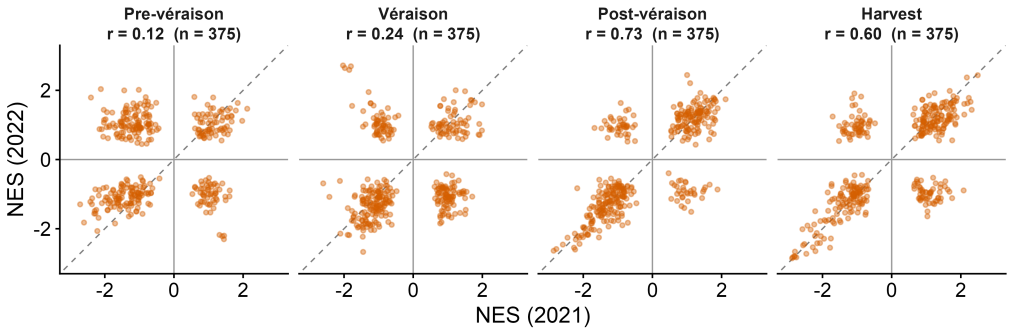
B



C



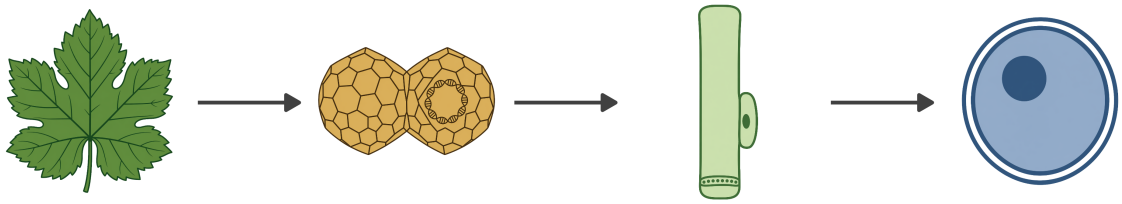
D



GRBV molecular disruption cascade

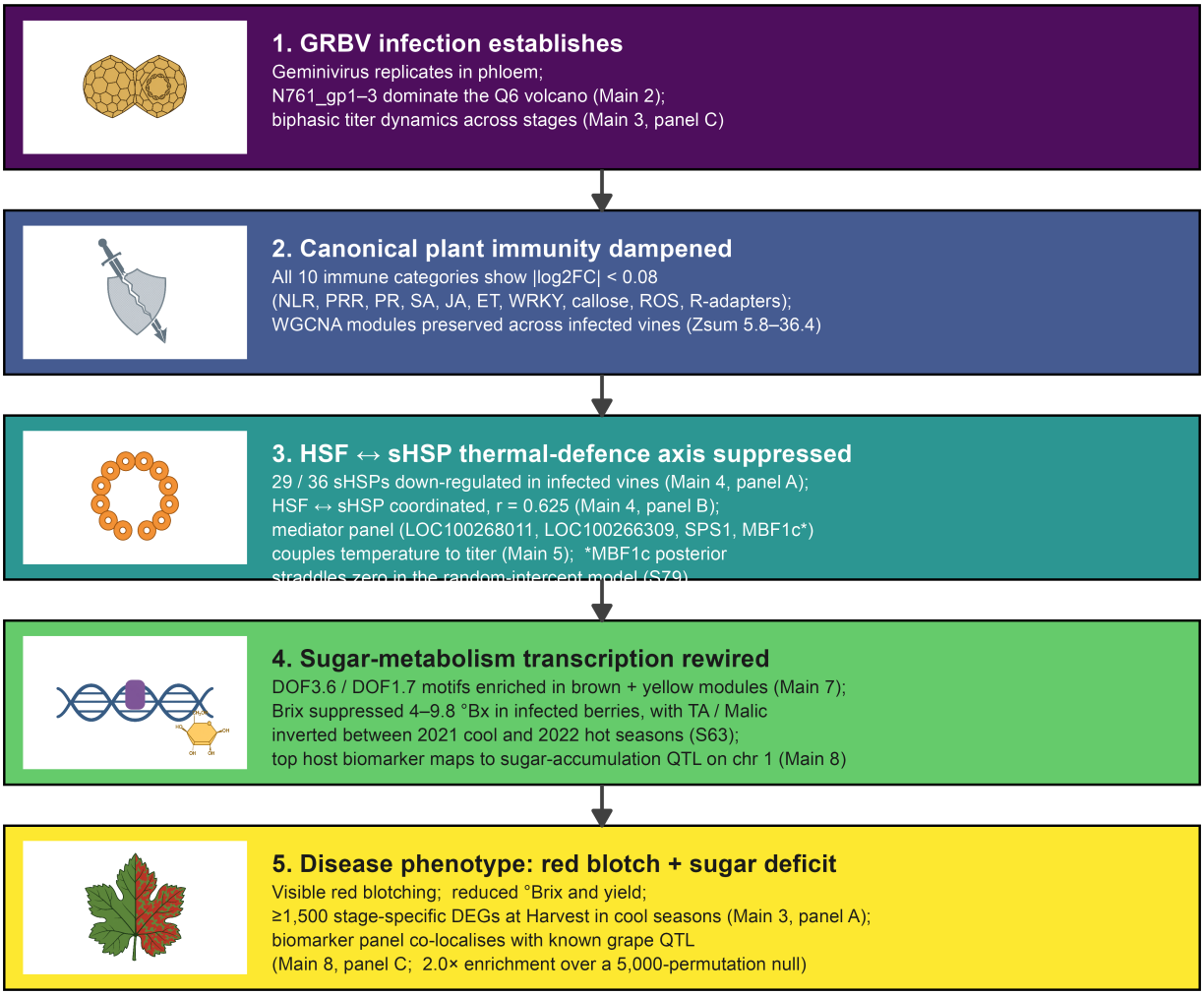
Viral establishment → silent immunity → thermal-defence dampening → sugar-pathway rewiring → disease phenotype

A Tissue-level view: where GRBV acts



Infected leaf | Geminivirus virion | Phloem sieve element | Host cell nucleus

B Mechanism cascade



ator panel (multi-pipeline): LOC100268011 RdRp1, LOC100266309, SPS1 (LOC100232974), MBF1c*; *MBF1c posterior softened by random-

Supplementary Figure S1. Global PCA overview

Status: Data current (kept as-is, restyle pending)

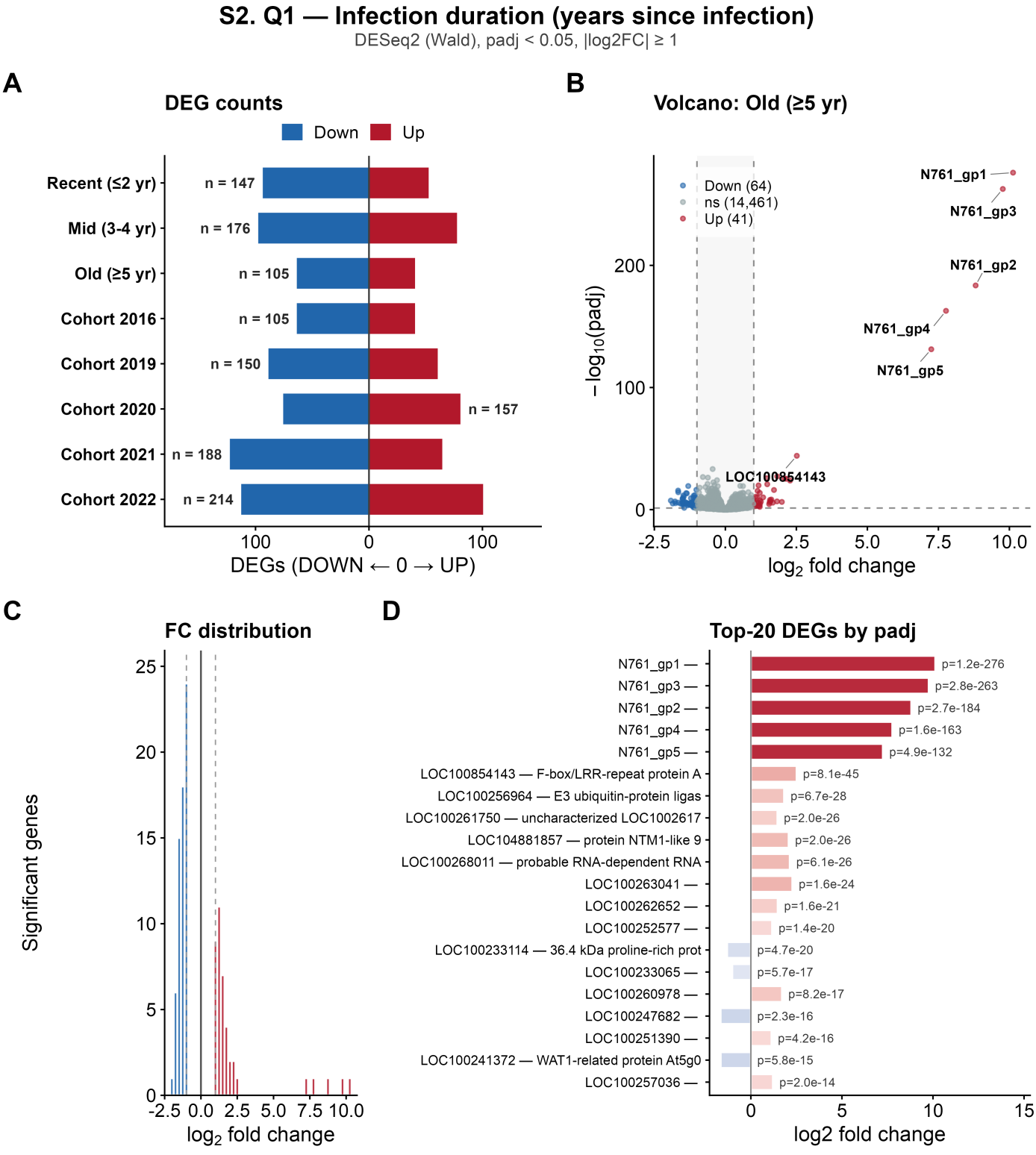
Note: See Main 1 (panel B)

[file not found: 01_global_pca_overview.png]

Supplementary Figure S2. Q1 infection-duration DEGs

Status: Regenerated under DESeq2, R/ggplot2, Arial

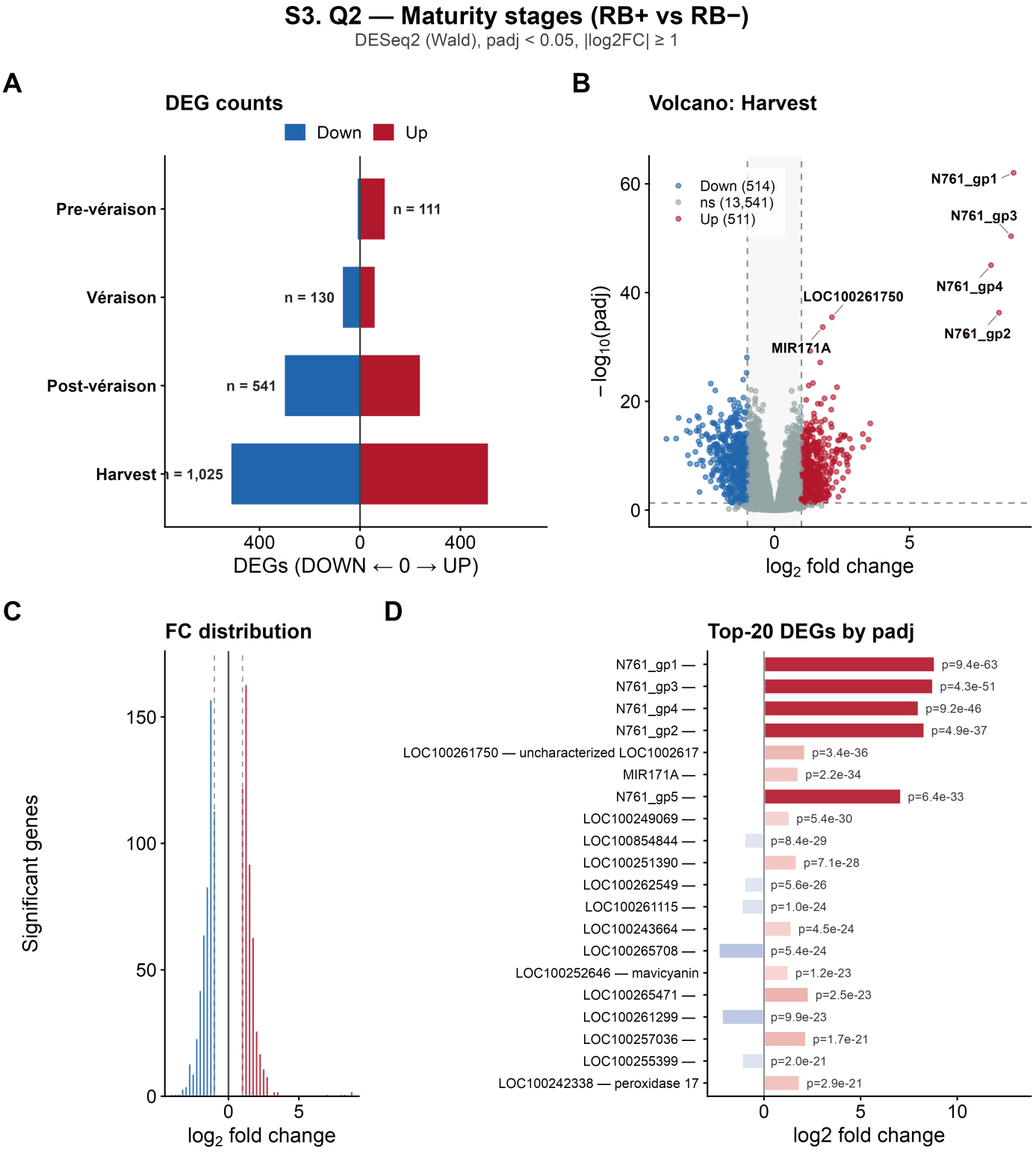
Note: See Main 2; refit with de_deseq2_extras Q1a/Q1b



Supplementary Figure S3. Q2 maturity-stage DEGs

Status: Regenerated under DESeq2, R/ggplot2, Arial

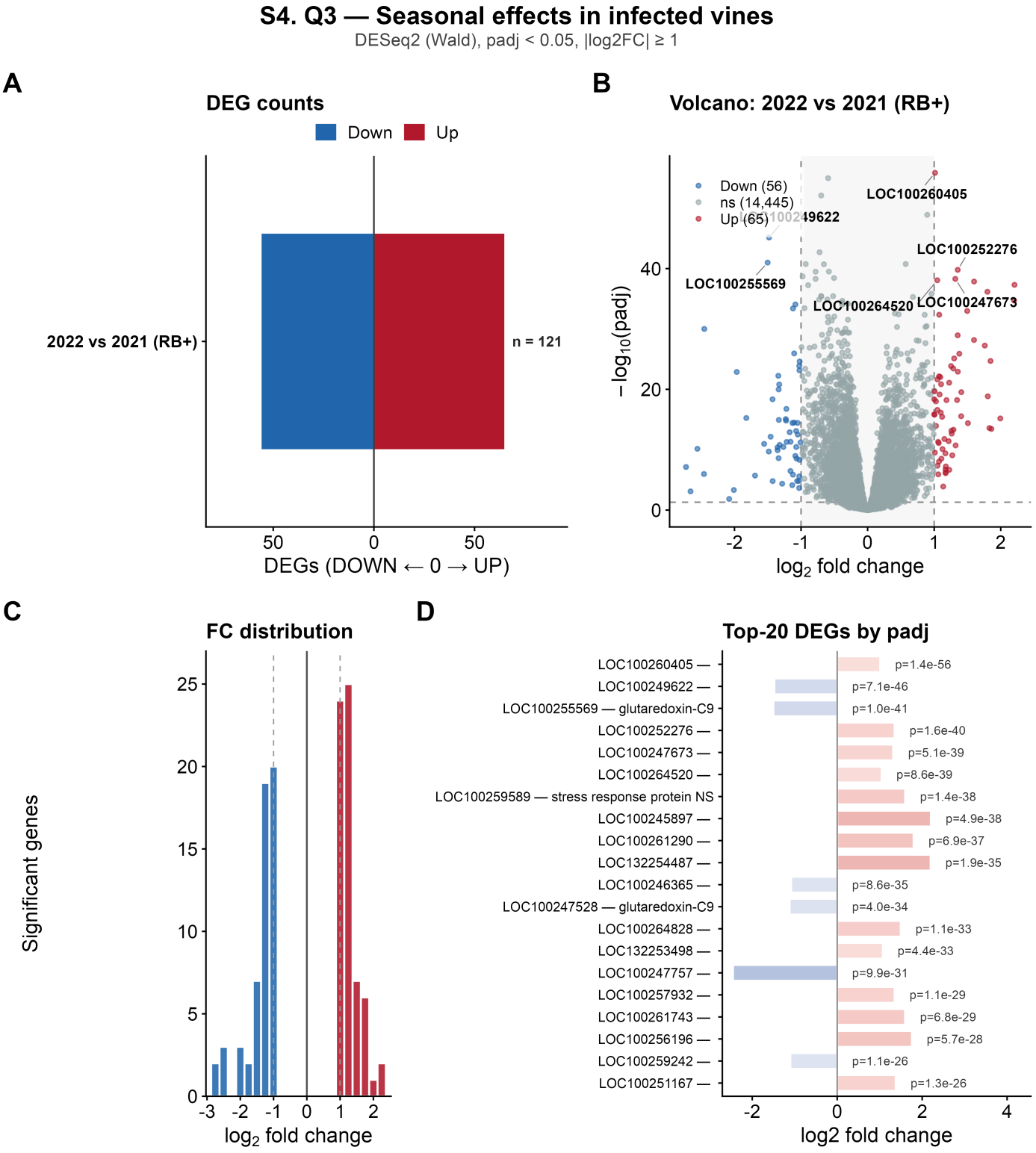
Note: See Main 2 + de_deseq2_voom Q2



Supplementary Figure S4. Q3 seasonal effects in RB+

Status: Regenerated under DESeq2, R/ggplot2, Arial

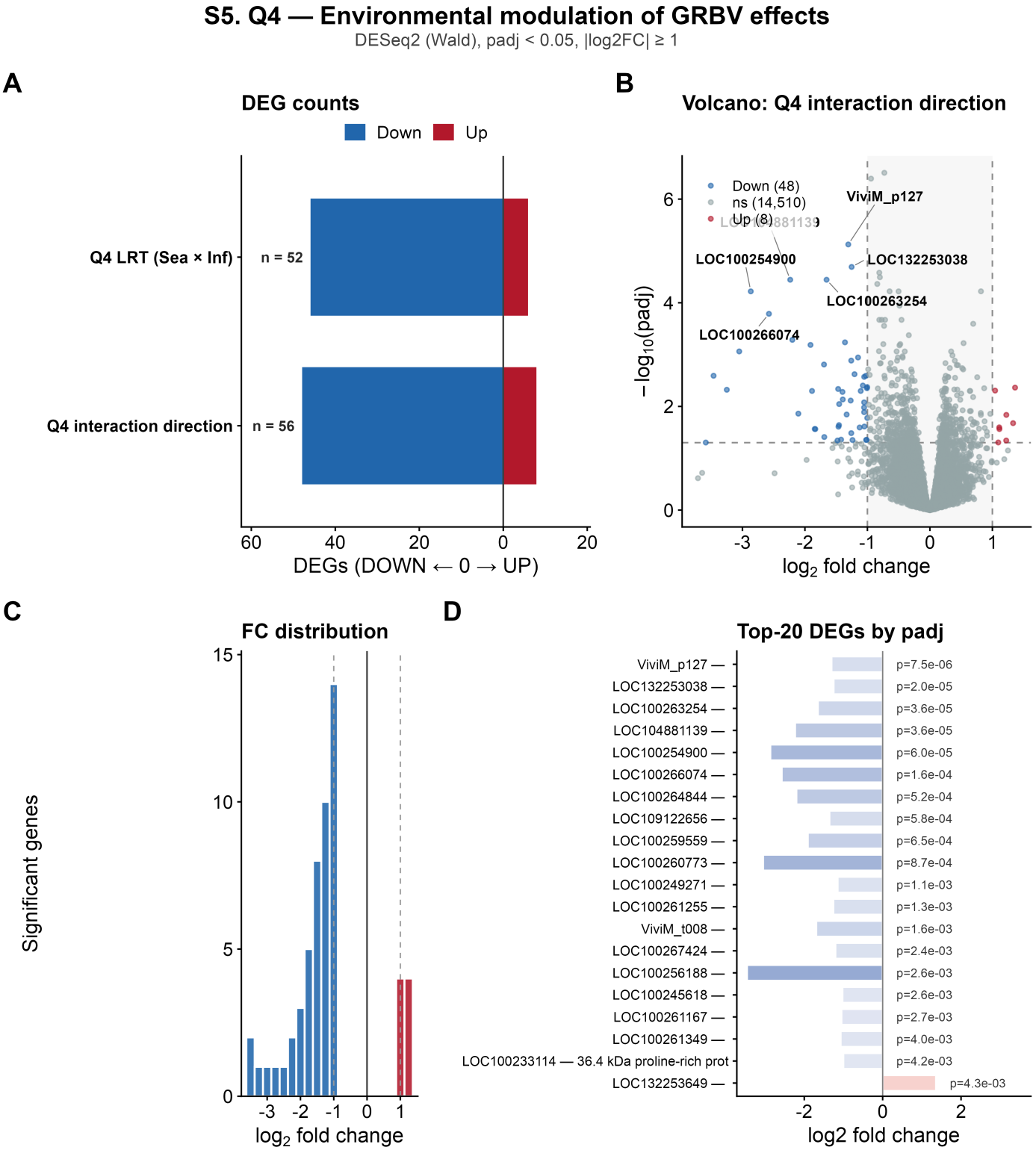
Note: See Main 2 + de_deseq2_voom Q3



Supplementary Figure S5. Q4 environmental modulation

Status: Regenerated under DESeq2, R/ggplot2, Arial

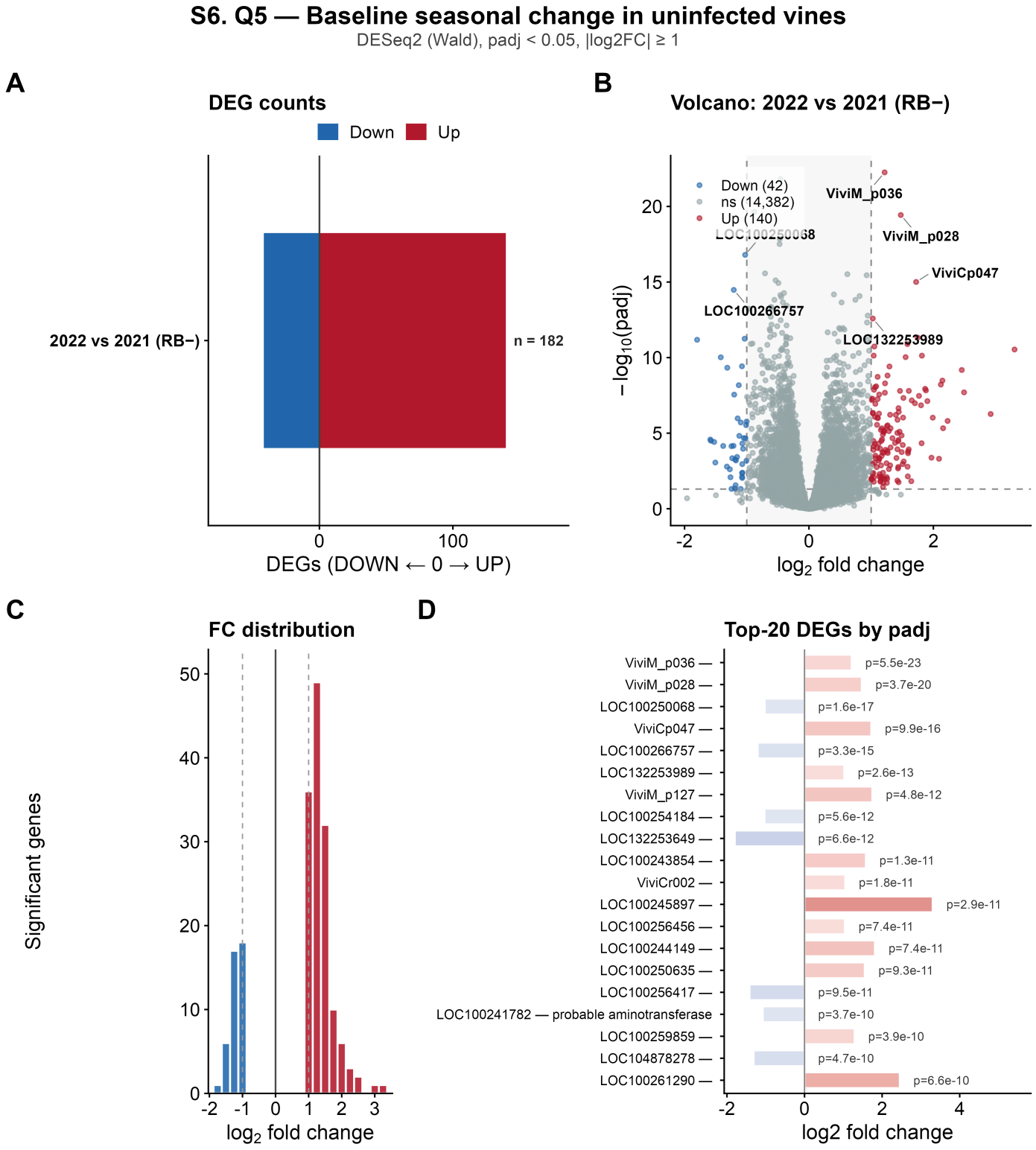
Note: See Main 2 + de_deseq2_extras Q4d



Supplementary Figure S6. Q5 baseline seasonal change in RB-

Status: Regenerated under DESeq2, R/ggplot2, Arial

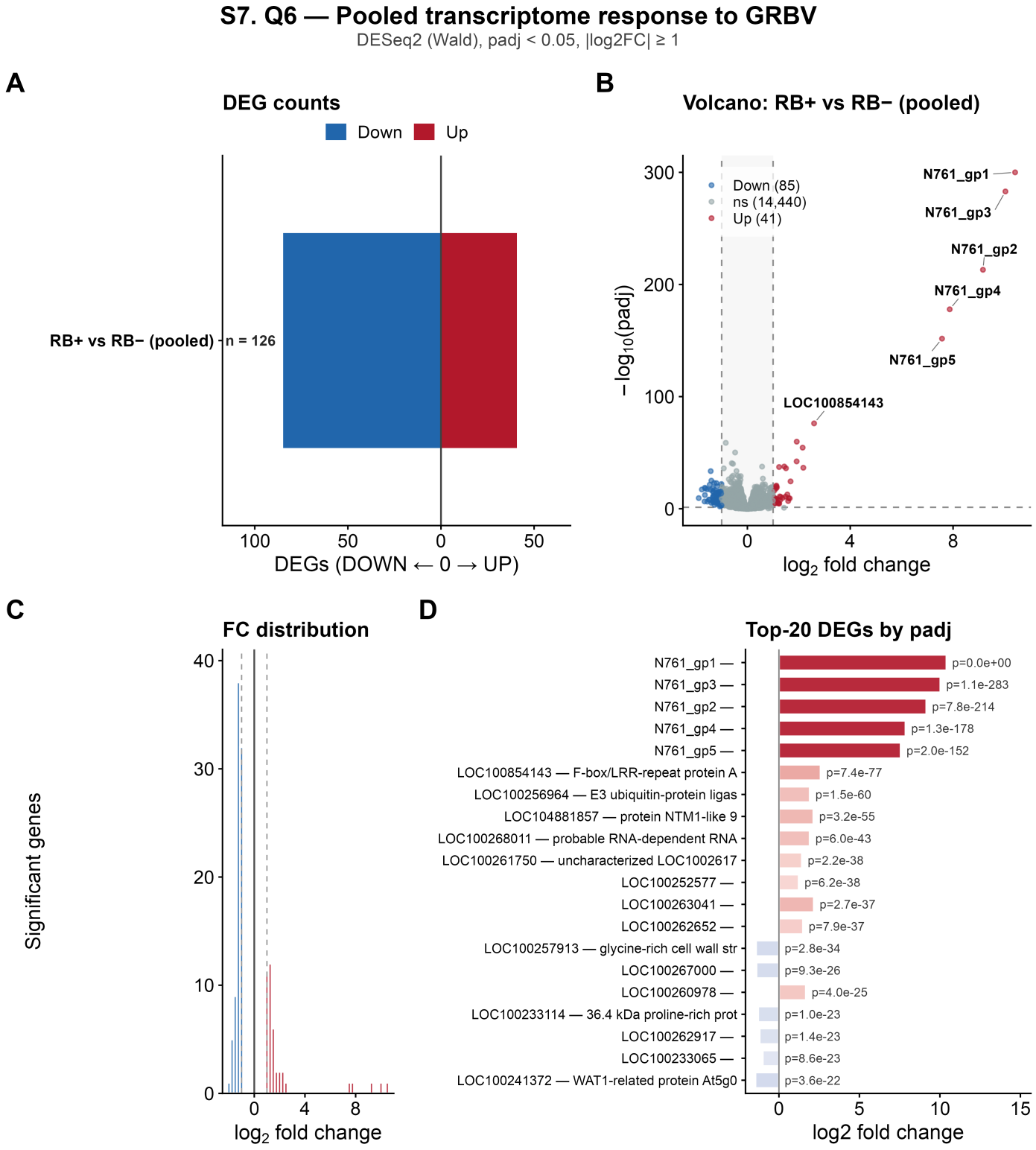
Note: See de_deseq2_voom Q5



Supplementary Figure S7. Q6 pooled transcriptome trajectory

Status: Regenerated under DESeq2, R/ggplot2, Arial

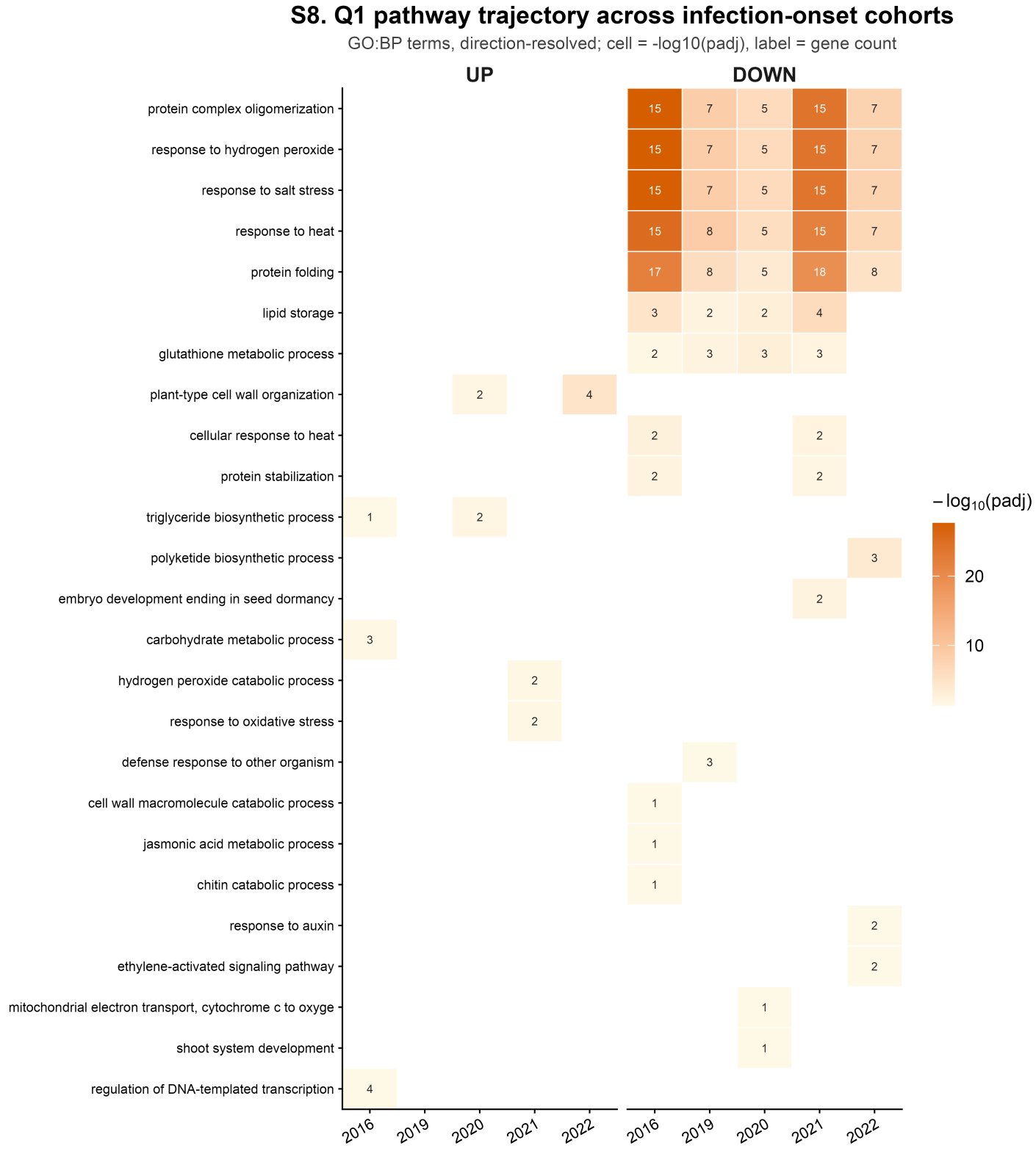
Note: See Main 2 panel B



Supplementary Figure S8. Q1 pathway enrichment

Status: Regenerated under DESeq2, R/ggplot2, Arial

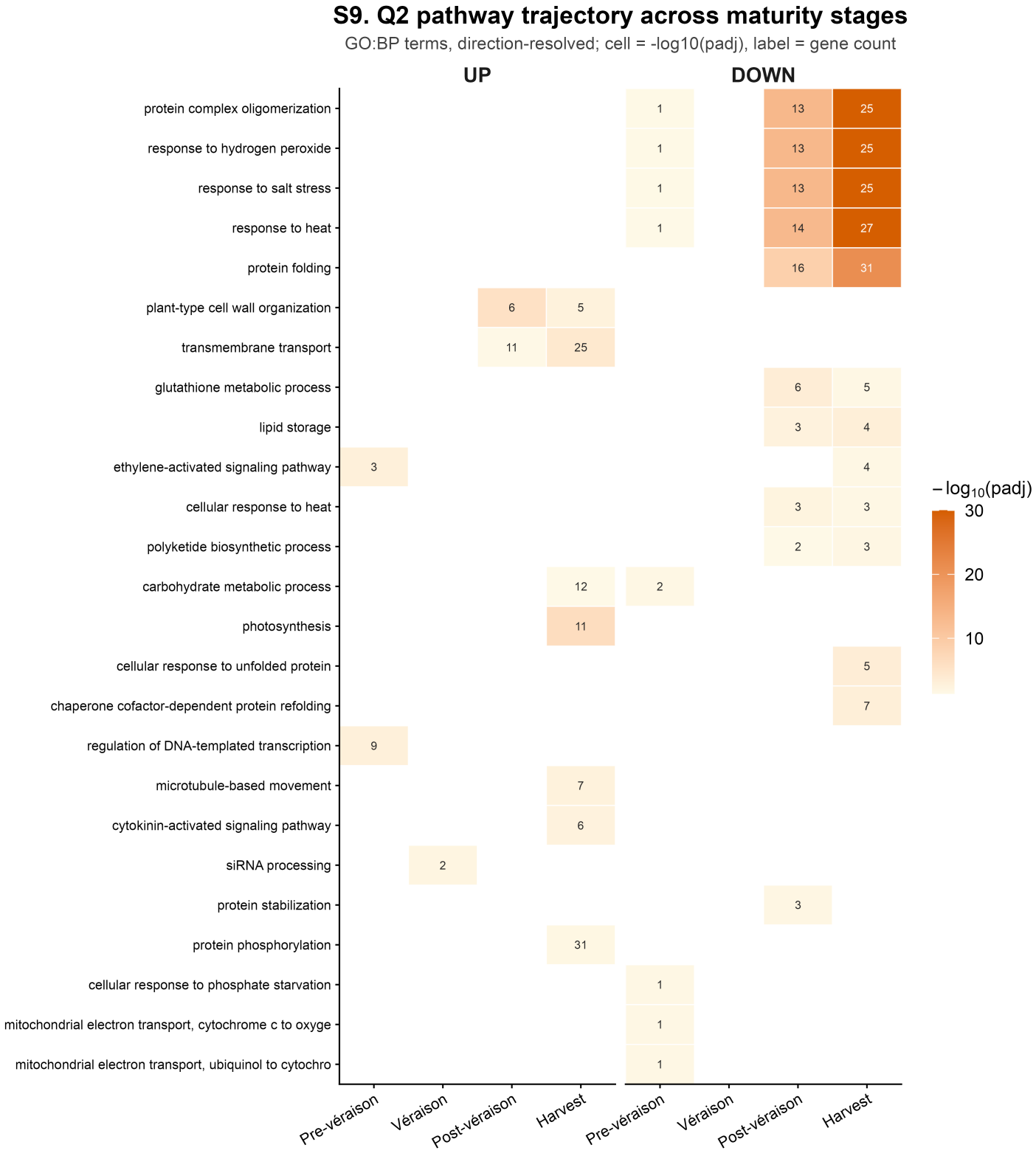
Note: Recompute on Q1a/Q1b DEG sets



Supplementary Figure S9. Q2 pathway enrichment

Status: Regenerated under DESeq2, R/ggplot2, Arial

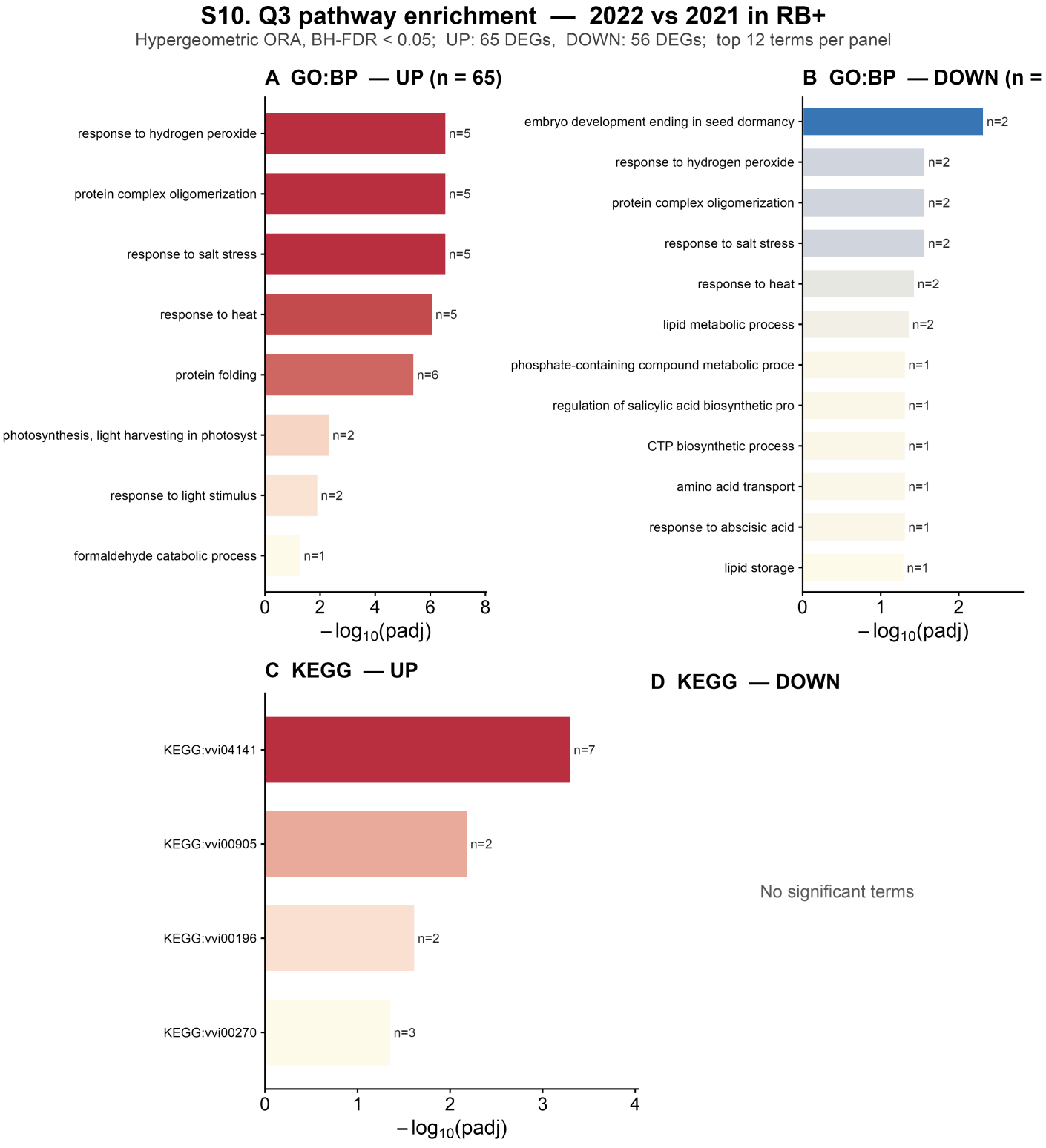
Note: Recompute on Q2 DEG sets



Supplementary Figure S10. Q3 pathway enrichment

Status: Regenerated under DESeq2, R/ggplot2, Arial

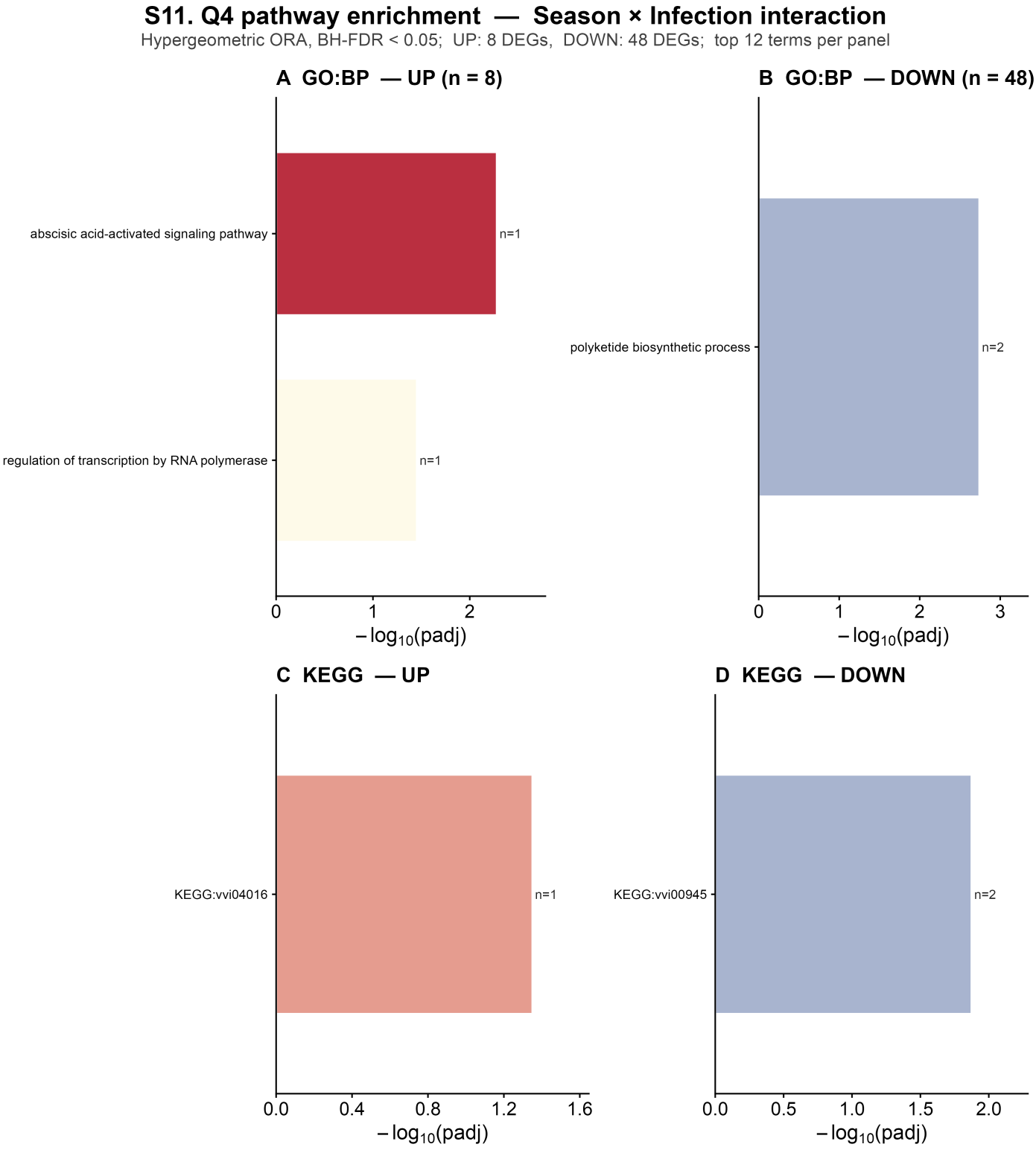
Note: Recompute on Q3 DEG sets



Supplementary Figure S11. Q4 pathway enrichment

Status: Regenerated under DESeq2, R/ggplot2, Arial

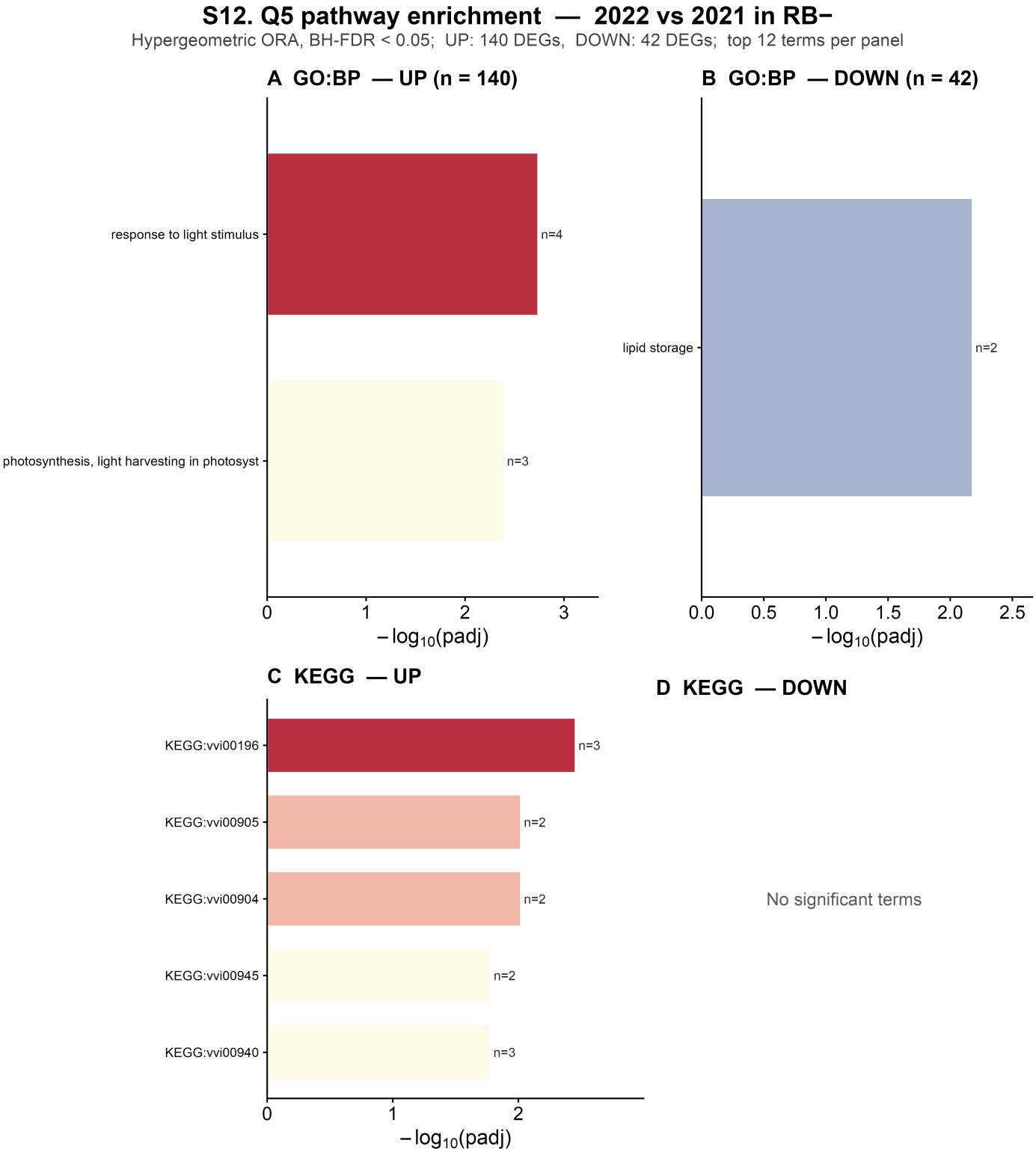
Note: Recompute on Q4d DEG sets



Supplementary Figure S12. Q5 pathway enrichment

Status: Regenerated under DESeq2, R/ggplot2, Arial

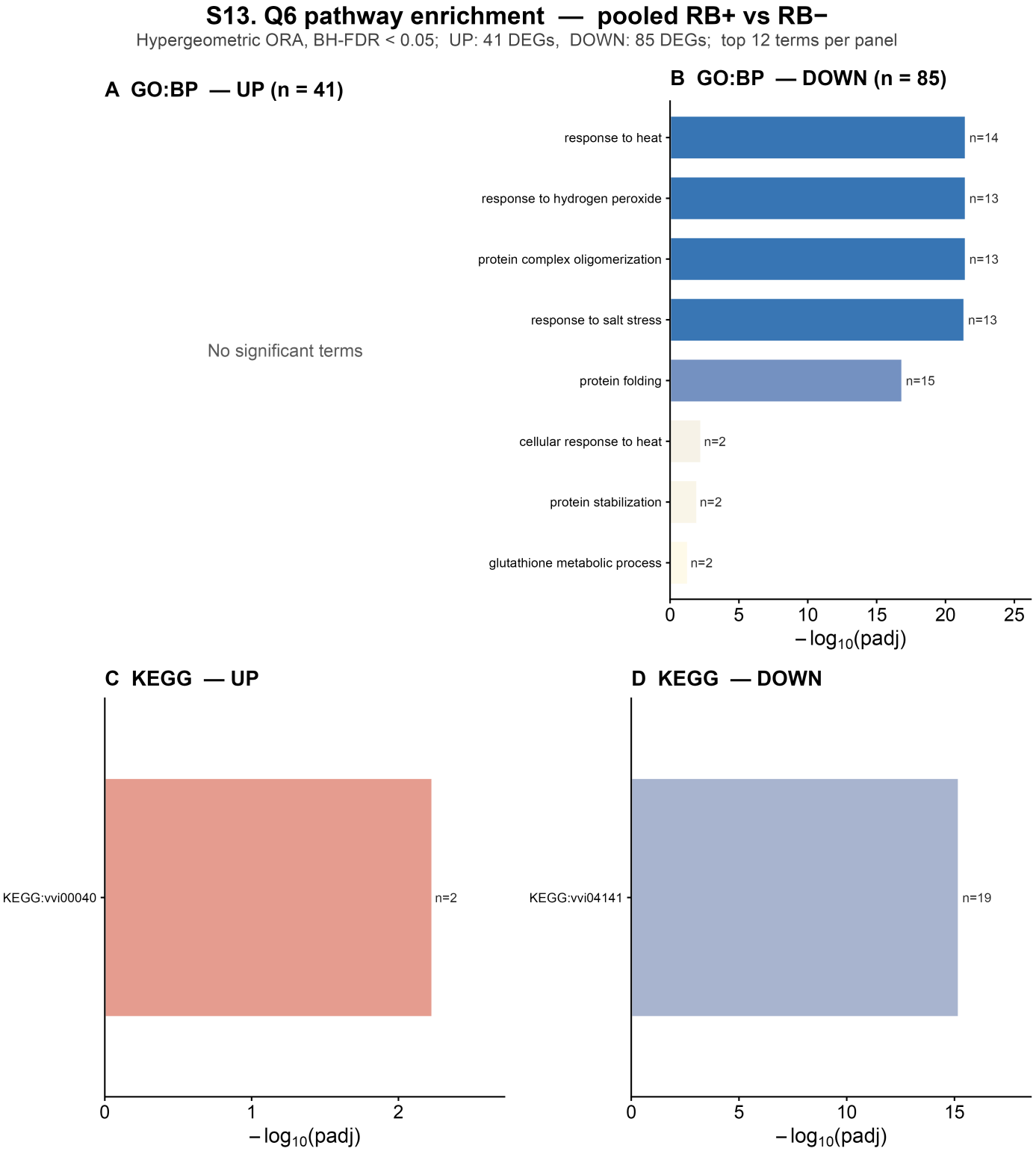
Note: Recompute on Q5 DEG sets



Supplementary Figure S13. Q6 pathway enrichment

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Recompute on Q6 DEG sets



Supplementary Figure S14. Master pathway landscape (all questions)

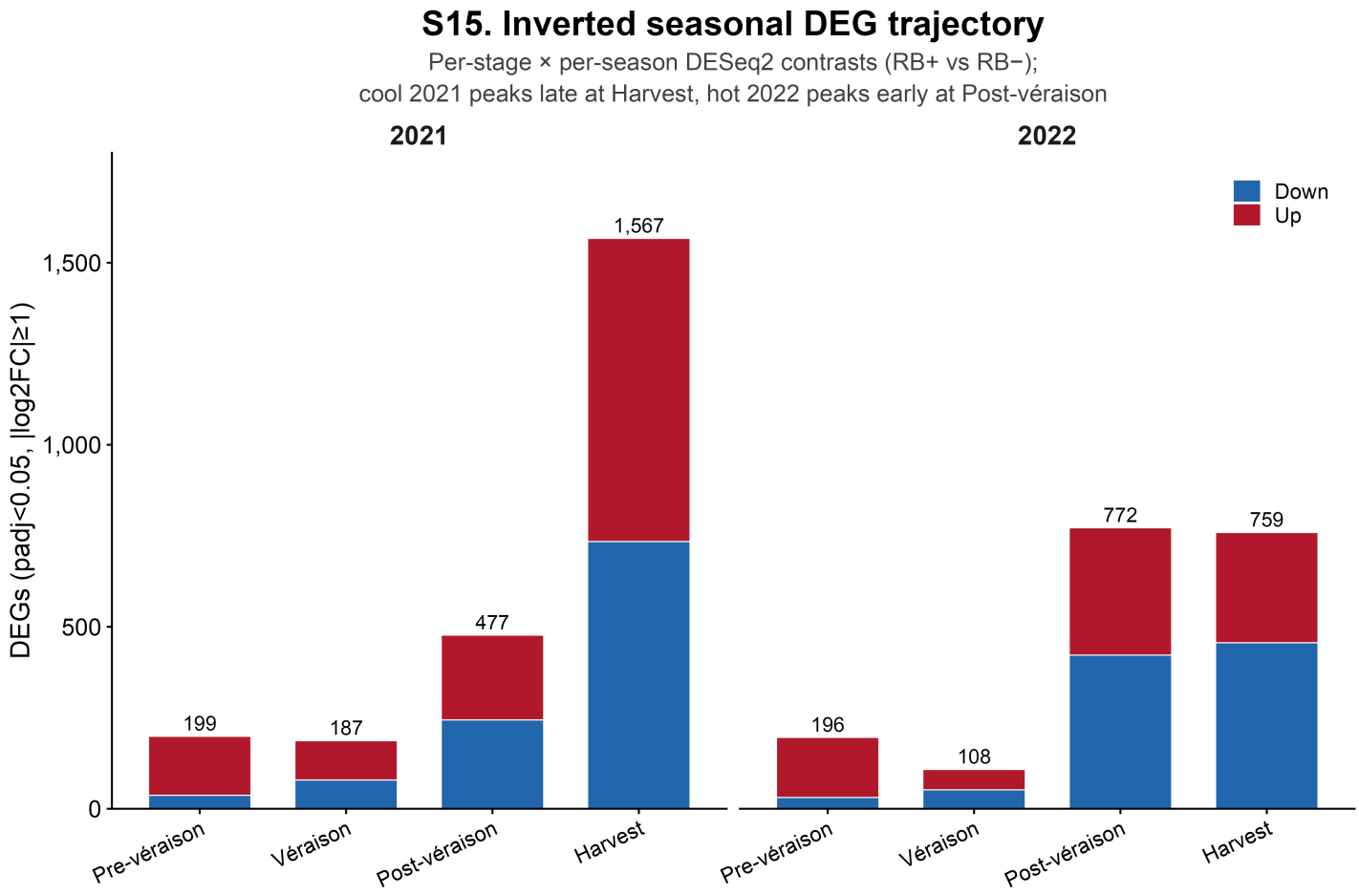
Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Aggregate of S8-S13

Supplementary Figure S15. Inverted seasonal trajectory (key)

Status: Regenerated under DESeq2, R/ggplot2, Arial

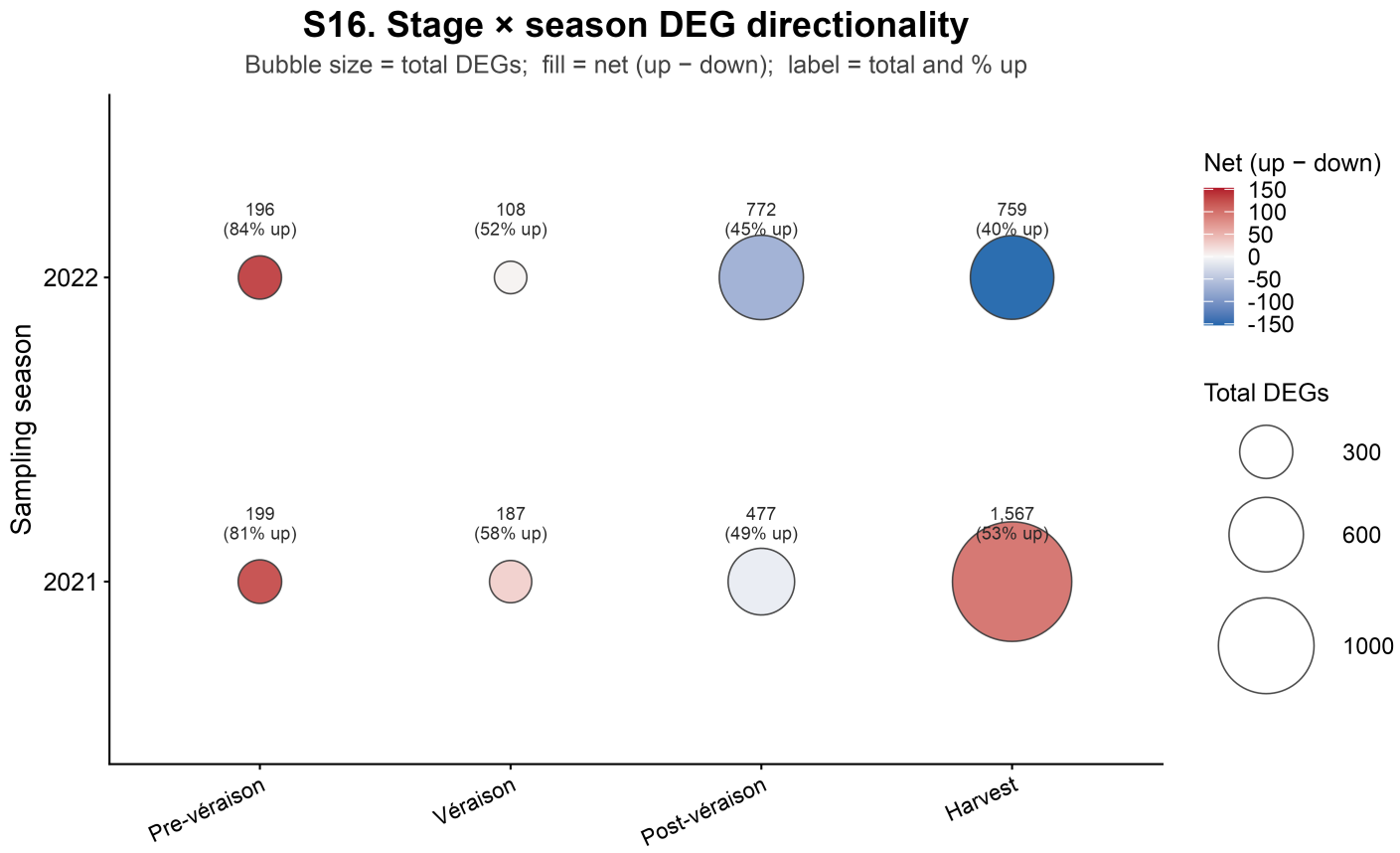
Note: See Main 3 panel A



Supplementary Figure S16. Stage × season pathway enrichment grid

Status: Regenerated under DESeq2, R/ggplot2, Arial

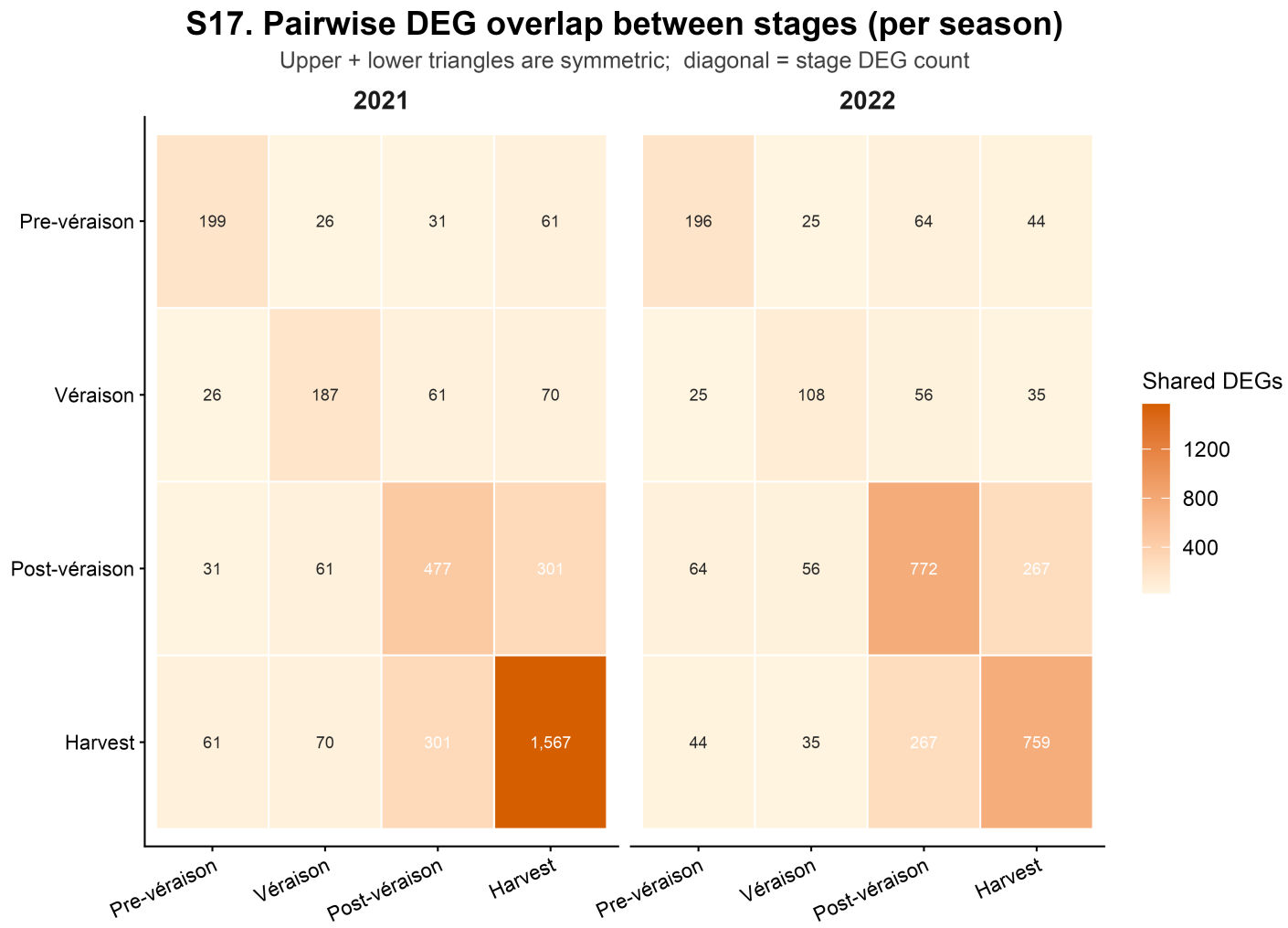
Note: Recompute from de_deseq2_extras Q2x



Supplementary Figure S17. Per-season pathway landscape heatmaps

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Recompute



Supplementary Figure S18. Stage transitions PCA

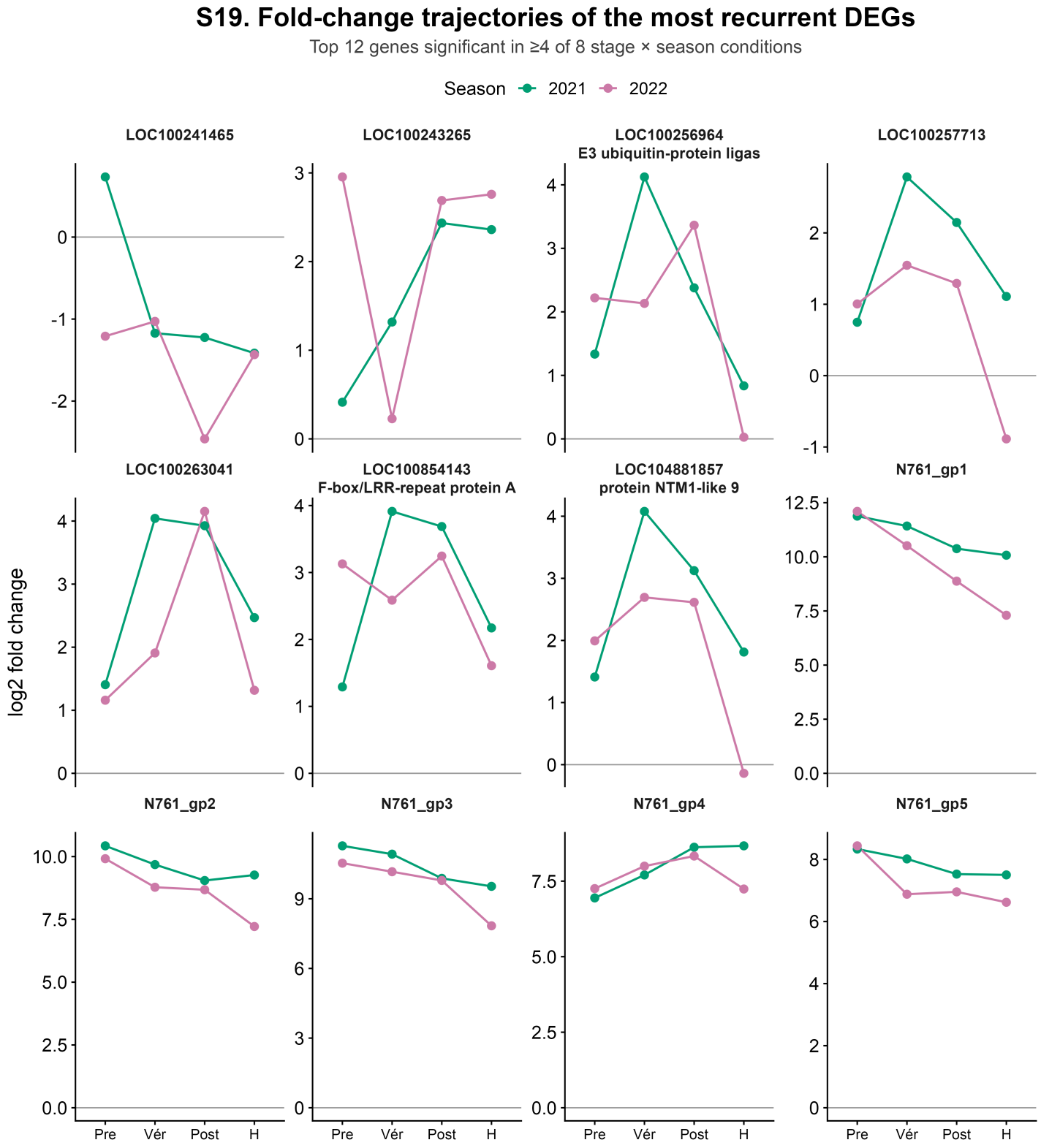
Status: Data current (kept as-is, restyle pending)

[file not found: 18_seasonal_maturity_transitions_PCA.png]

Supplementary Figure S19. Pathway detail per stage

Status: Regenerated under DESeq2, R/ggplot2, Arial

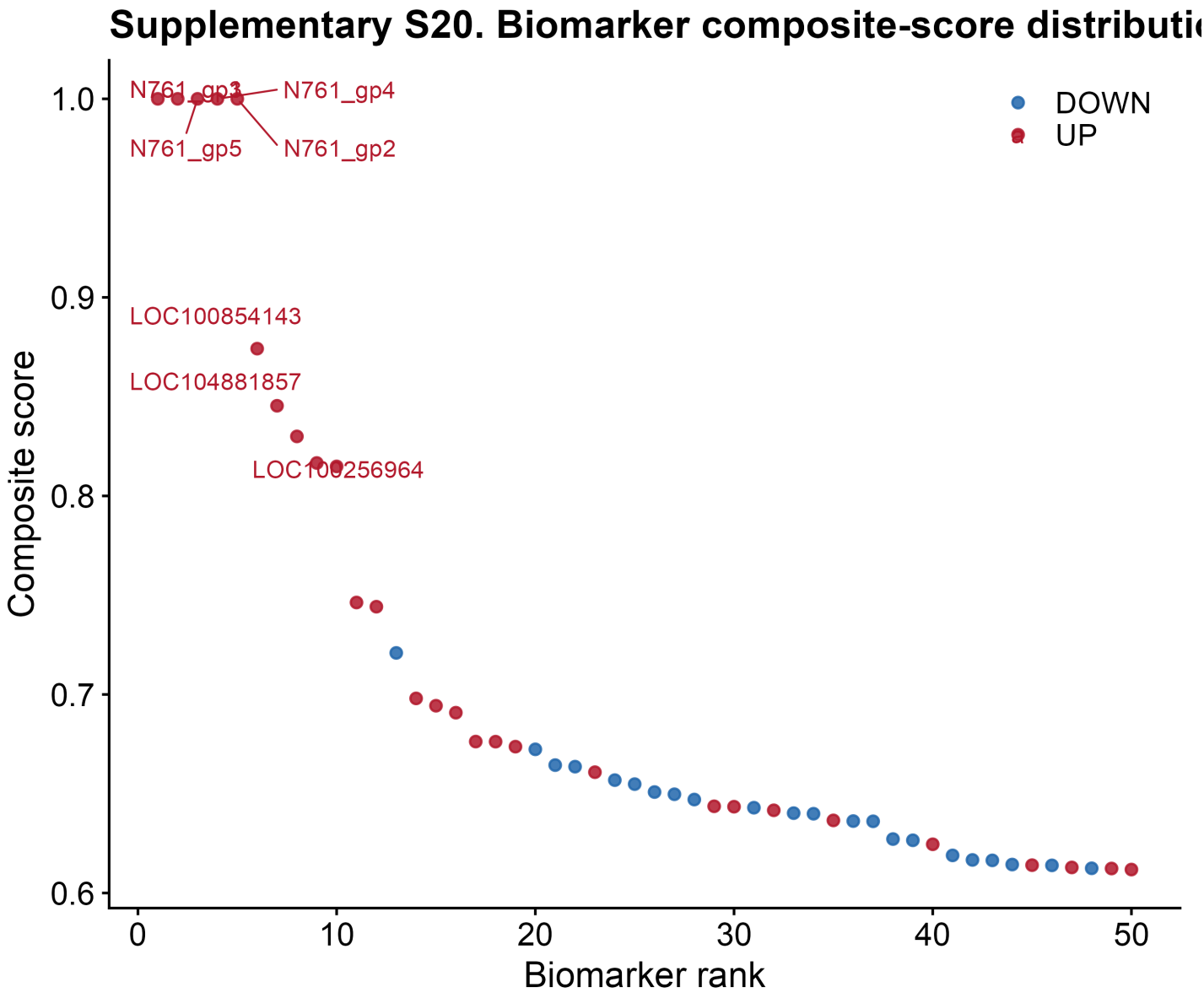
Note: Recompute



Supplementary Figure S20. Biomarker overview (scores, AUC, panel)

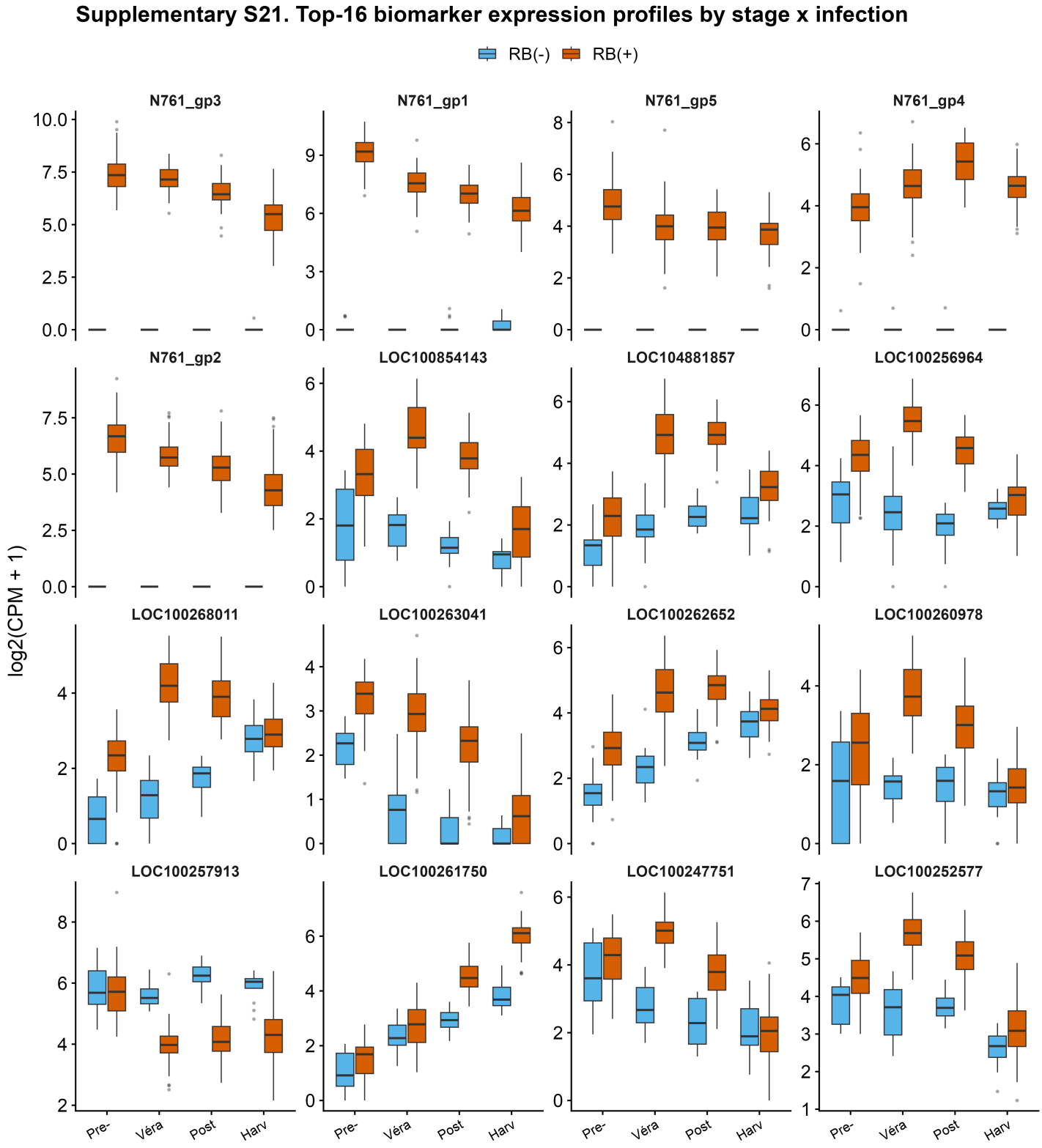
Status: Reframed in Main composite (see note)

Note: See Main 6



Supplementary Figure S21. Top-16 biomarker expression profiles

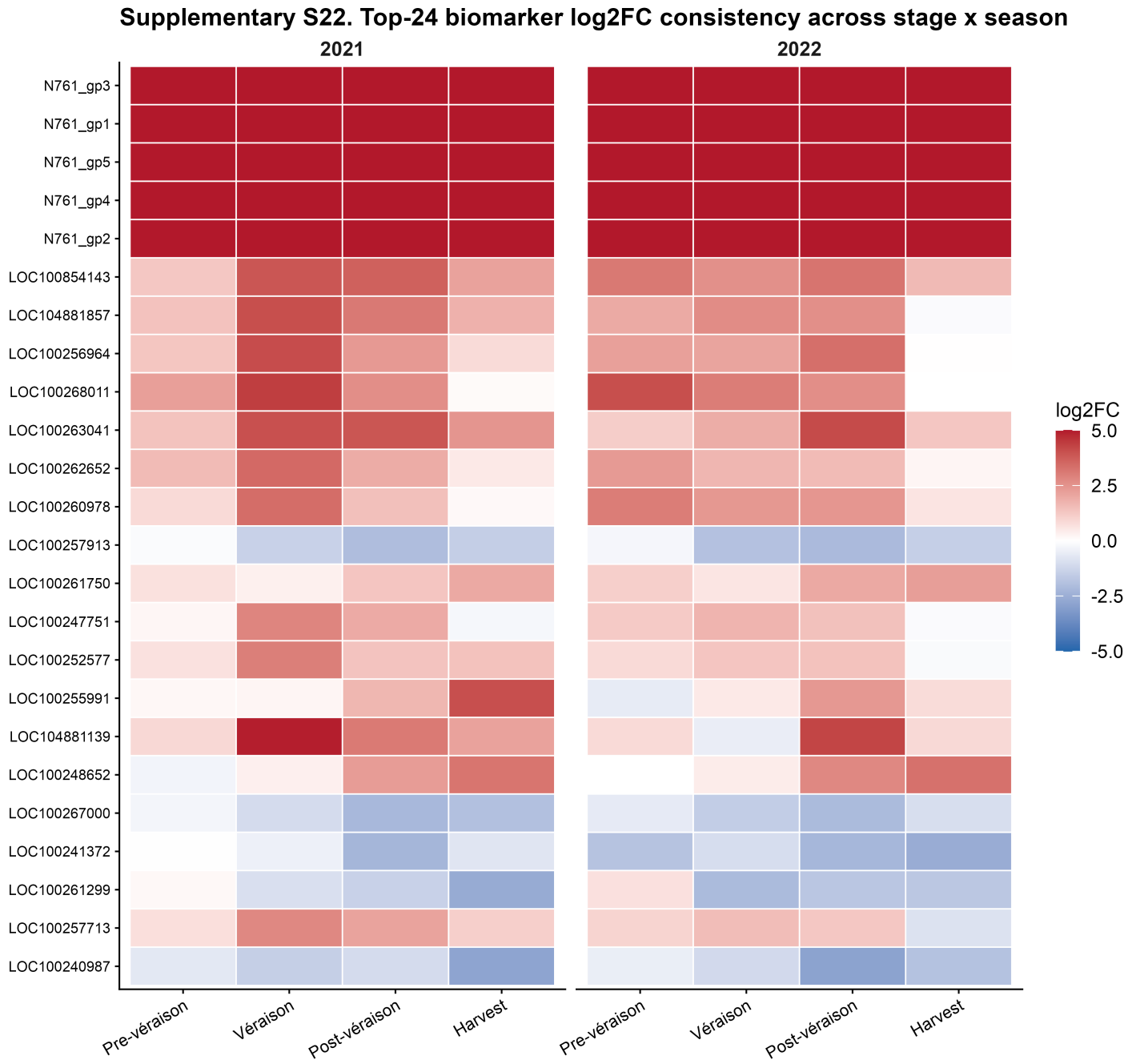
Status: Data current (kept as-is, restyle pending)



Supplementary Figure S22. Biomarker FC consistency + AUC stability

Status: Reframed in Main composite (see note)

Note: See Main 6 panel A/B



Supplementary Figure S23. Pipeline analysis schematic

Status: Data current (kept as-is, restyle pending)

[file not found: 23_analysis_schematic.png]

Supplementary Figure S24. Biomarker positions on chromosome map

Status: Data current (kept as-is, restyle pending)

Note: See Main 8 panel B

[file not found: 24_biomarker_QTL_chromosome_map.png]

Supplementary Figure S25. QTL overlap summary (no permutation null)

Status: Reframed in Main composite (see note)

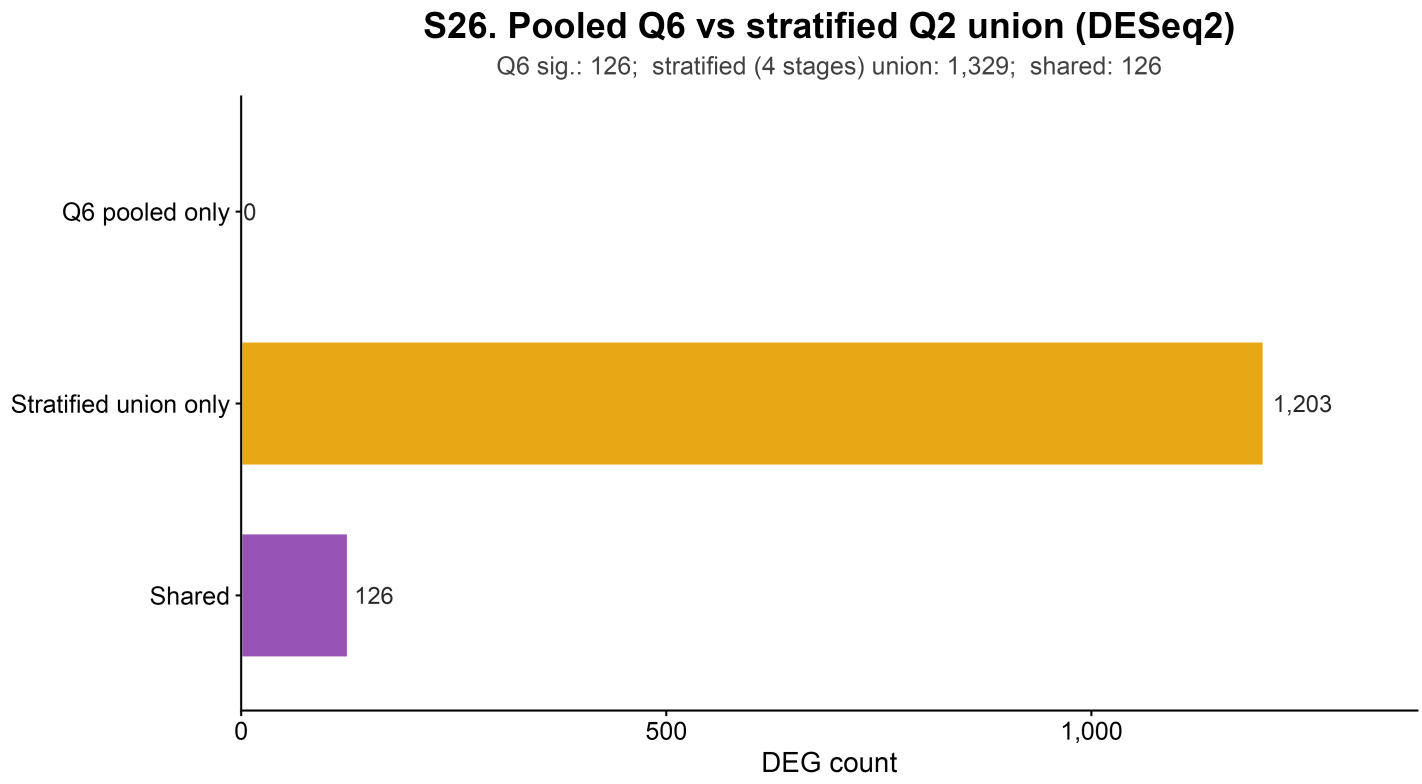
Note: See Main 8 panel C

[file not found: 25_biomarker_QTL_crossref_summary.png]

Supplementary Figure S26. Pooled vs stratified DE concordance

Status: Regenerated under DESeq2, R/ggplot2, Arial

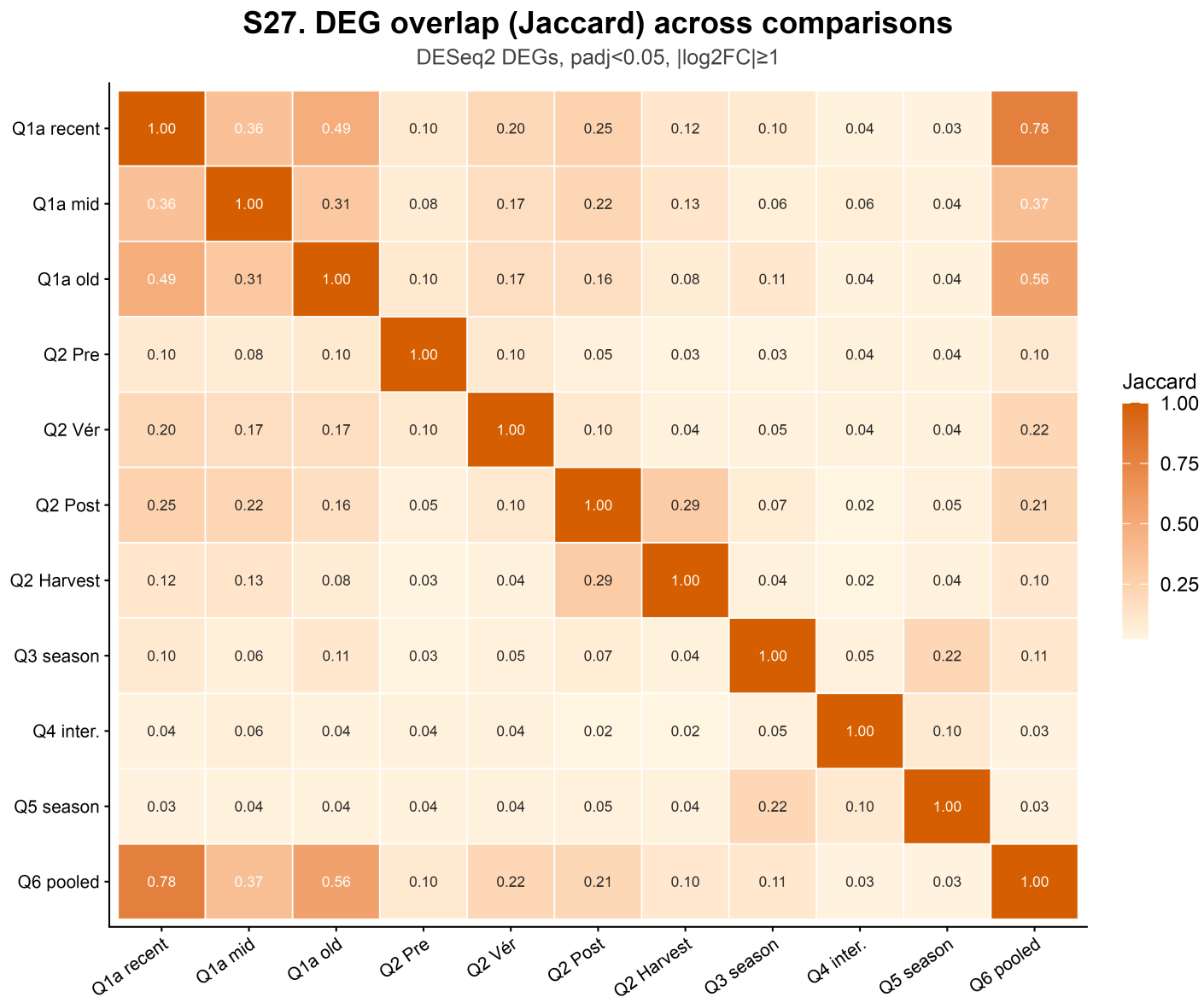
Note: Recompute with de_deseq2_voom



Supplementary Figure S27. DEG overlap across questions

Status: Regenerated under DESeq2, R/ggplot2, Arial

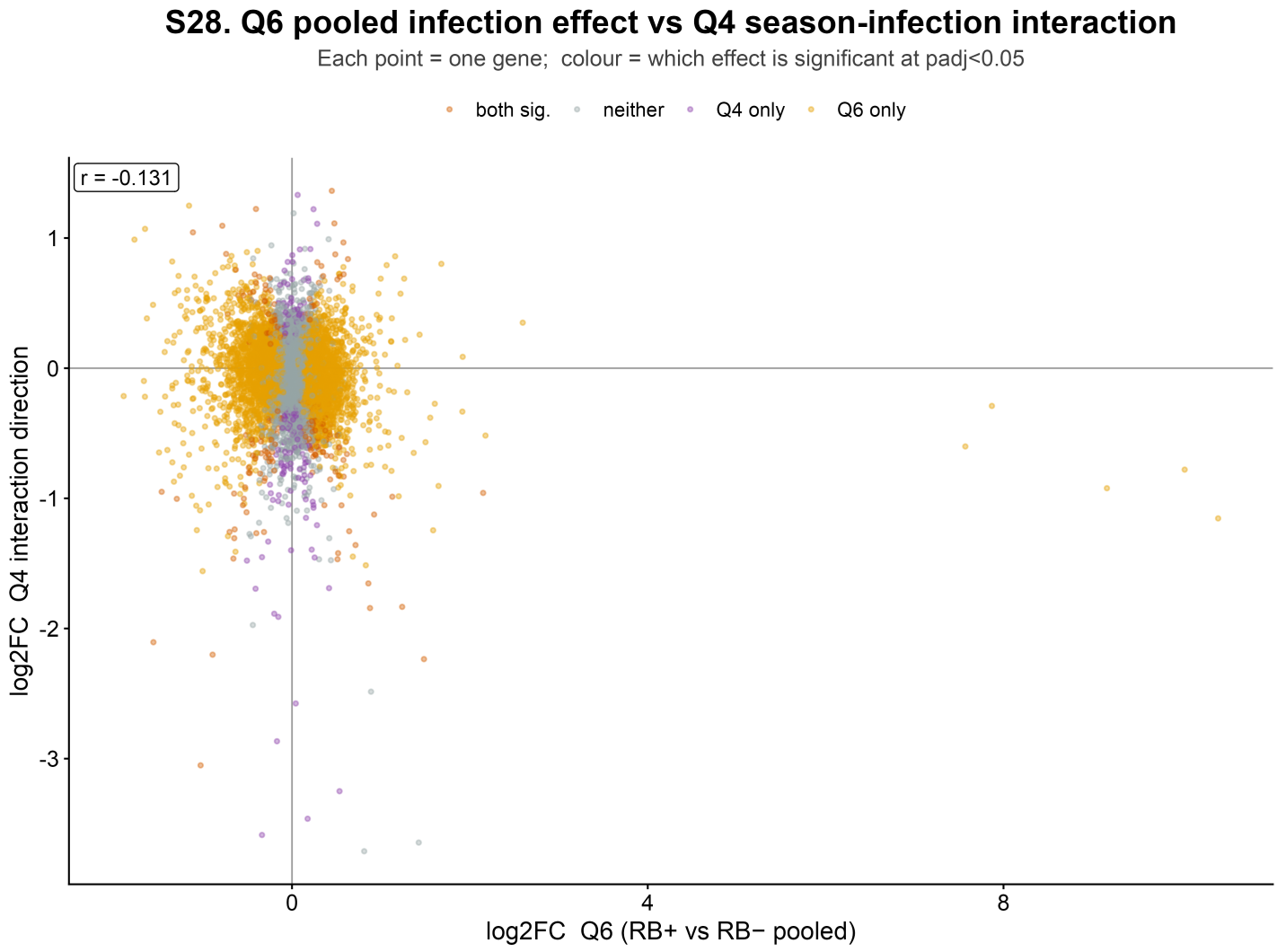
Note: Recompute with de_redo/ + de_extras



Supplementary Figure S28. Cross-question DEG cross-reference

Status: Regenerated under DESeq2, R/ggplot2, Arial

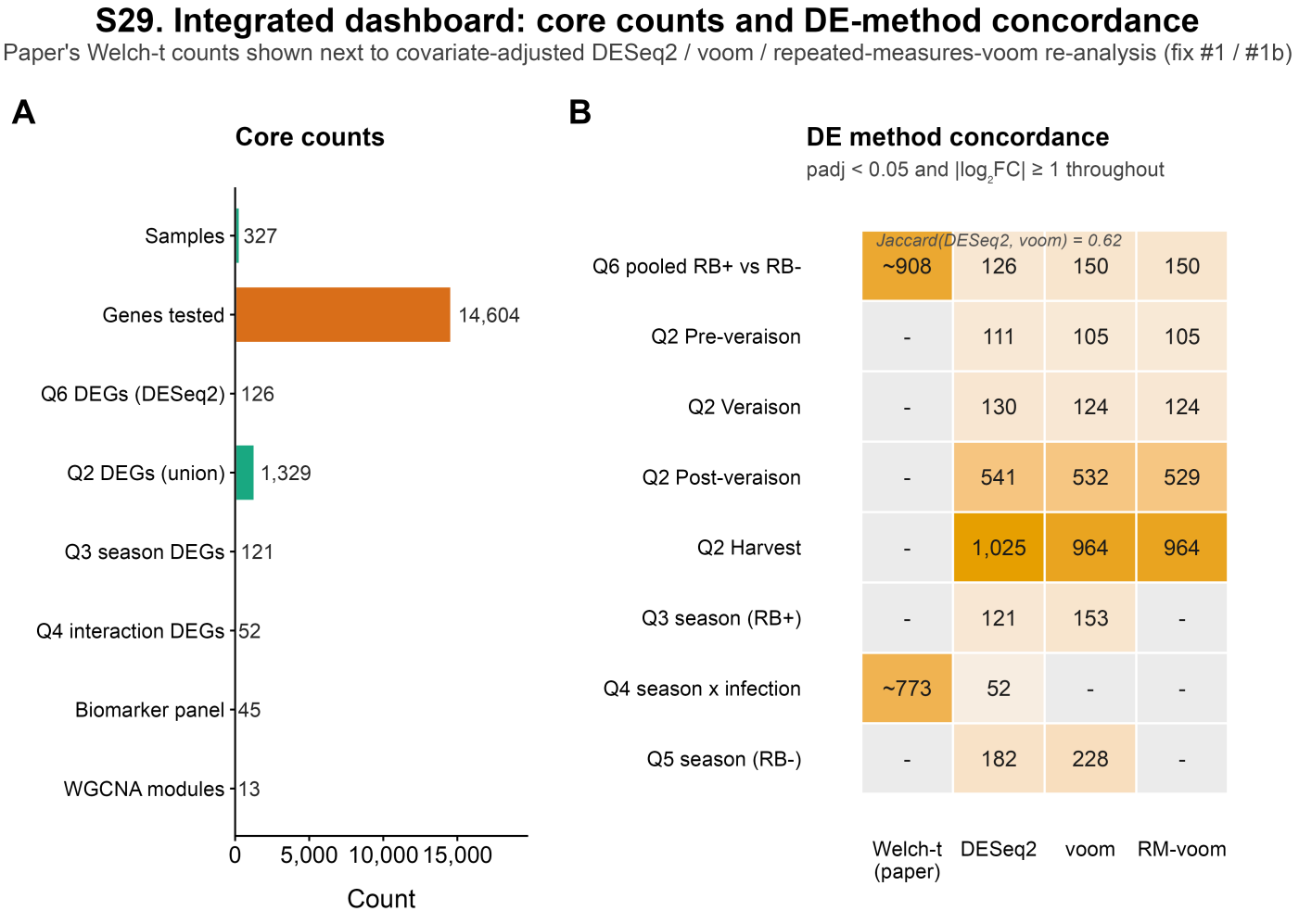
Note: Recompute



Supplementary Figure S29. Integrated findings dashboard

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Recompute



Supplementary Figure S30. Weather overview at sampling

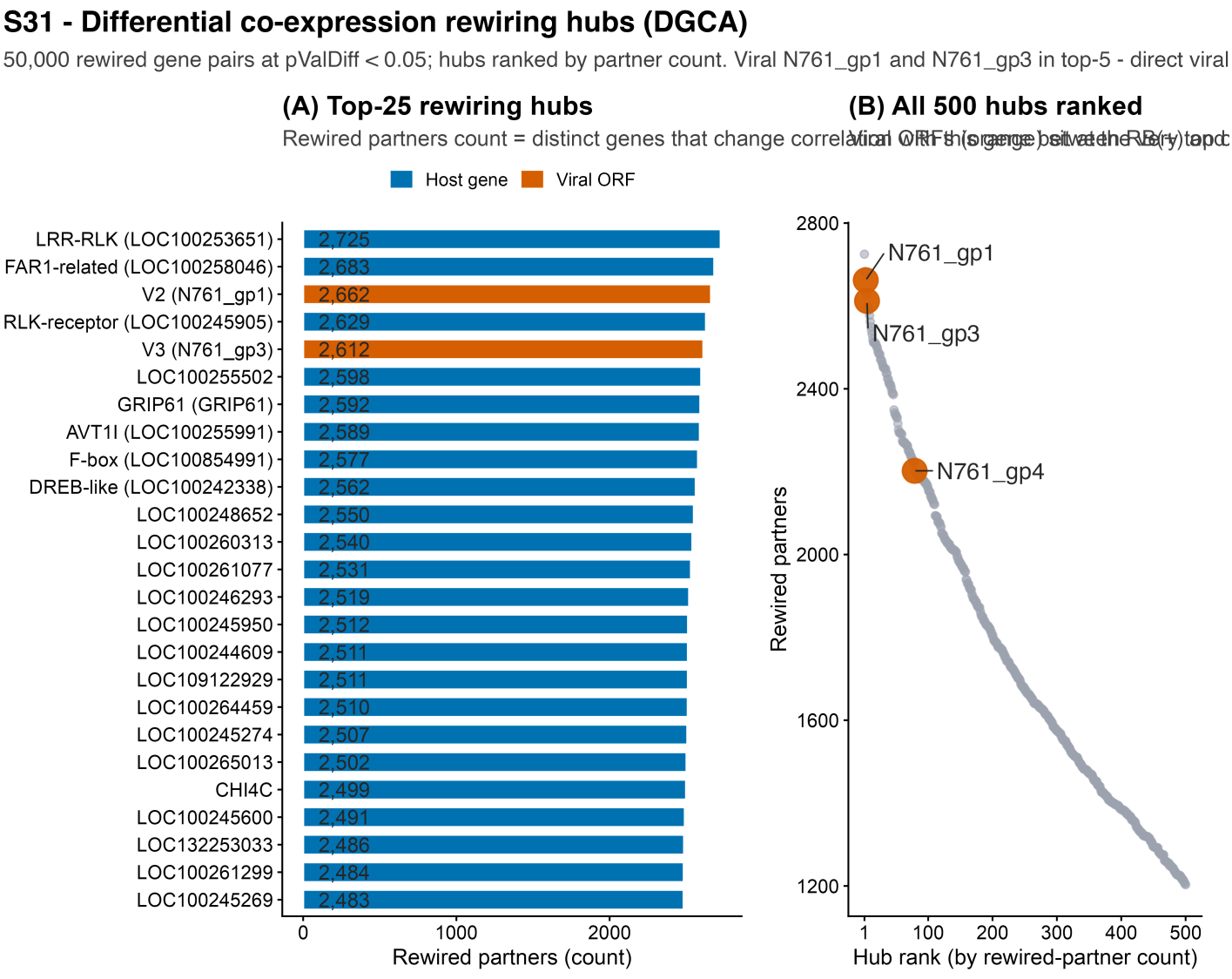
Status: Data current (kept as-is, restyle pending)

[file not found: 30_weather_overview_sampling.png]

Supplementary Figure S31. Viral titer dynamics

Status: Data current (kept as-is, restyle pending)

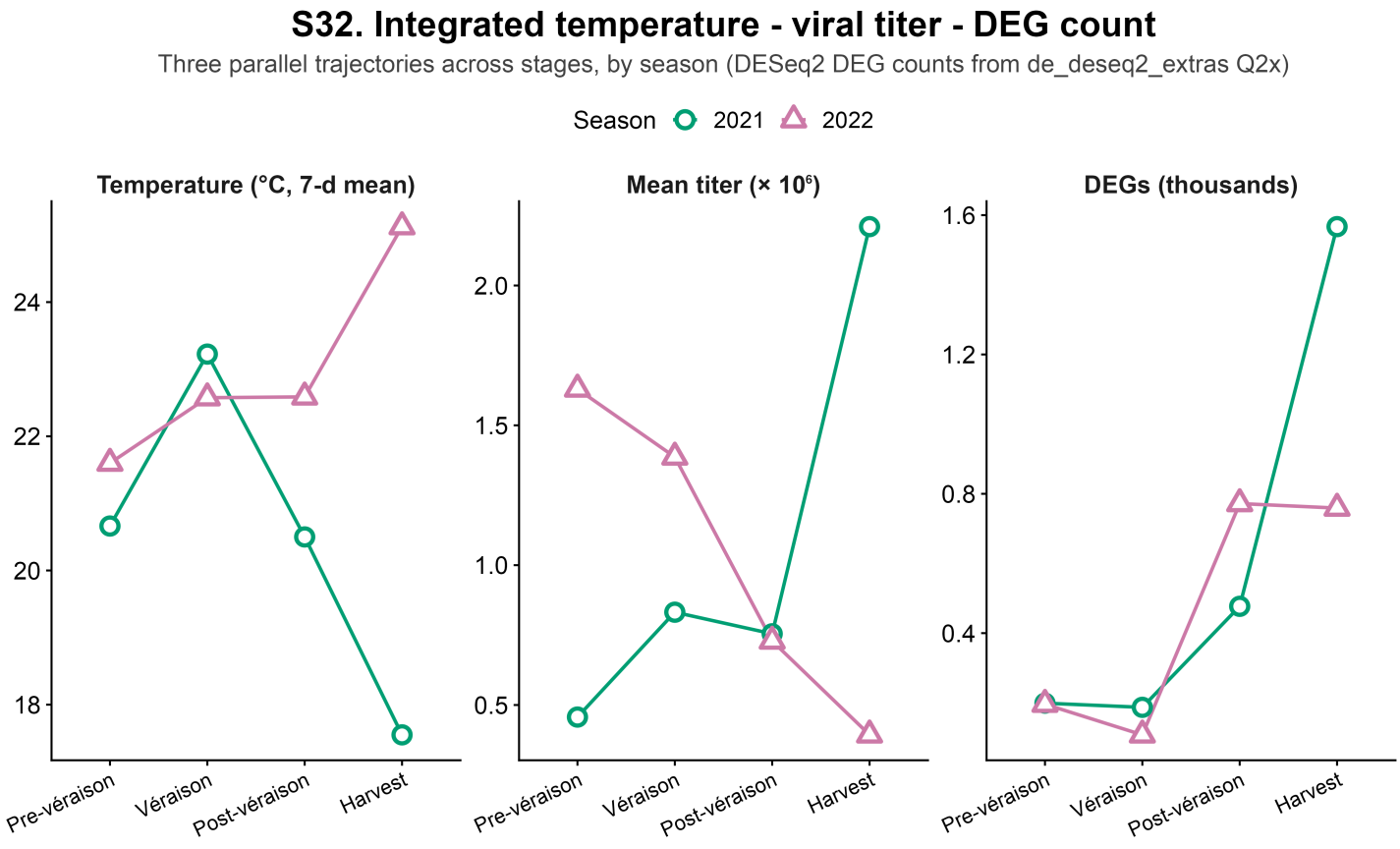
Note: See Main 3 panel C



Supplementary Figure S32. Integrated temperature-titer-DEG

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Recompute with de_deseq2_extras



Supplementary Figure S33. Temperature-titer correlations

Status: Data current (kept as-is, restyle pending)

S33 - WGCNA module preservation under GRBV infection

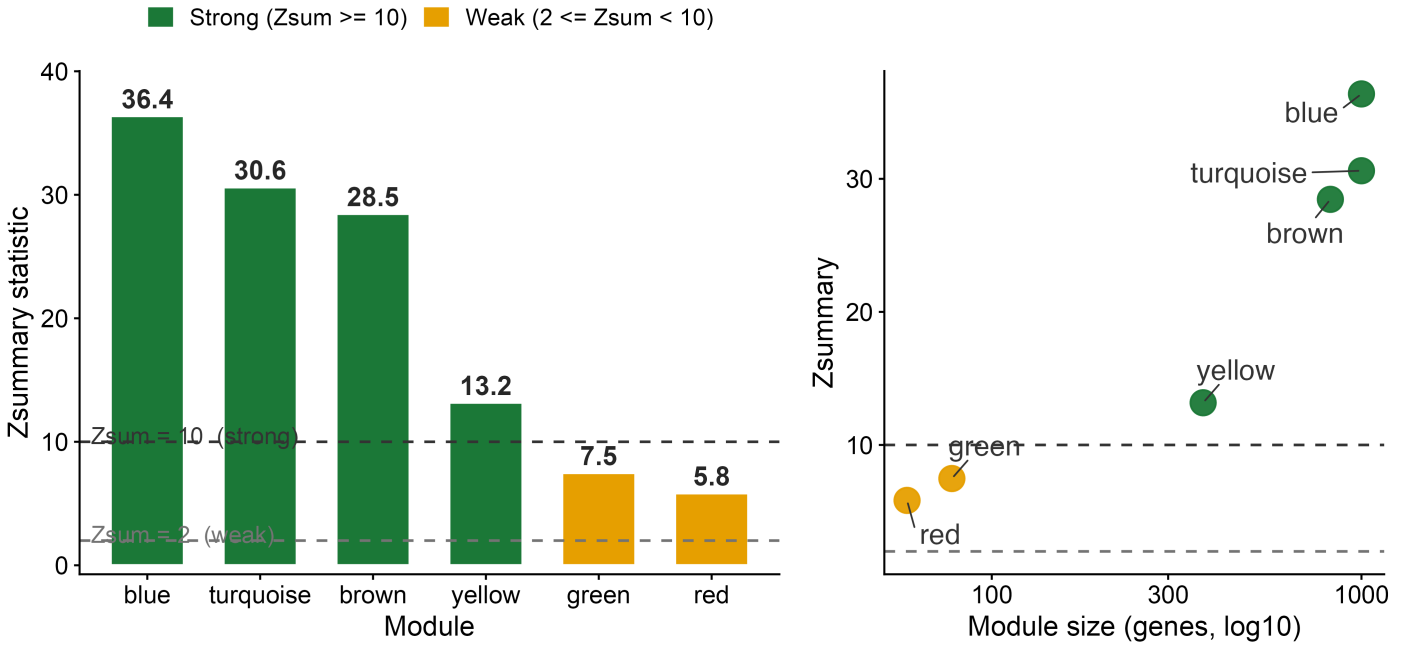
6-module preservation build (power = 6, corType = pearson); distinct from the 13-module main-paper build (power = 30, corType

(A) Module preservation Zsummary

All 6 modules from the preservation build (power=6, pearson) pres

(B) Module size vs preservation st

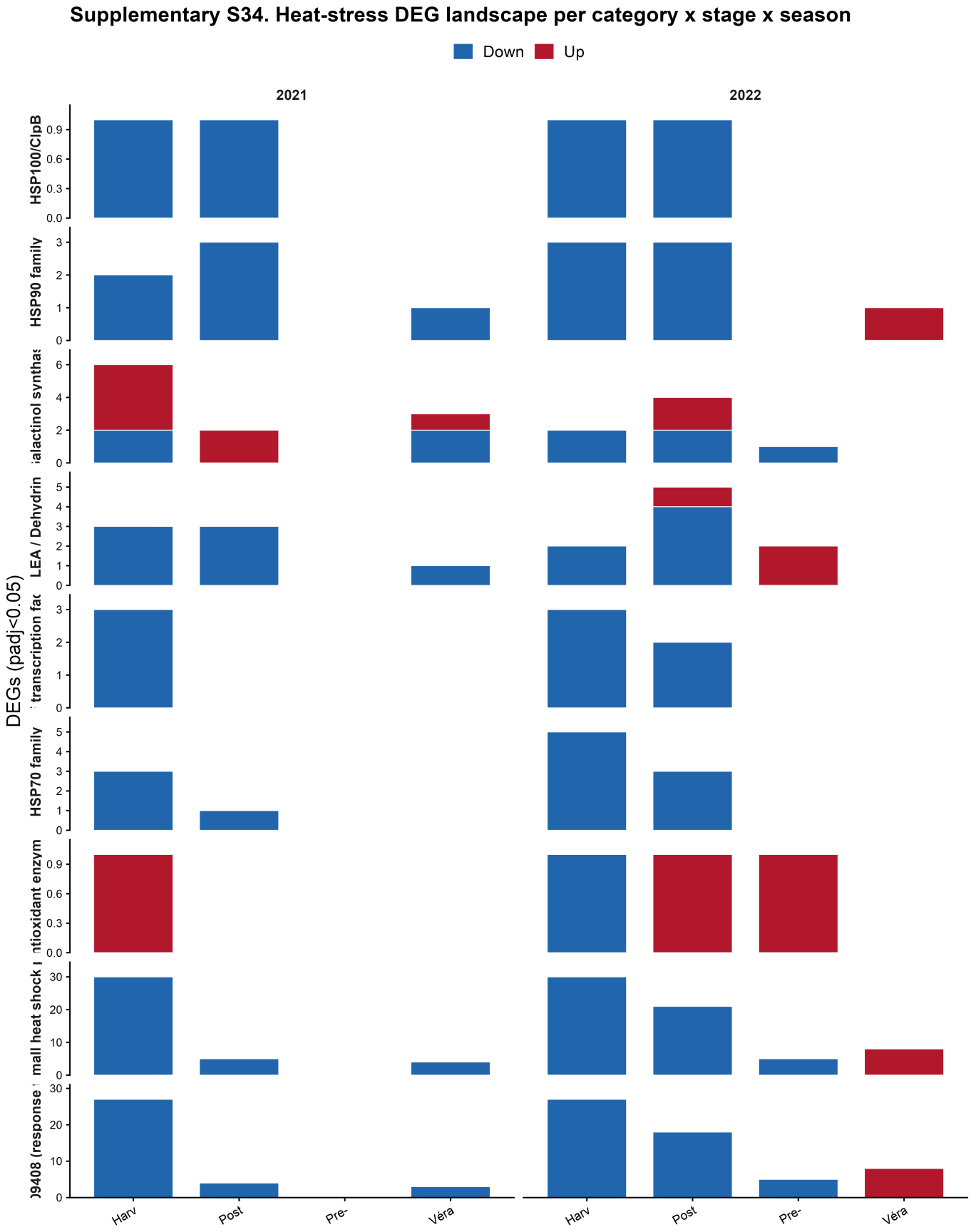
Larger modules tend toward higher Zsum;



Supplementary Figure S34. Heat-gene landscape across categories

Status: Data current (kept as-is, restyle pending)

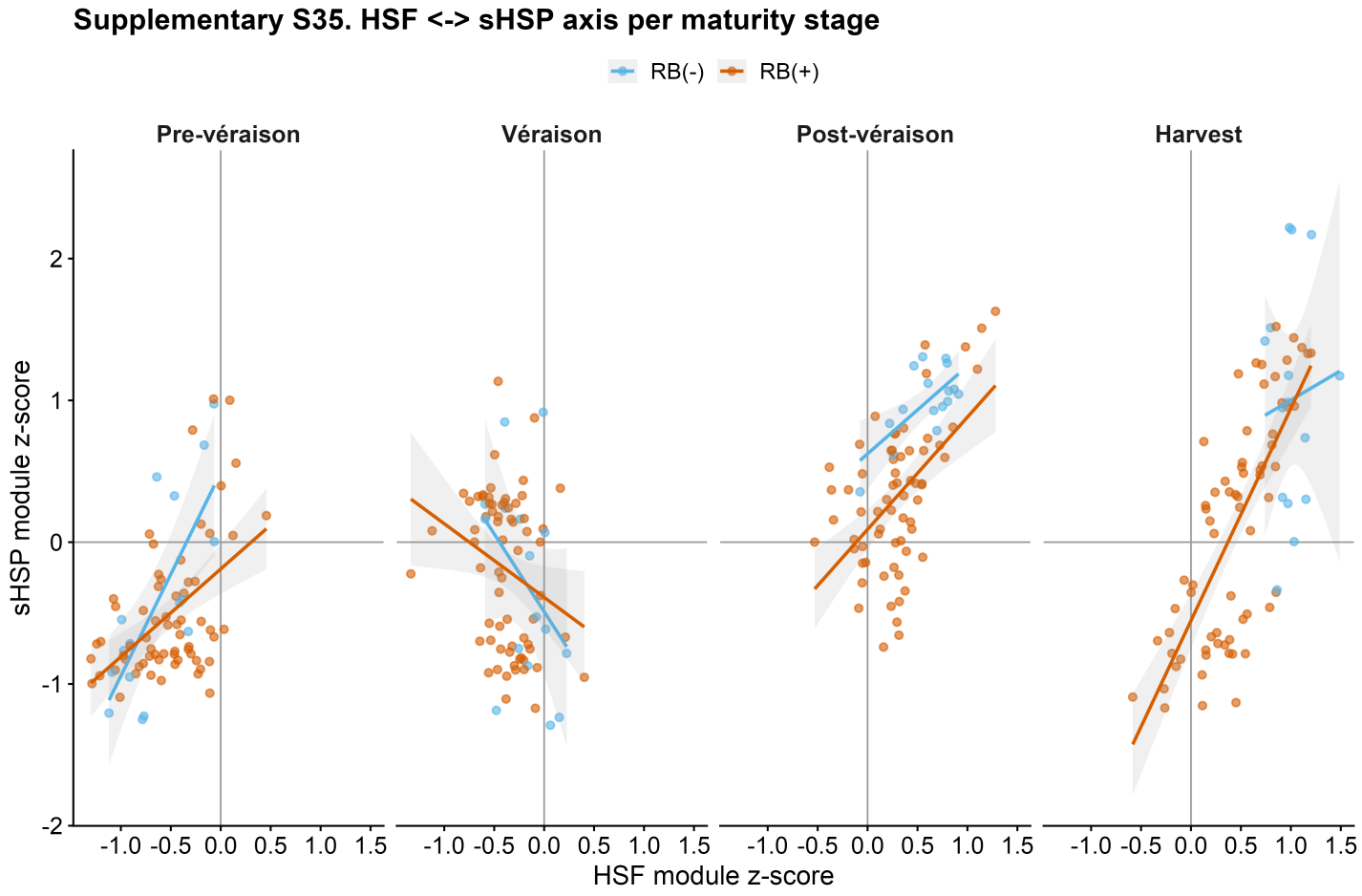
Note: See Main 4 panel A



Supplementary Figure S35. Heat vs virus disentanglement (HSF-sHSP axis)

Status: Data current (kept as-is, restyle pending)

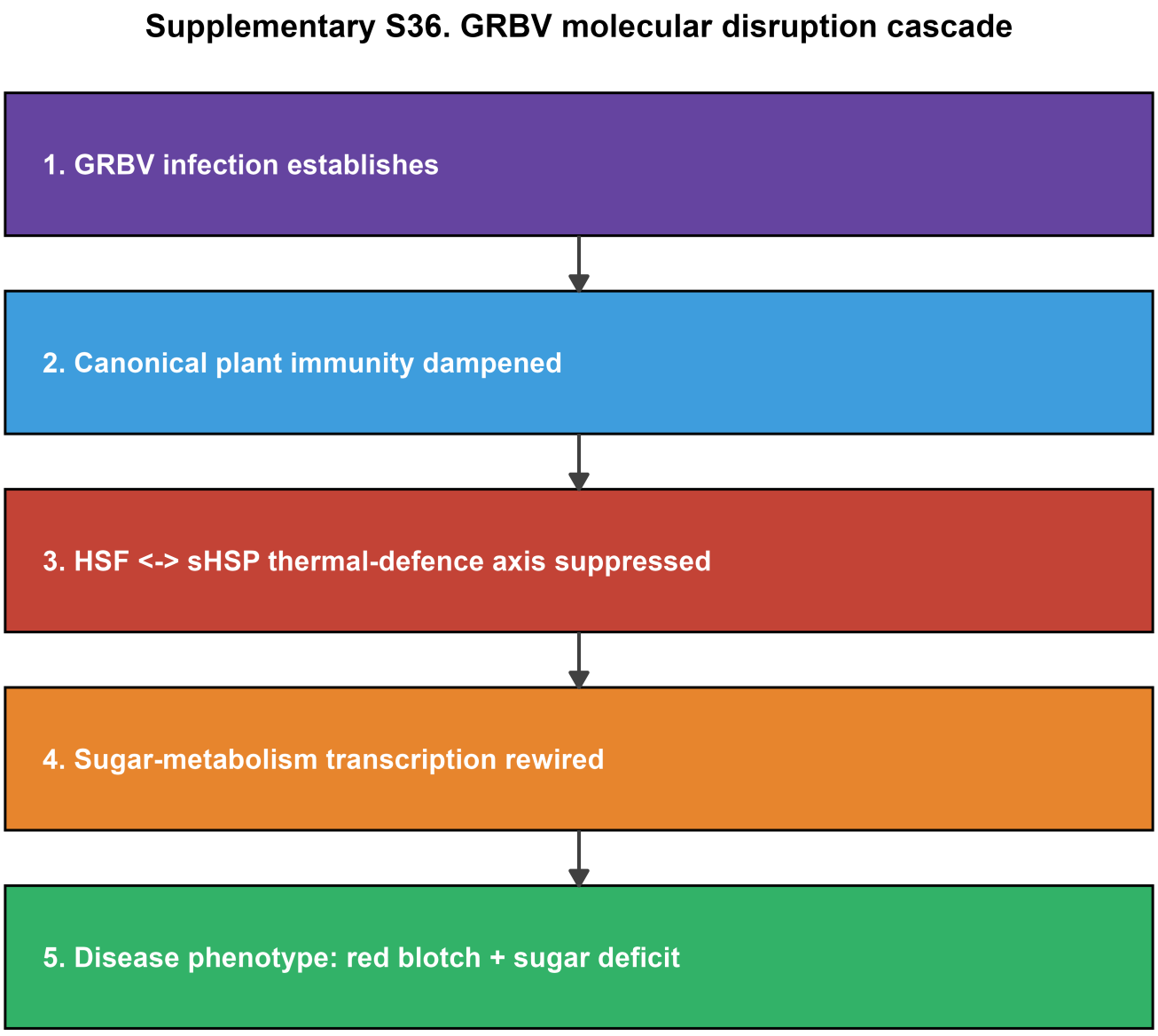
Note: See Main 4 panel B



Supplementary Figure S36. Integrated biology infographic

Status: Data current (kept as-is, restyle pending)

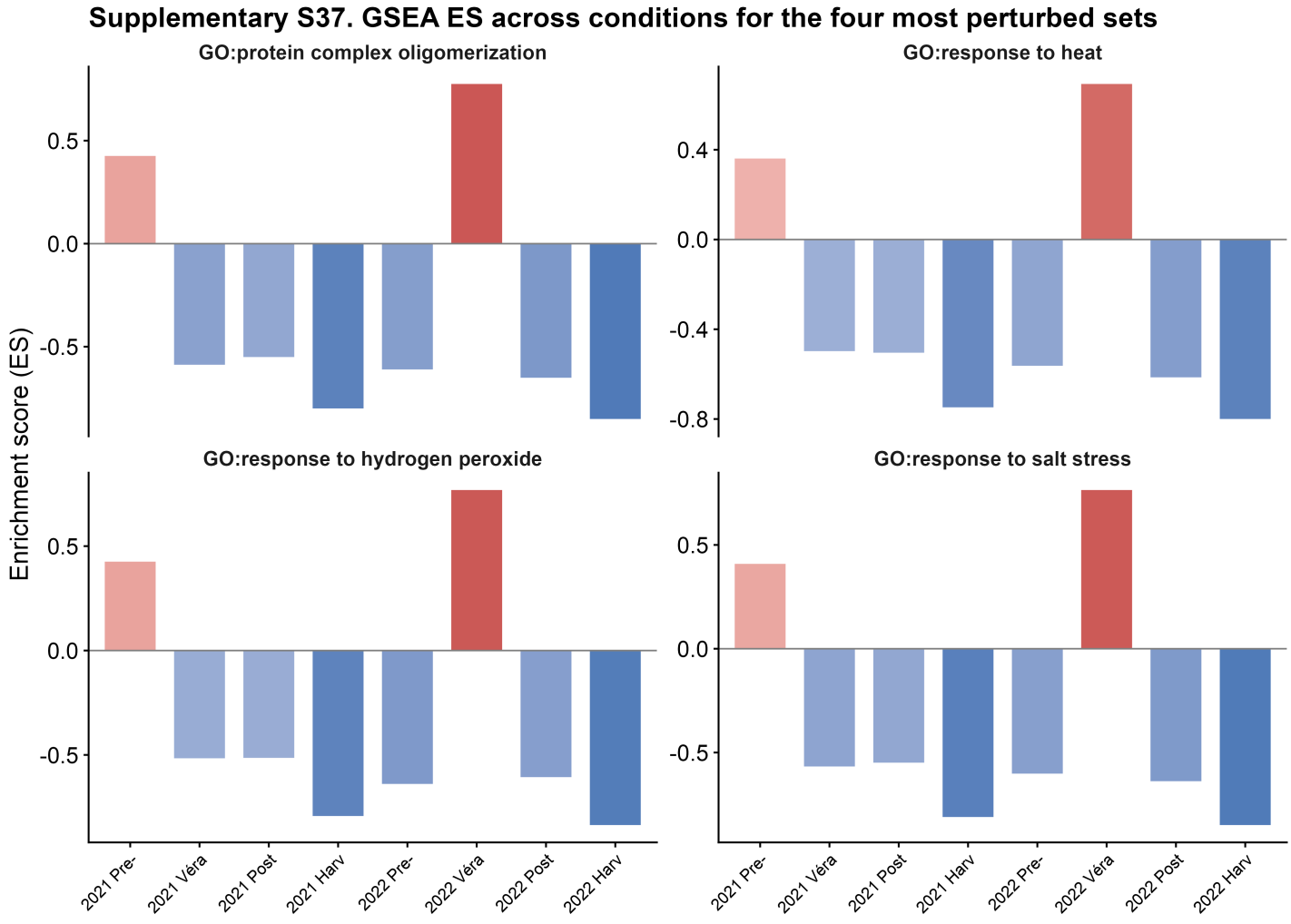
Note: See Main 10 (cleaner cascade)



Robust temperature -> titer mediators: MBF1c (LOC100249249), SPS1 (LOC100232974), uncharacterised LOC100266309

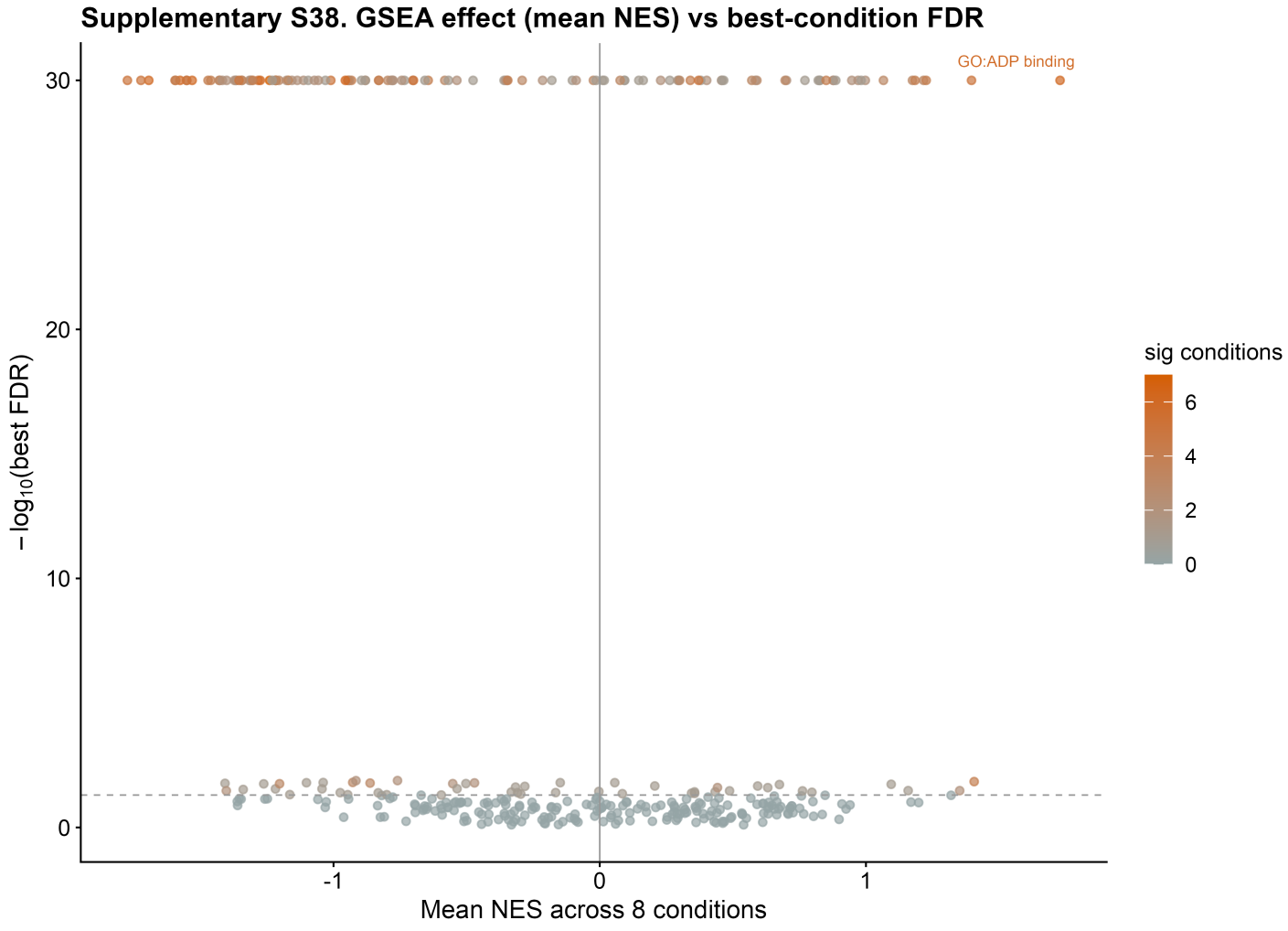
Supplementary Figure S37. GSEA running enrichment plots

Status: Data current (kept as-is, restyle pending)



Supplementary Figure S38. GSEA vs Fisher ORA concordance

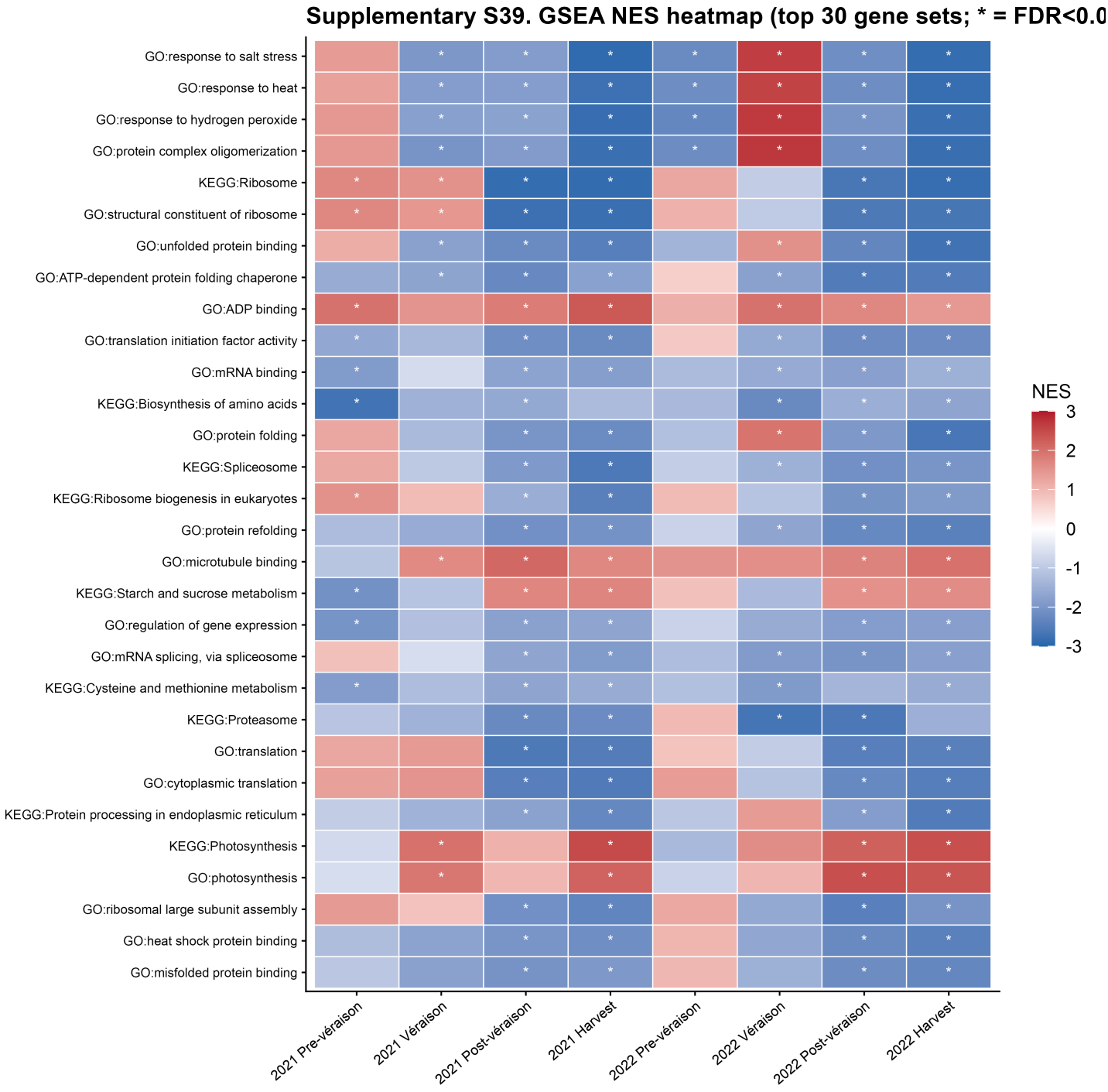
Status: Data current (kept as-is, restyle pending)



Supplementary Figure S39. GSEA NES heatmap

Status: Data current (kept as-is, restyle pending)

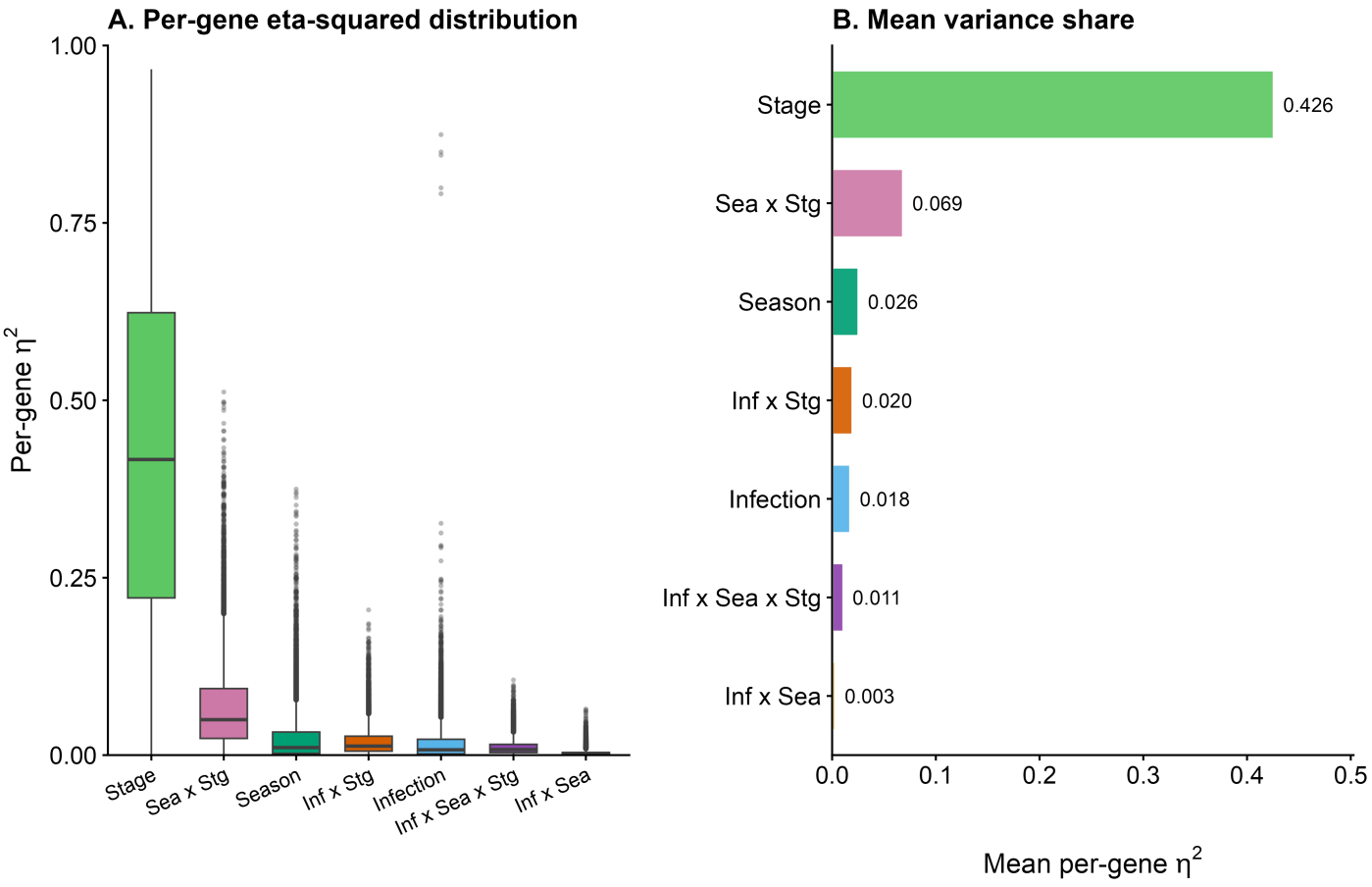
Note: See Main 9 panel A



Supplementary Figure S40. ANOVA η² variance decomposition

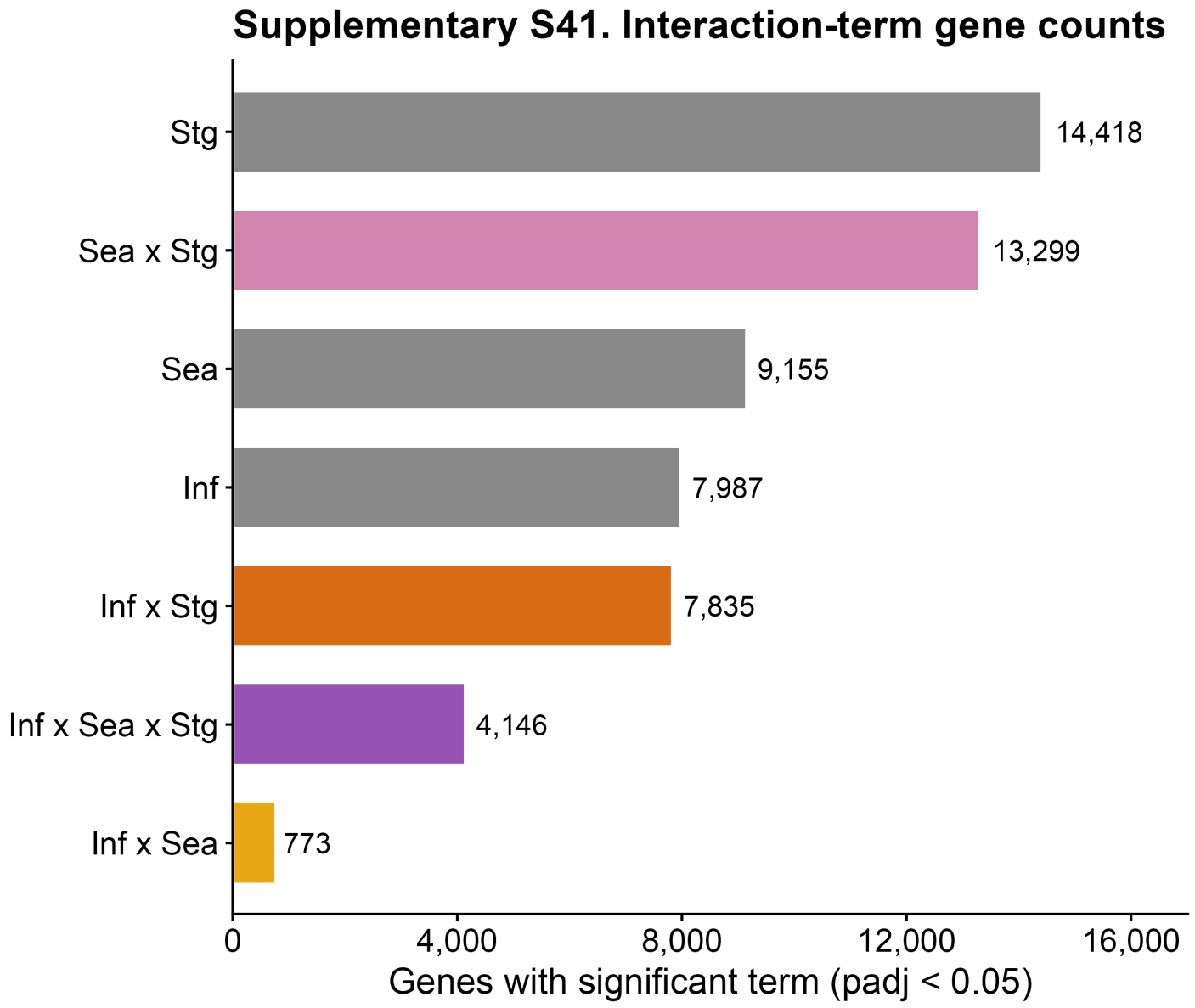
Status: Data current (kept as-is, restyle pending)

Note: See Main 1 panels C/D



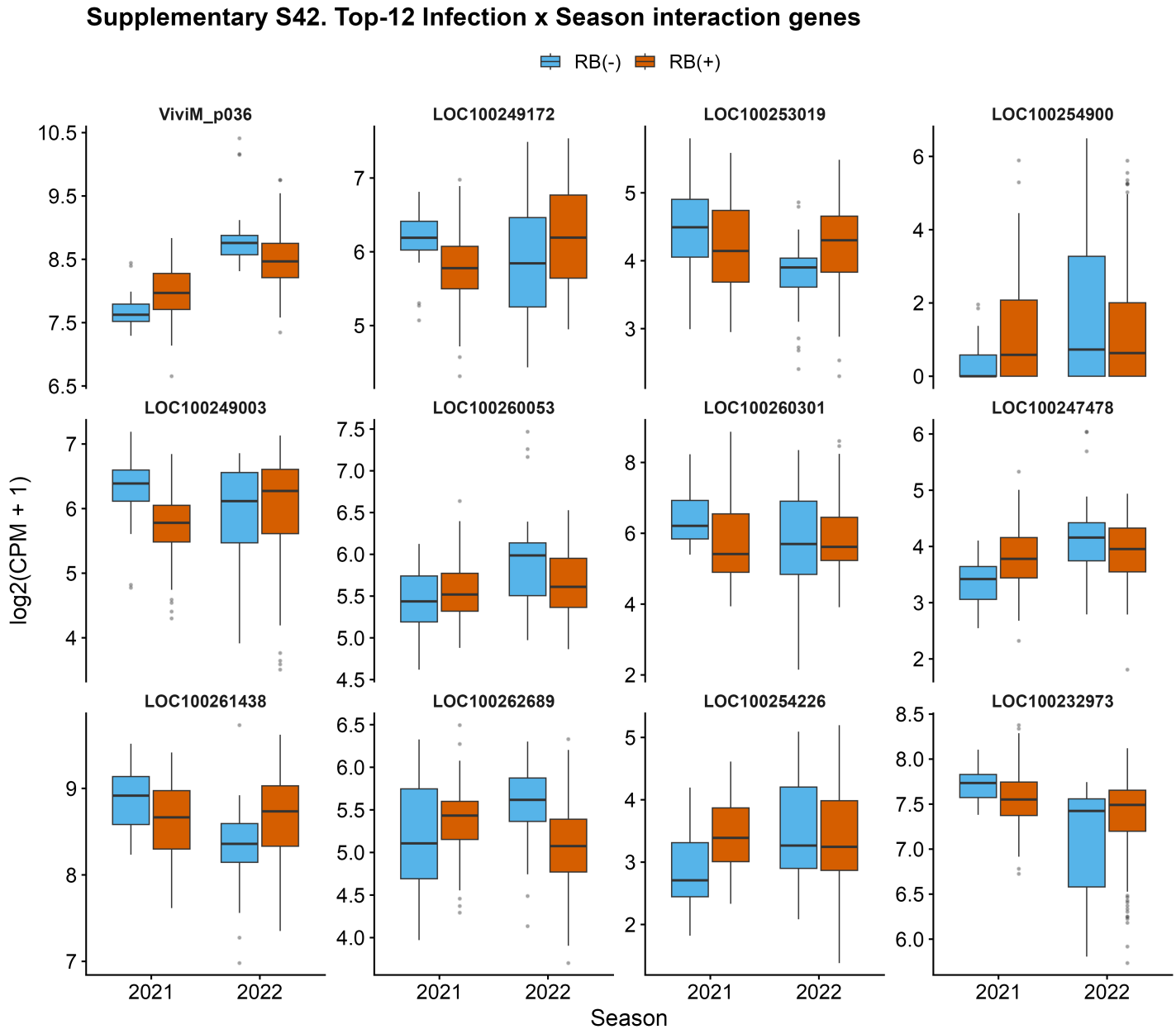
Supplementary Figure S41. Interaction gene-set counts and overlap

Status: Data current (kept as-is, restyle pending)



Supplementary Figure S42. Top Inf × Sea interaction profiles

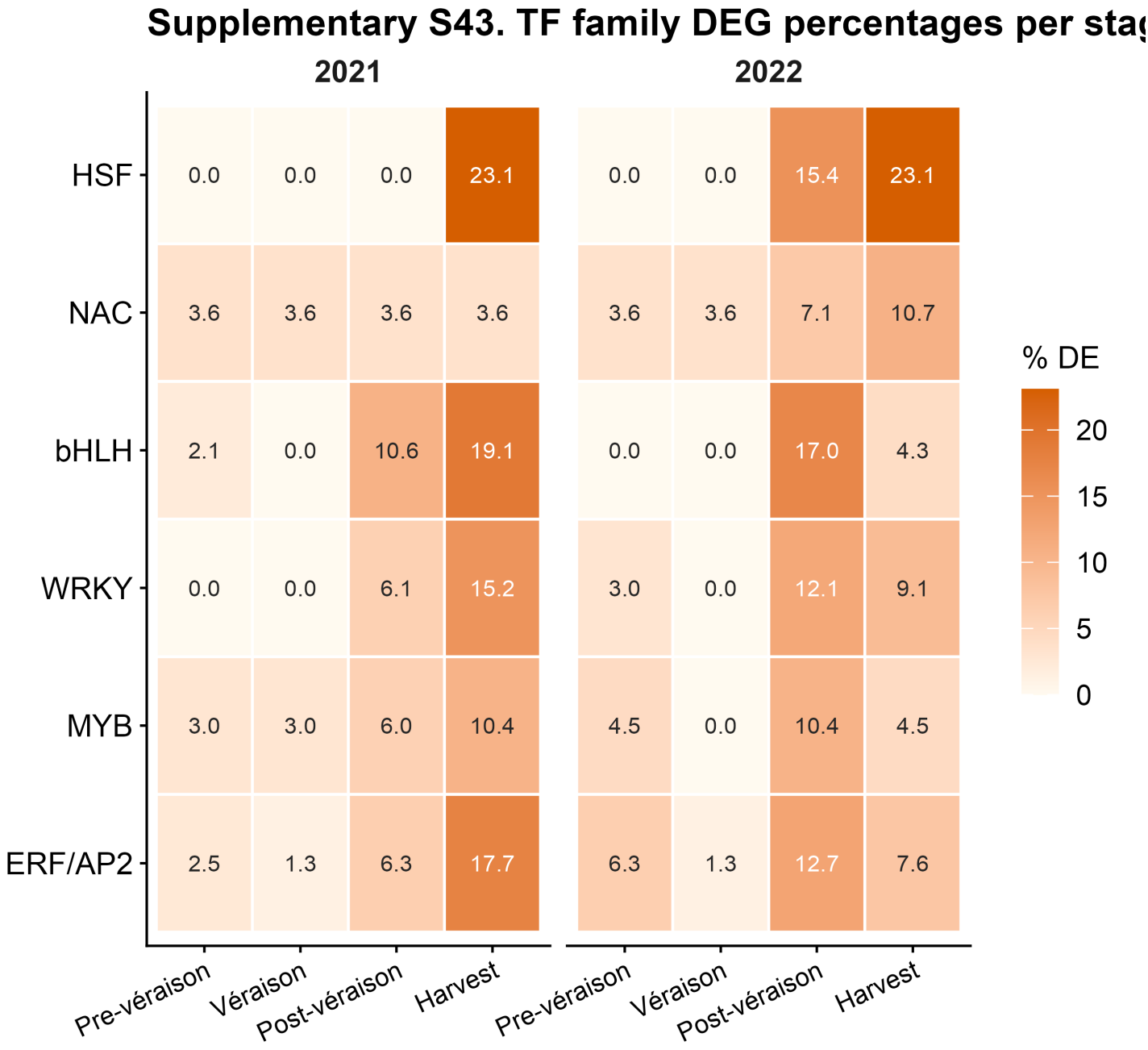
Status: Data current (kept as-is, restyle pending)



Supplementary Figure S43. TF family %DE and mean FC heatmap

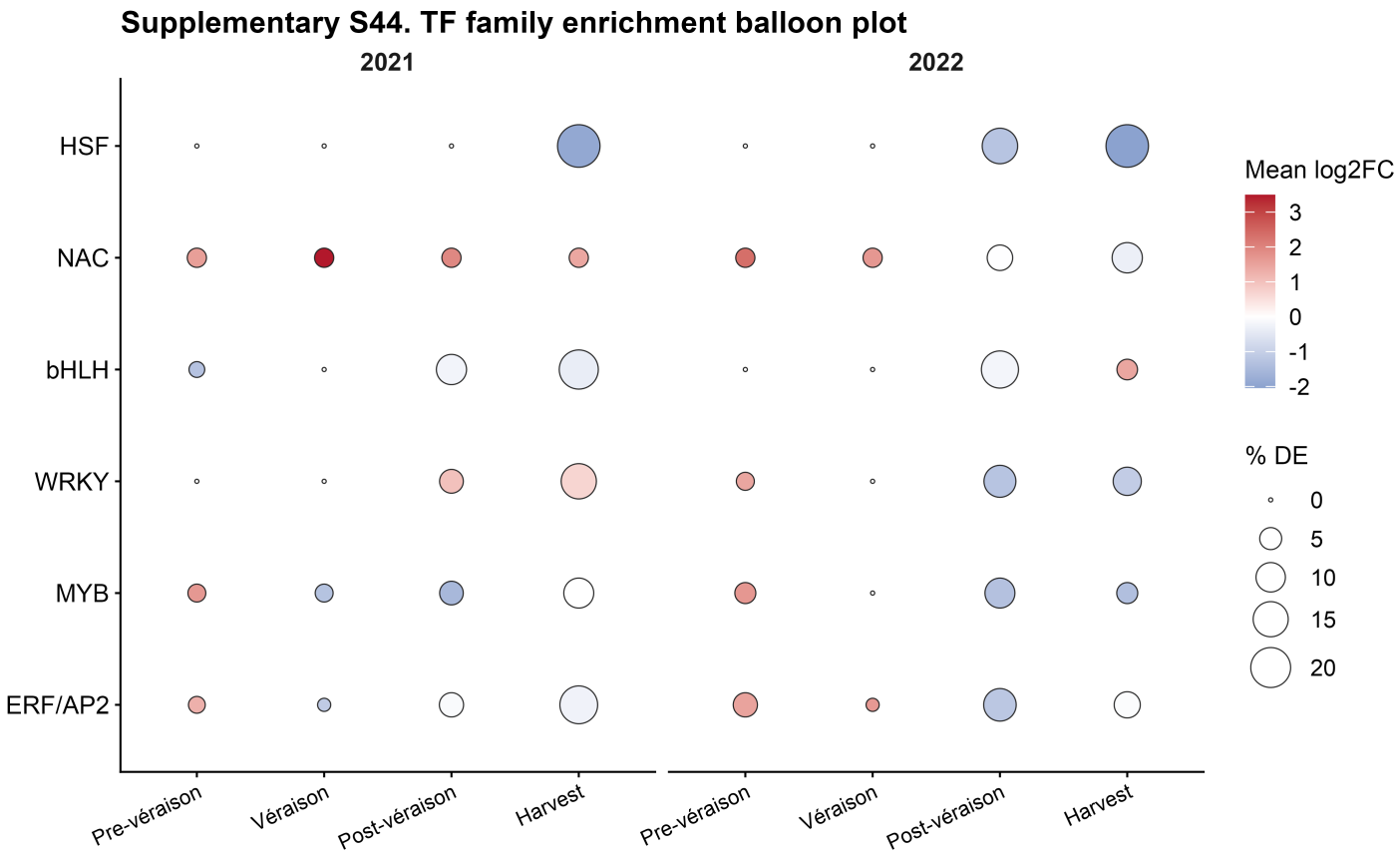
Status: Data current (kept as-is, restyle pending)

Note: See Main 7 panel B



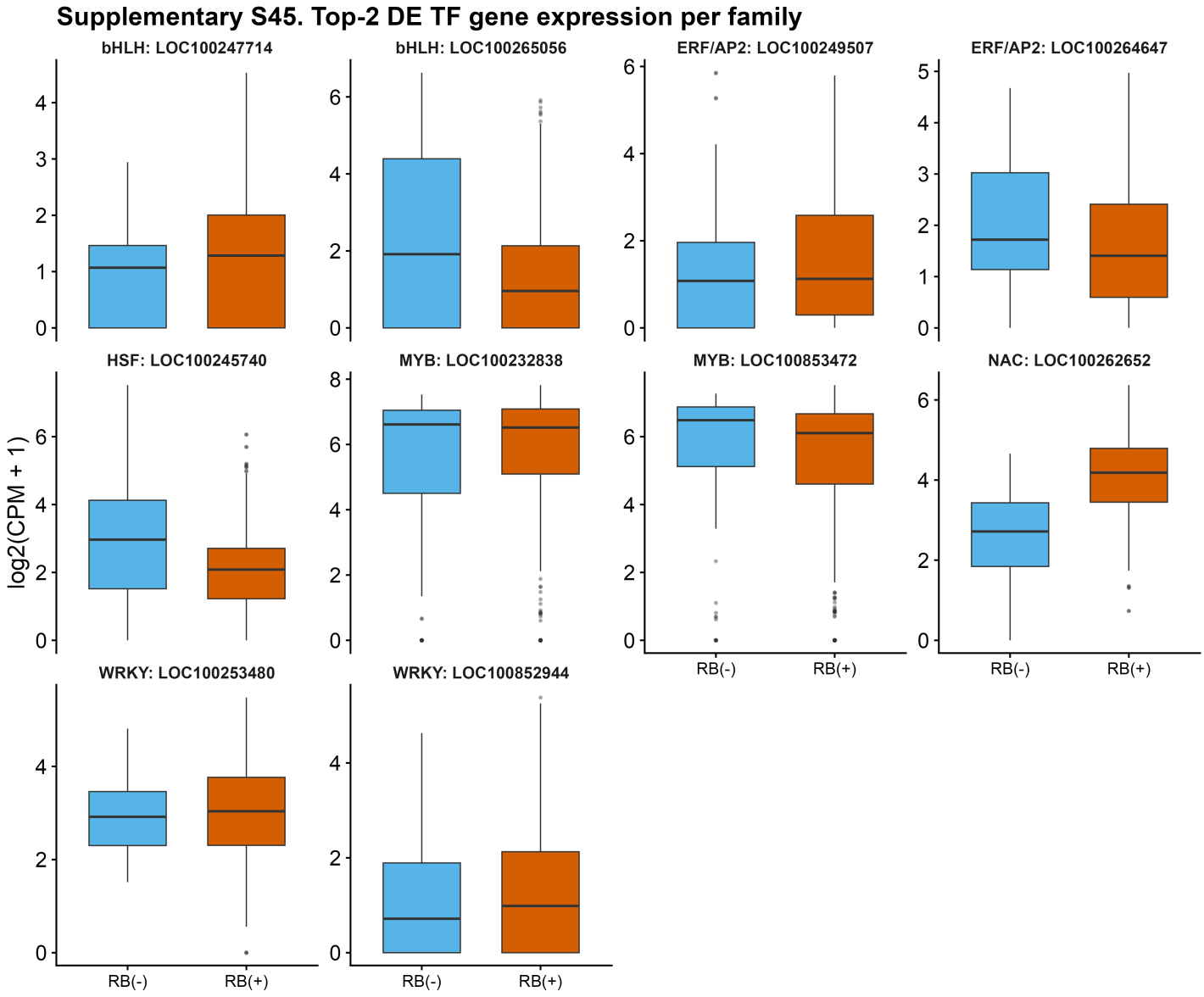
Supplementary Figure S44. TF family DEG enrichment balloon plot

Status: Data current (kept as-is, restyle pending)



Supplementary Figure S45. Top TF gene expression boxplots

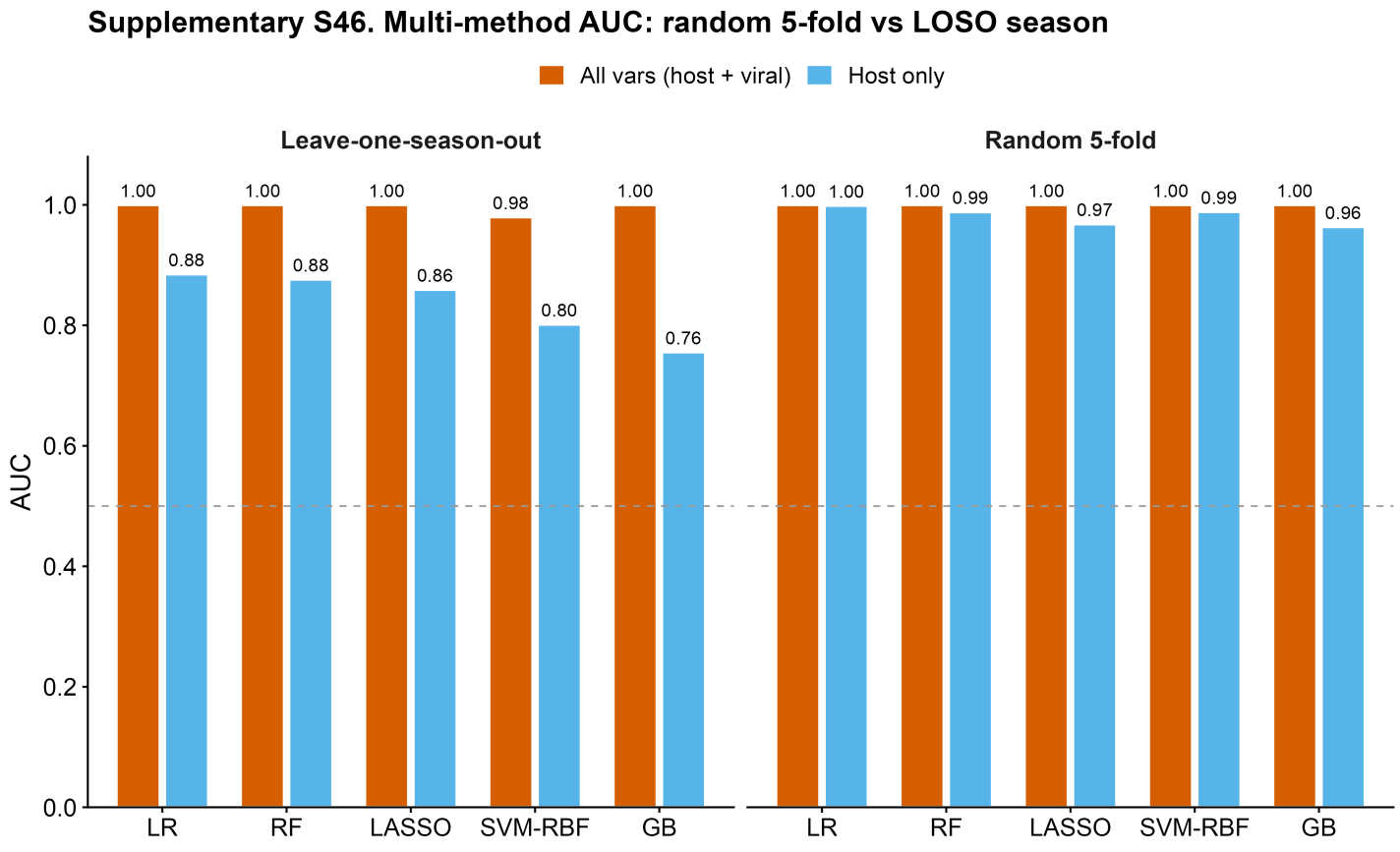
Status: Data current (kept as-is, restyle pending)



Supplementary Figure S46. Multi-method ML AUC comparison

Status: Reframed in Main composite (see note)

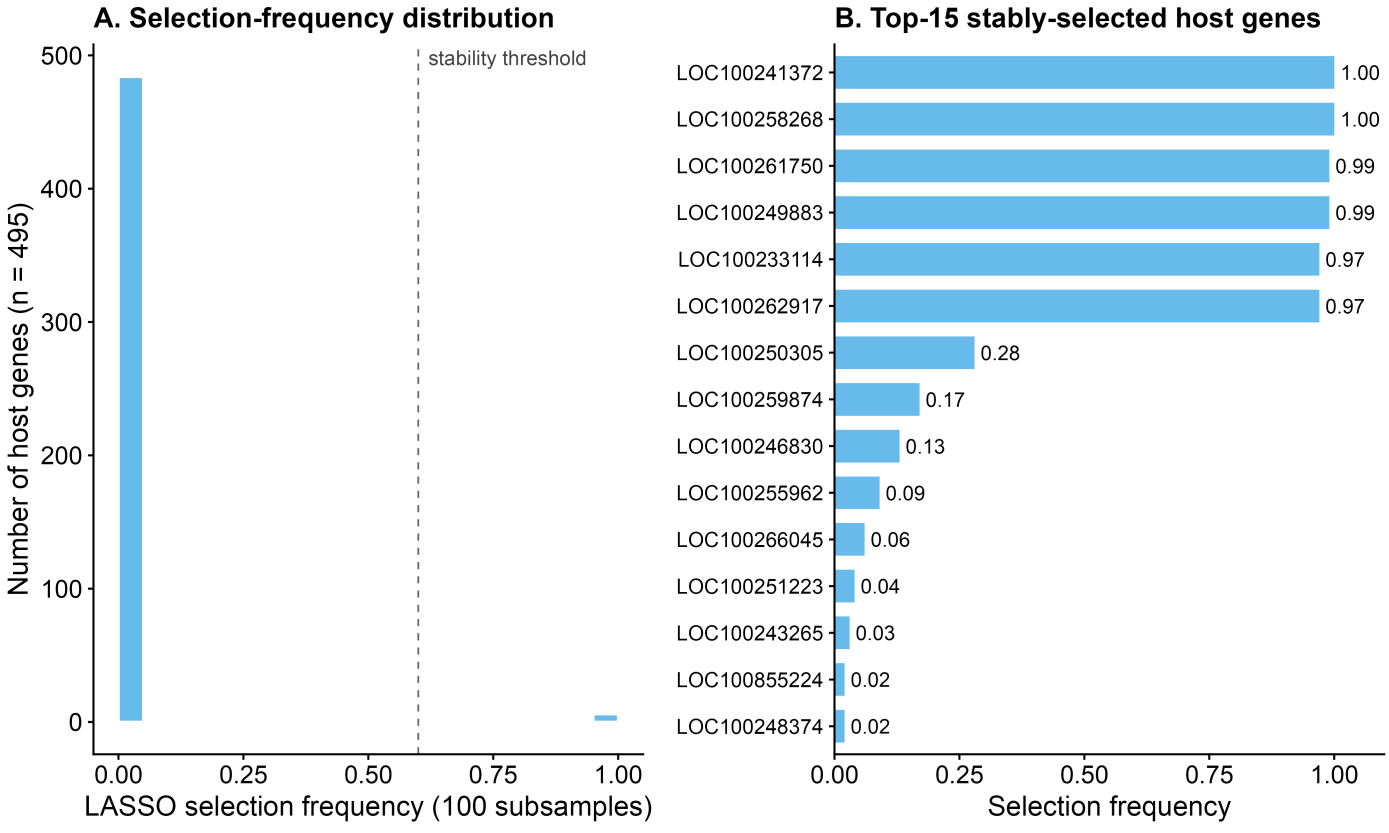
Note: See Main 6



Supplementary Figure S47. LASSO stability selection (100 subsamples)

Status: Reframed in Main composite (see note)

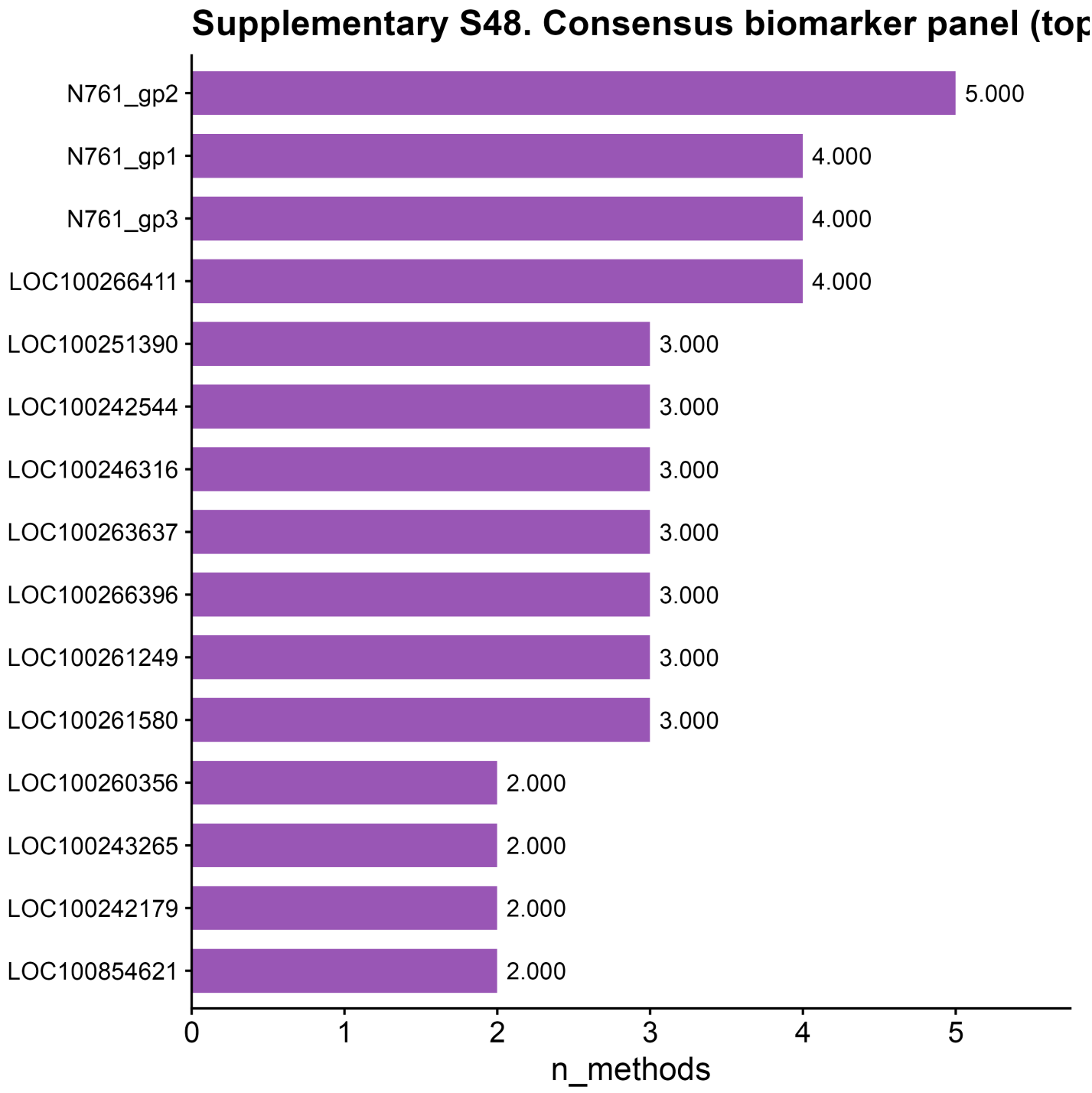
Note: See Main 6 panel C



Supplementary Figure S48. Consensus biomarker panel + ROC

Status: Reframed in Main composite (see note)

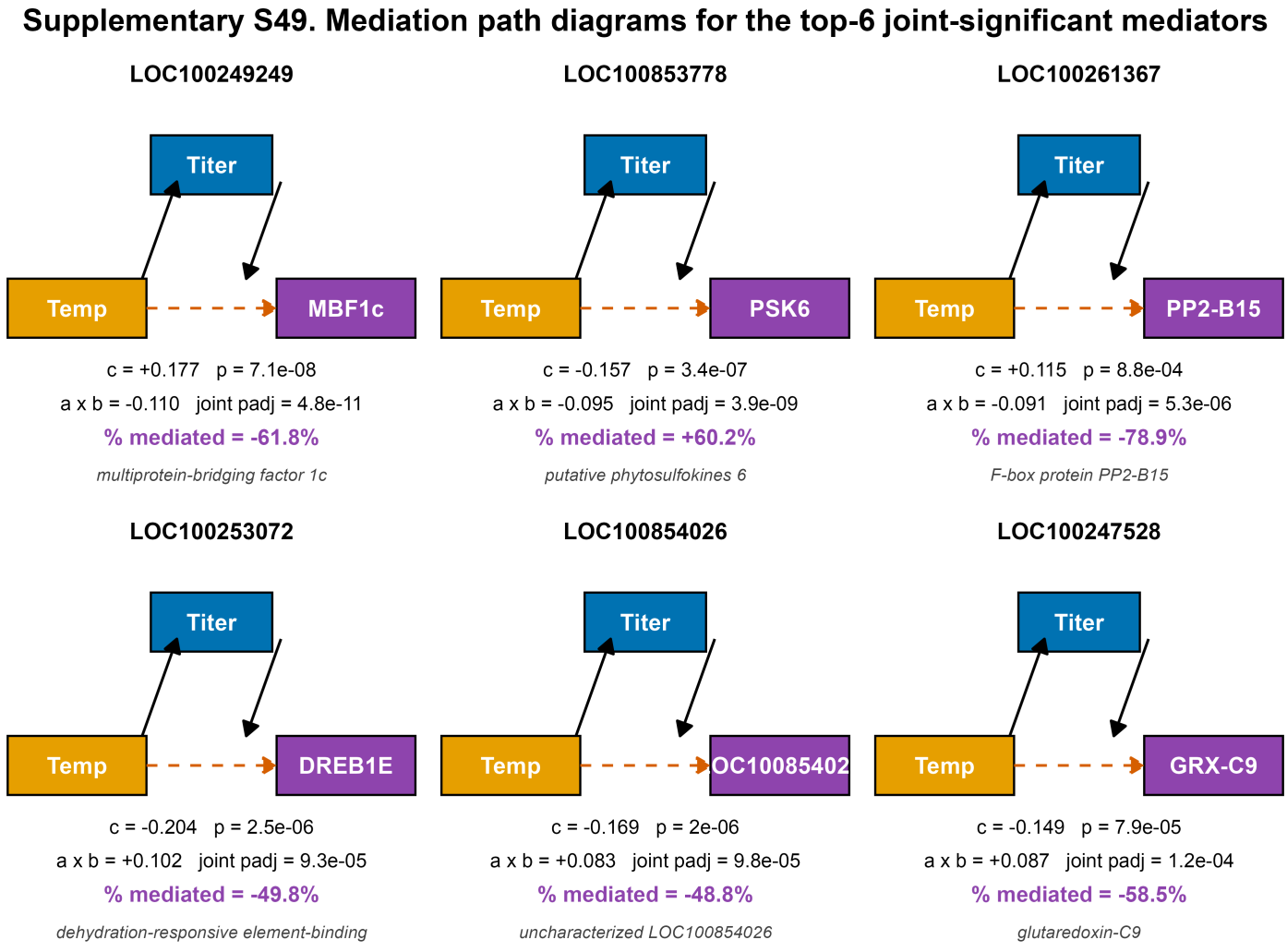
Note: See Main 6



Supplementary Figure S49. Mediation path diagrams (top 6)

Status: Data current (kept as-is, restyle pending)

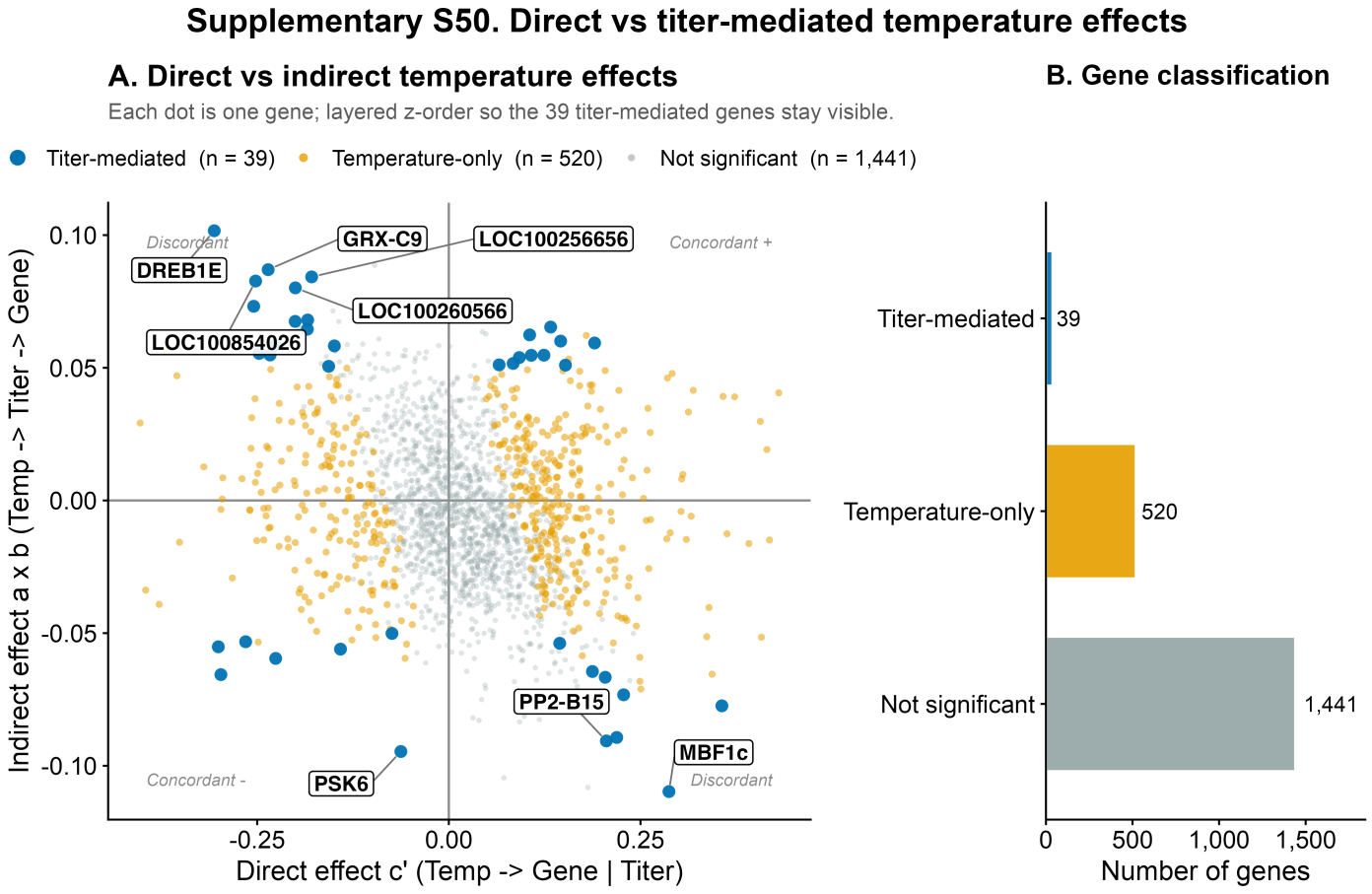
Note: See Main 5 panel A (MBF1c)



Supplementary Figure S50. Direct vs indirect effect scatter

Status: Data current (kept as-is, restyle pending)

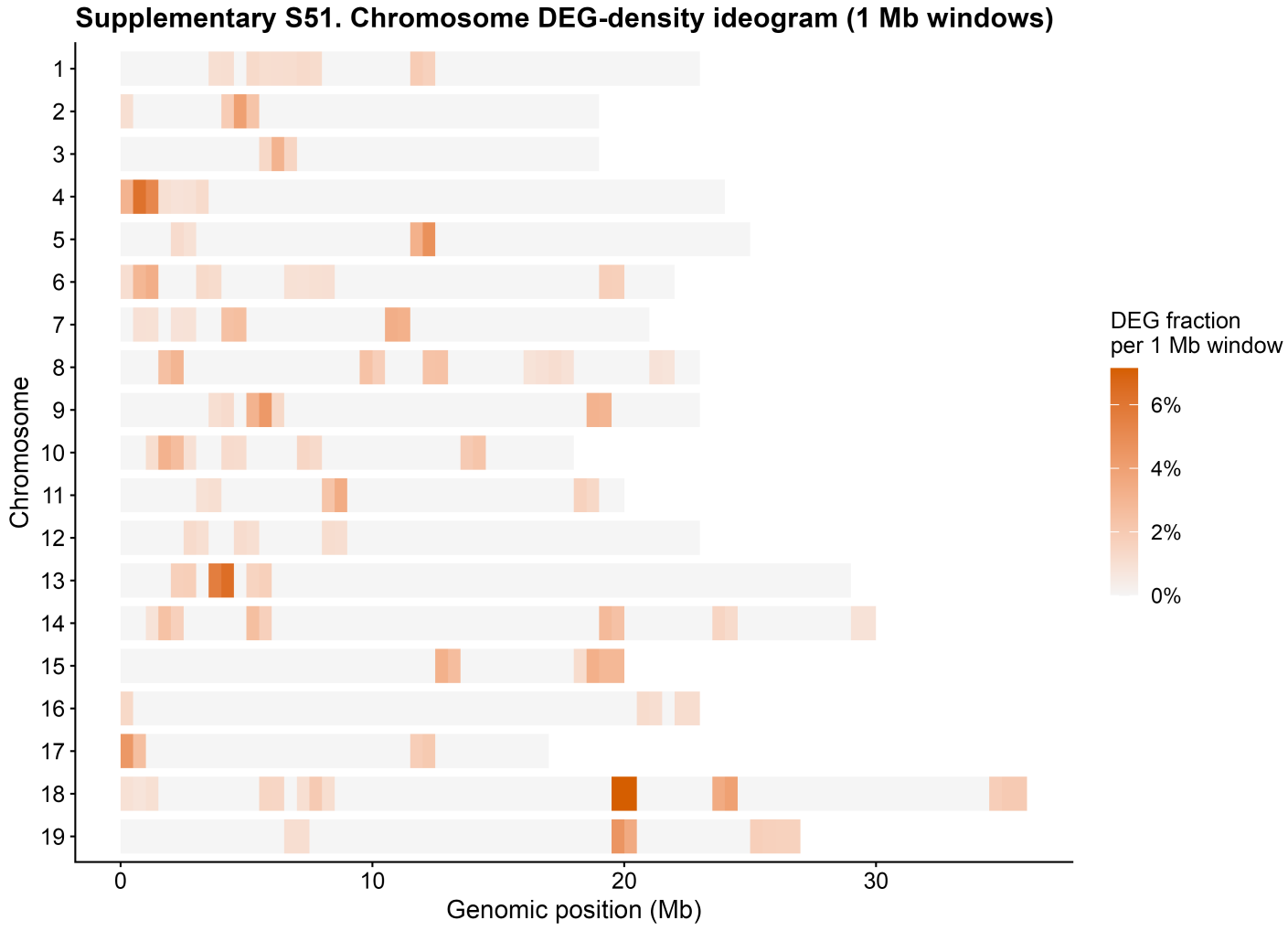
Note: See Main 5 panels B-D



Mediation: Temperature -> Viral titer -> Gene expression on the top 2,000 variable genes.
Concordant quadrants = titer reinforces the temperature effect on the gene; Discordant = titer opposes it.
Top-8 titer-mediated genes labelled by ascending p-value on path b (titer -> gene | temperature).

Supplementary Figure S51. Chromosome DEG-density ideogram

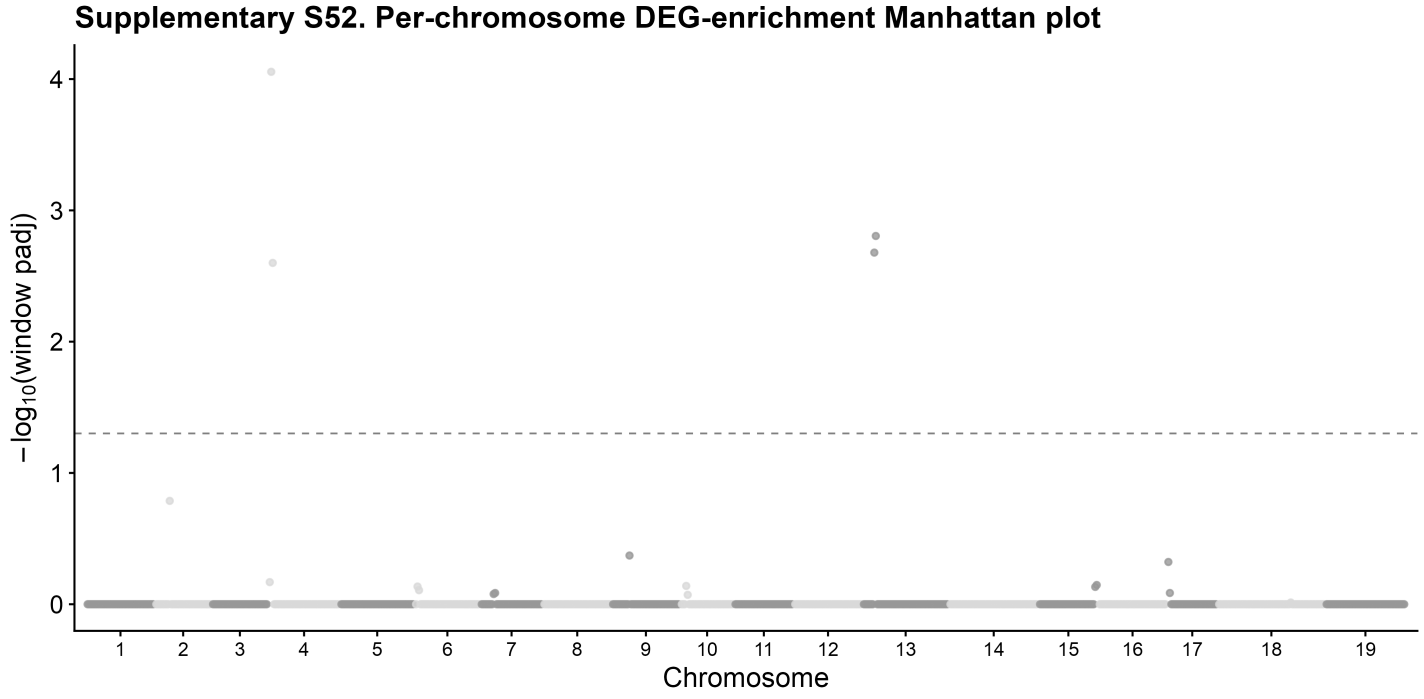
Status: Data current (kept as-is, restyle pending)



Supplementary Figure S52. DEG enrichment Manhattan

Status: Data current (kept as-is, restyle pending)

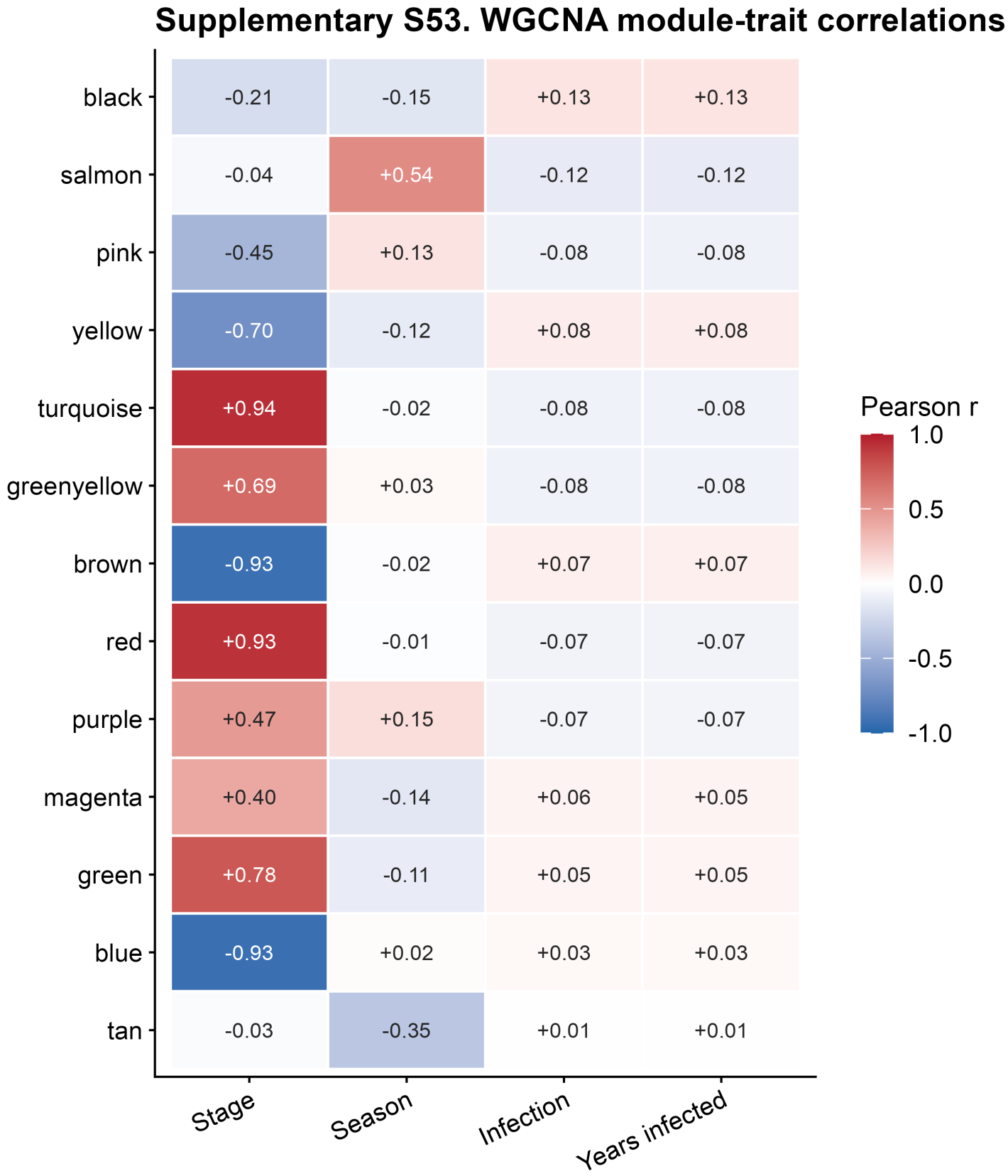
Note: See Main 8 panel A



Supplementary Figure S53. WGCNA module-trait correlation heatmap

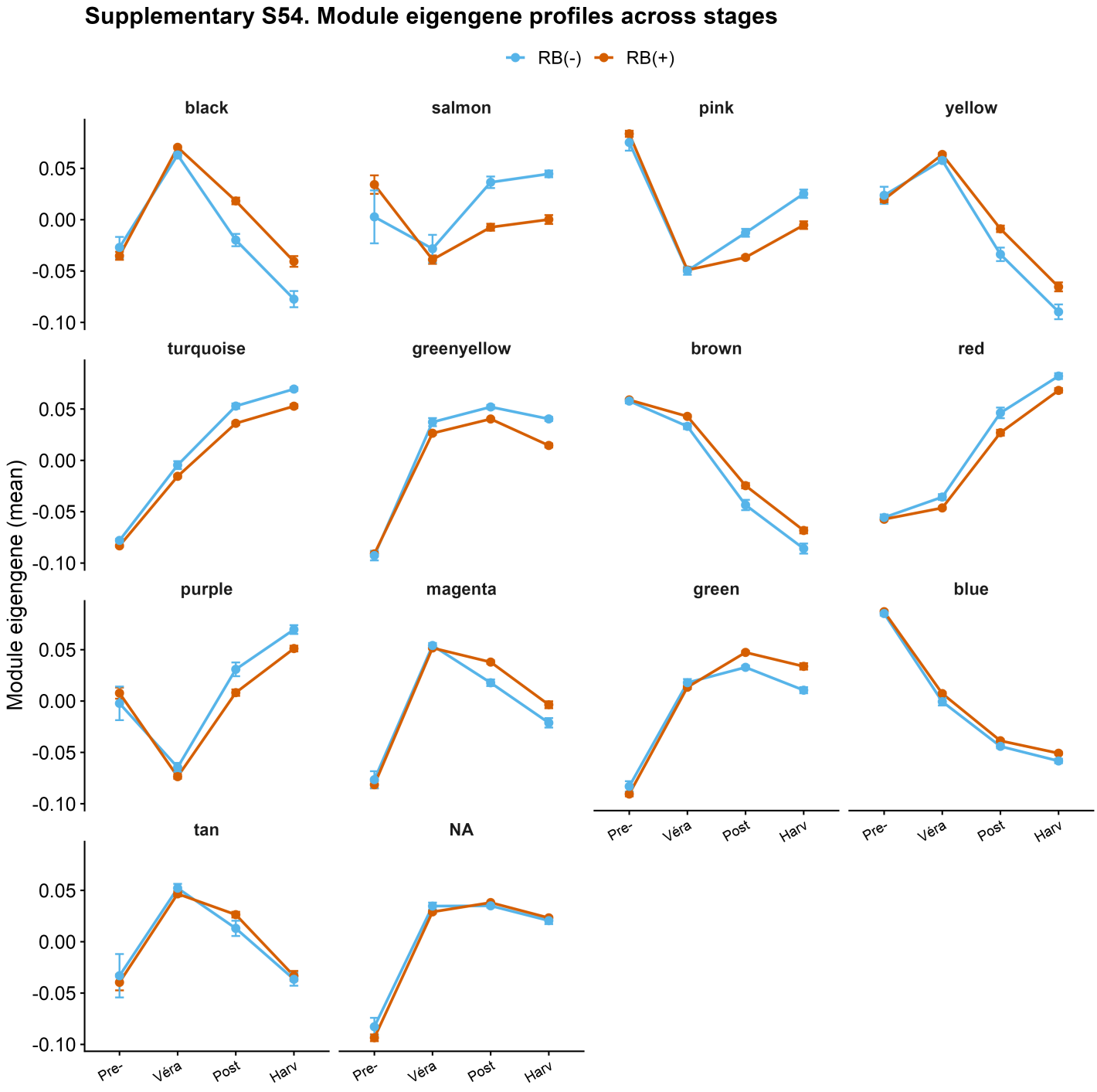
Status: Data current (kept as-is, restyle pending)

Note: See Main 7 panel A



Supplementary Figure S54. WGCNA module eigengene profiles

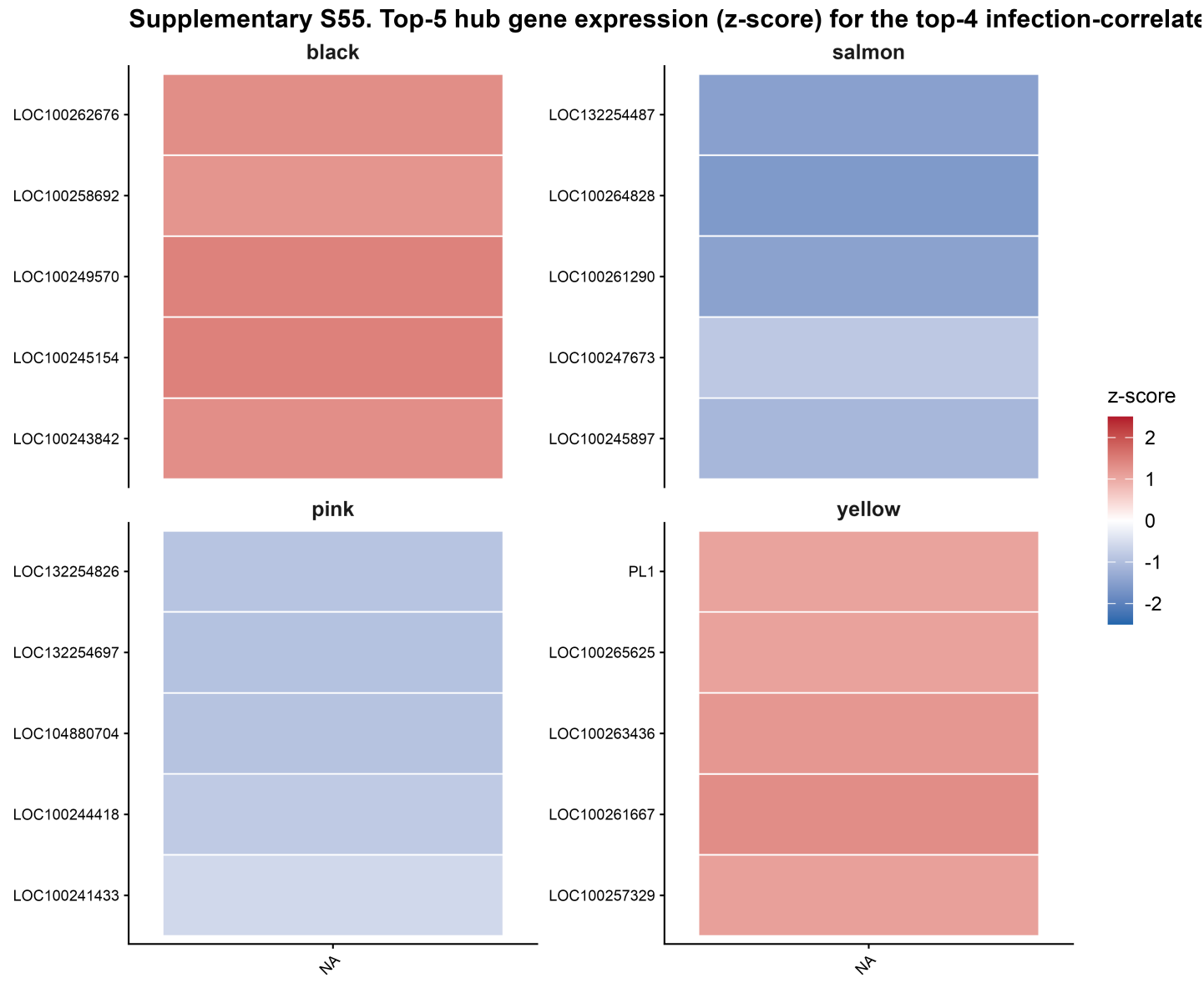
Status: Data current (kept as-is, restyle pending)



Supplementary Figure S55. WGCNA hub gene expression heatmaps

Status: Data current (kept as-is, restyle pending)

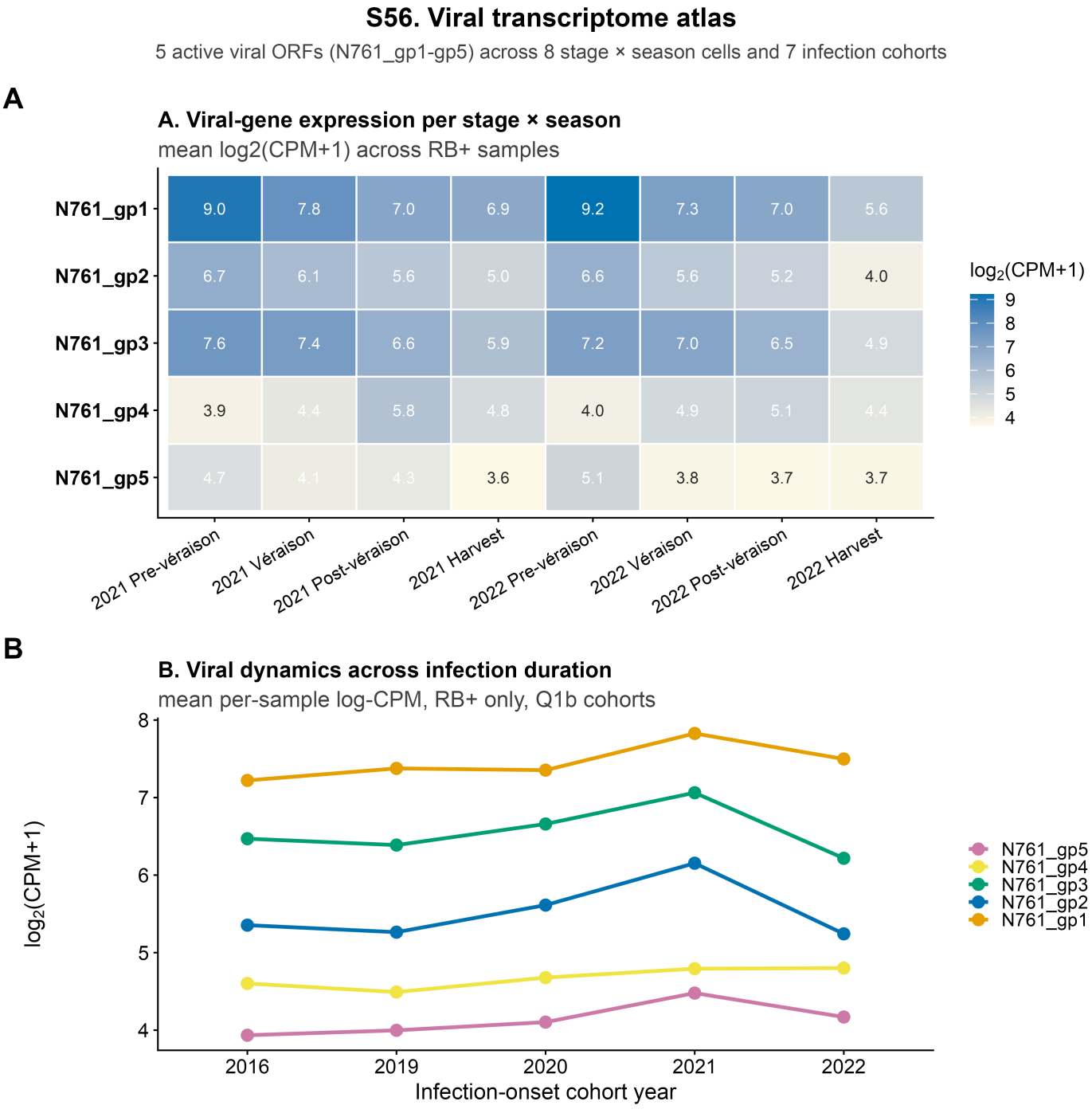
Note: See Main 7 panel C



Supplementary Figure S56. Viral transcriptome atlas (5 ORFs × 8 cells × cohorts)

Status: Regenerated under DESeq2, R/ggplot2, Arial

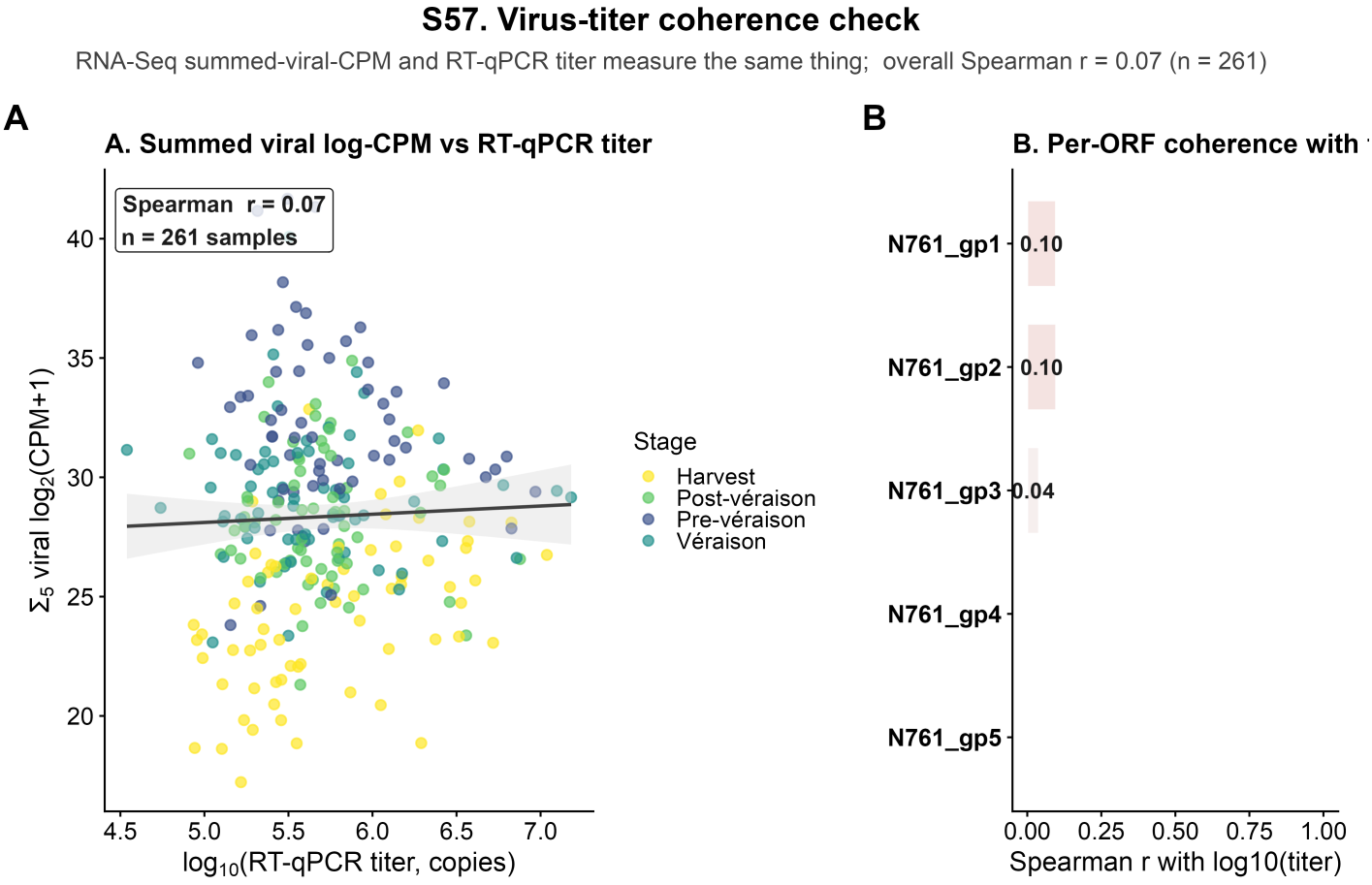
Note: Per-ORF log-CPM across stage/season/cohort



Supplementary Figure S57. Virus-titer coherence check

Status: Regenerated under DESeq2, R/ggplot2, Arial

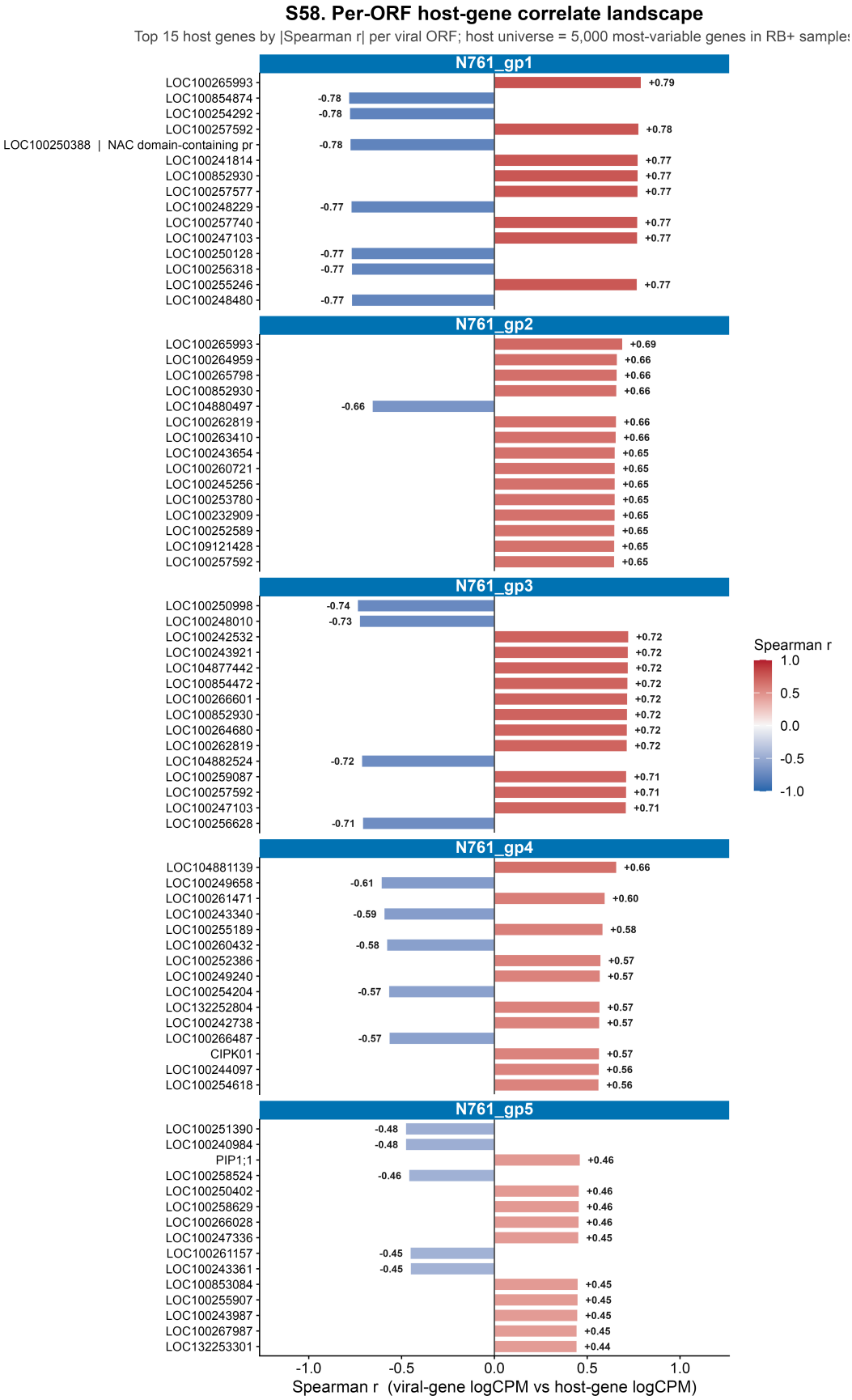
Note: RNA-Seq CPM vs RT-qPCR titer, r = 0.07 overall



Supplementary Figure S58. Per-ORF host-gene correlate landscape

Status: Regenerated under DESeq2, R/ggplot2, Arial

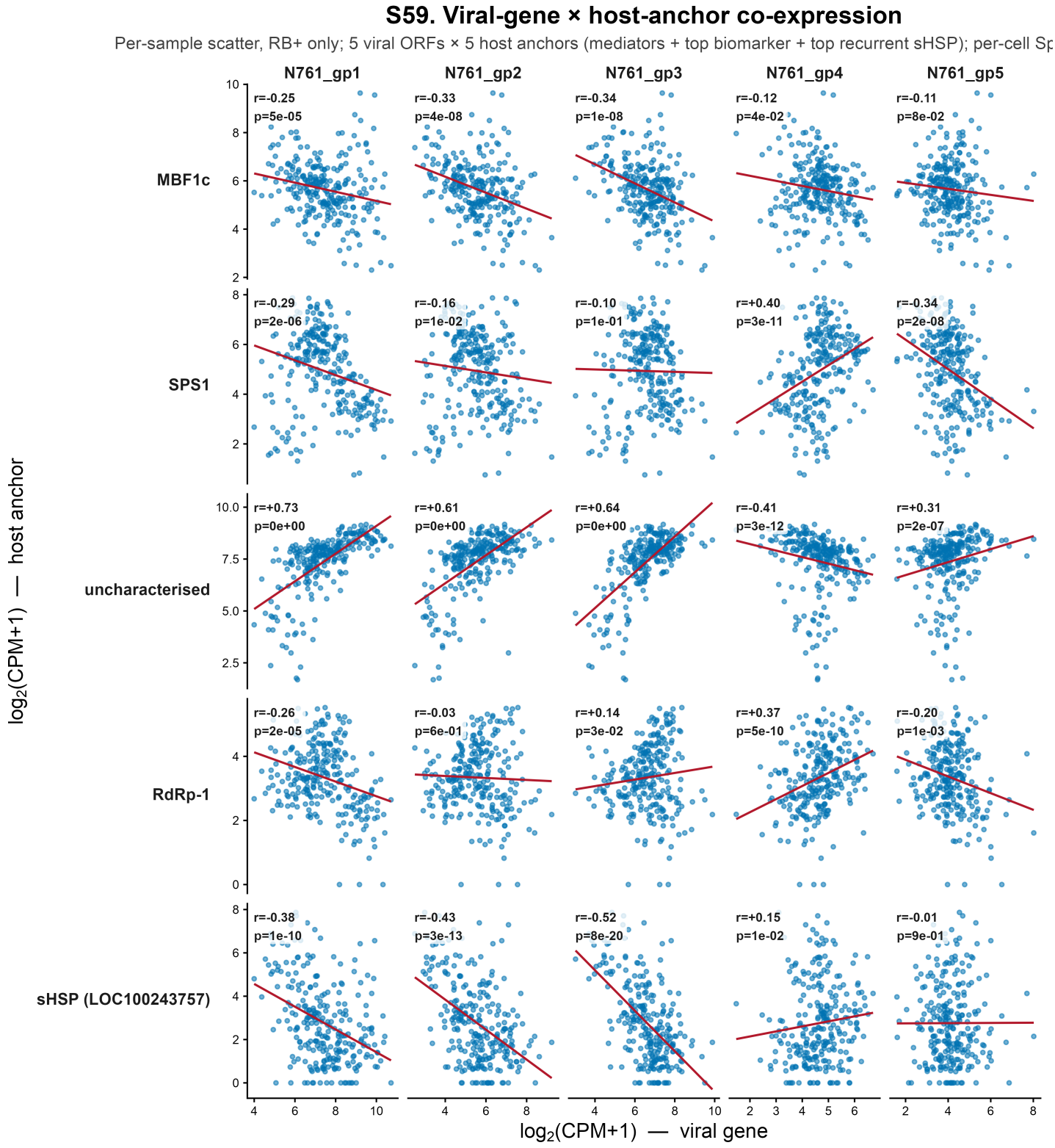
Note: Top 15 host Spearman correlates per viral gene



Supplementary Figure S59. Viral-host co-expression grid

Status: Regenerated under DESeq2, R/ggplot2, Arial

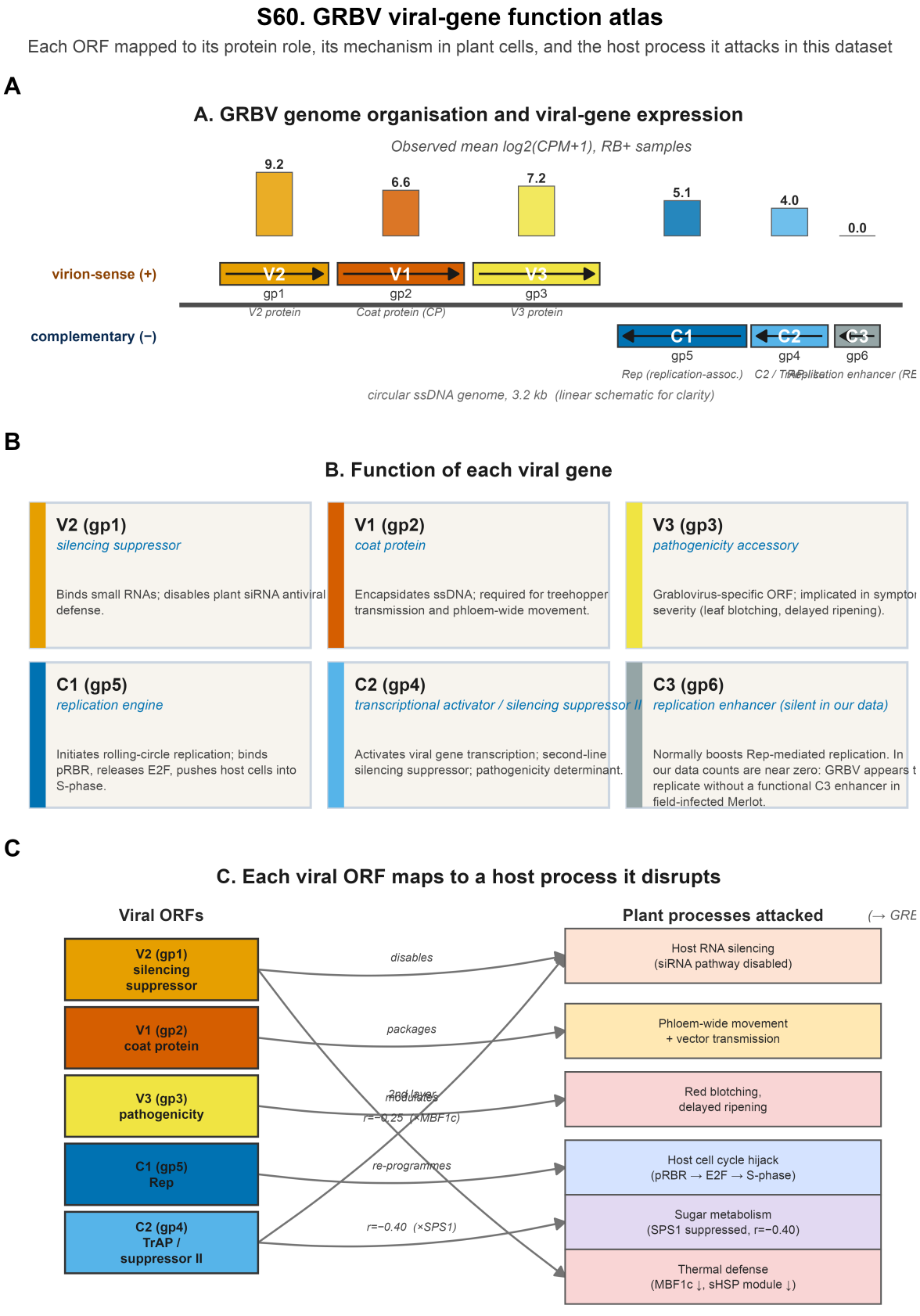
Note: 5 viral ORFs × 5 host anchors (MBF1c, SPS1, etc.)



Supplementary Figure S60. Viral-gene function atlas (biology explainer)

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Genome schematic + function cards + virus-host impact map



F coordinates approximated from GRBV isolate CF214 (NCBI NC_022002). Expression values from N761_gp1-gp5 in counts_matrix.txt. Correlations from S

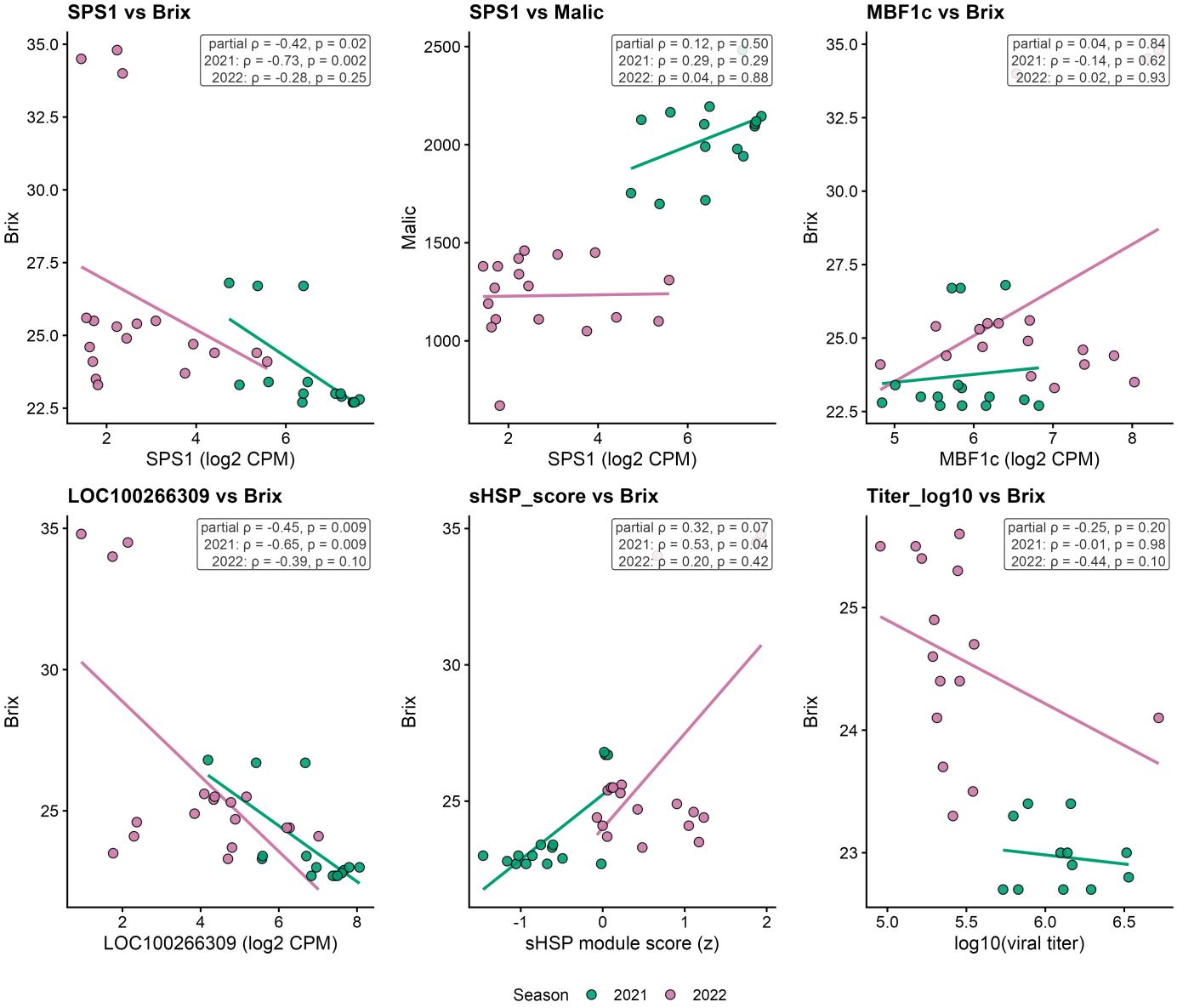
Supplementary Figure S61. Harvest biochem x transcriptome validation

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Per-season + partial rho for SPS1, MBF1c, LOC100266309, sHSP, titer vs Brix / Malic

Fig. S61 — Phenotypic validation of transcriptomic findings at Harvest

Per-sample expression (log2 CPM) or module score vs juice chemistry; n=33 paired samples. Partial rho controls for season.



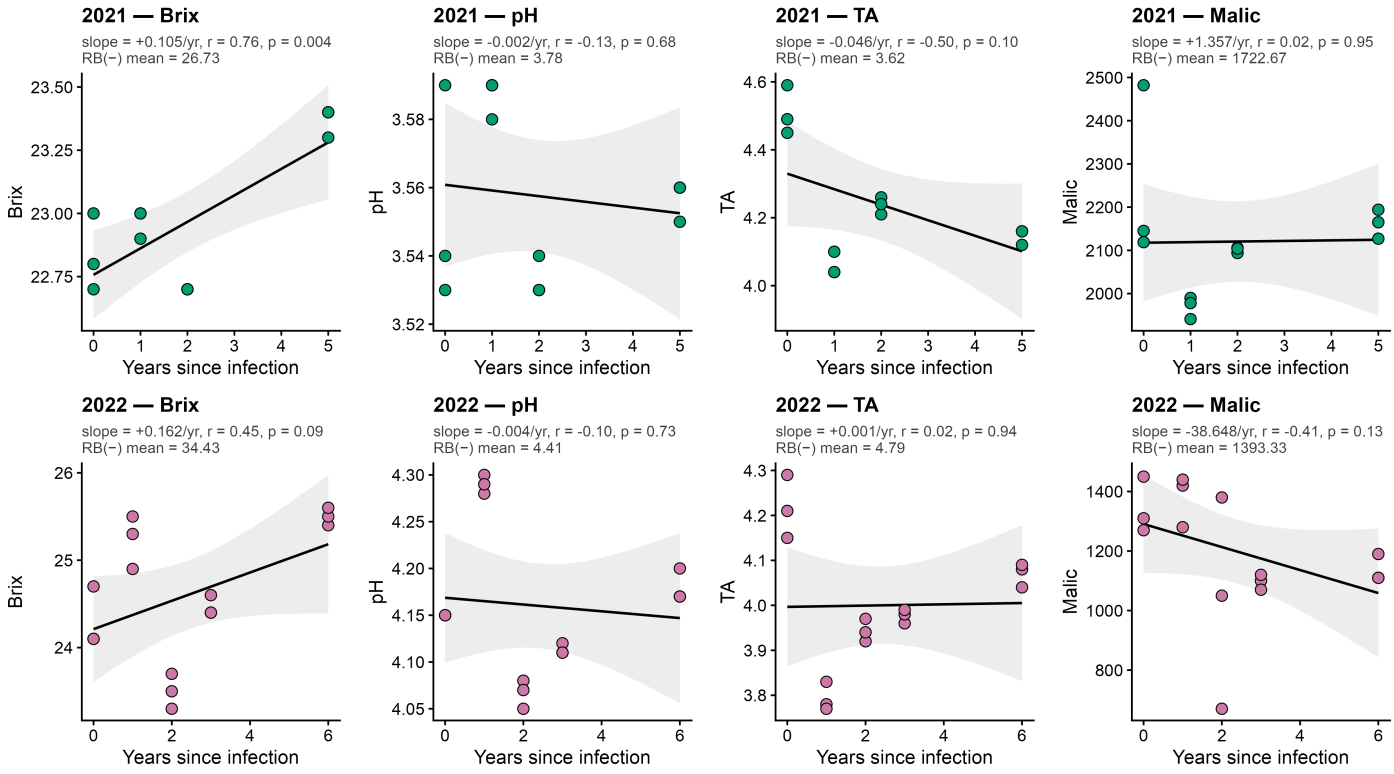
Supplementary Figure S62. Harvest biochem vs years-since-infection

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Brix / pH / TA / Malic vs YSI per season; 2021 Brix recovery +0.10 °Bx / yr (r=0.76, p=0.0045)

Fig. S62 — Cohort trajectory at Harvest: biochemistry vs years-since-infection

RB(+) samples only; each panel auto-scales to RB(+) data. RB(−) season mean noted in per-panel subtitle.



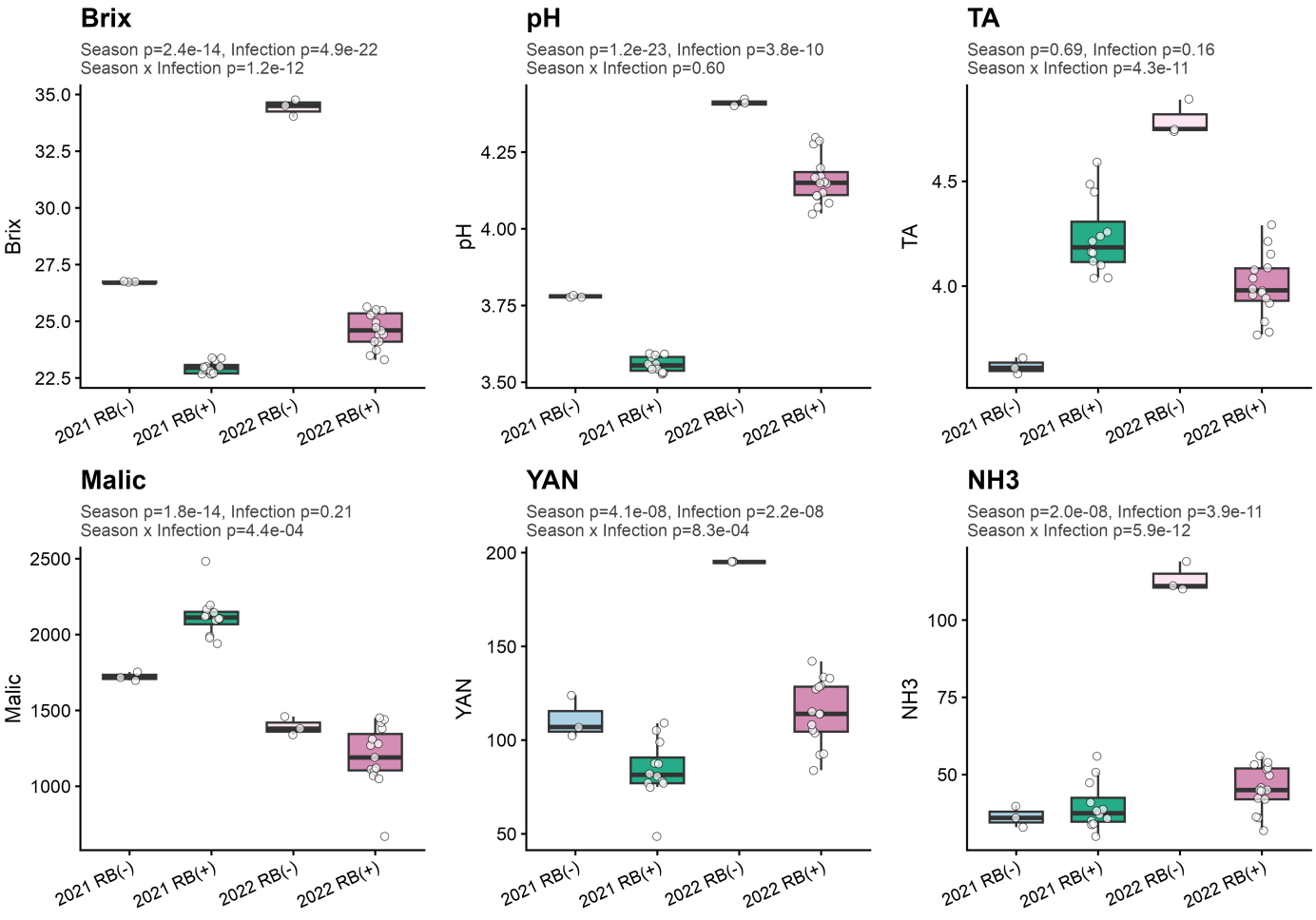
Supplementary Figure S63. Season x Infection ANOVA on juice chemistry

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: TA interaction η²=0.77; cool 2021 +0.62 g/L vs hot 2022 -0.79 g/L; Brix 2.6x stronger in 2022

Fig. S63 — Season x Infection ANOVA on harvest juice chemistry

Two-way ANOVA on n=33 samples; boxes show median + IQR, dots show individual samples.



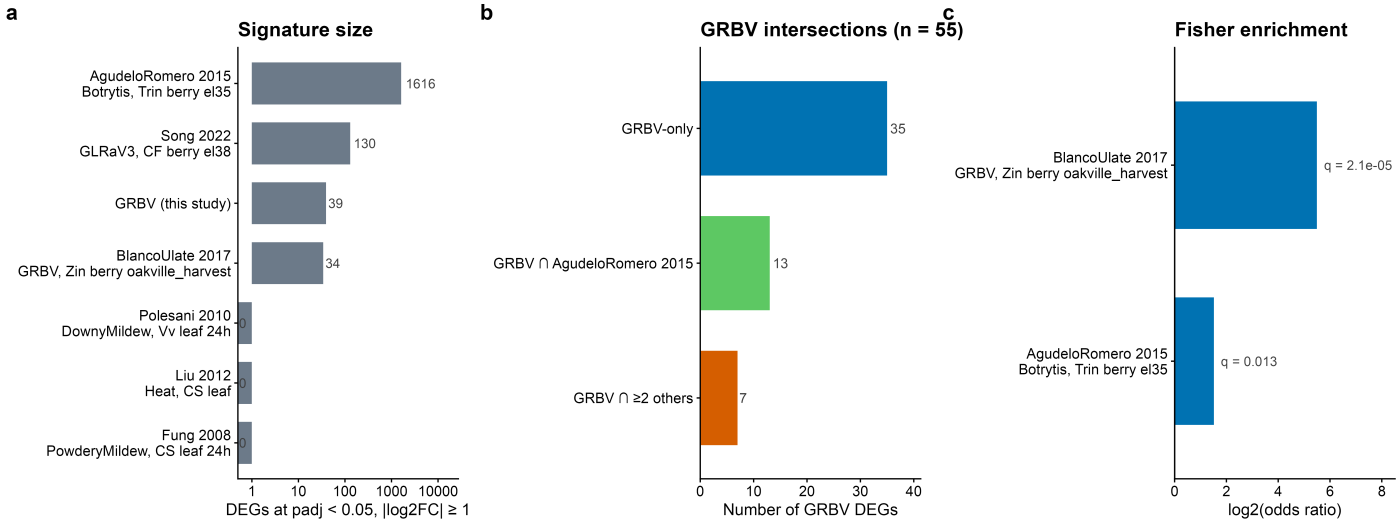
Supplementary Figure S64. Tier 1 DEG overlap: size, intersections, enrichment

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Signature sizes (log-x), GRBV intersections (35 unique / 13 Botrytis-shared / 7 multi), Fisher q=2.1e-5 vs Blanco-Ulate Zinfandel

Fig. S64 — Tier 1 DEG overlap: signature size, intersections, enrichment

6-paper comparative panel vs GRBV (this study); n = 55 GRBV DEGs.



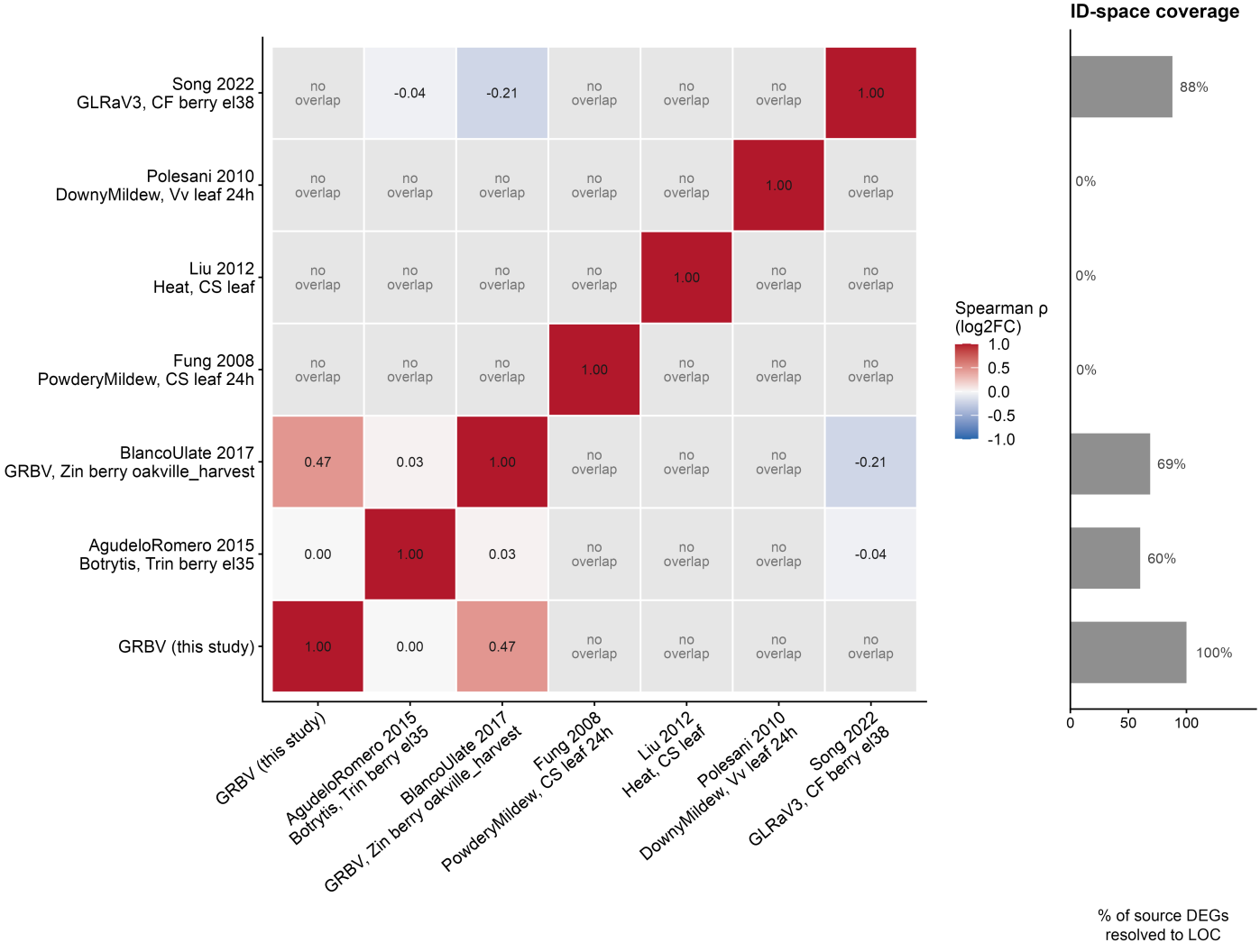
Supplementary Figure S65. Tier 1 log2FC Spearman concordance heatmap

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: GRBV (Merlot) x Blanco-Ulate GRBV (Zinfandel) rho=0.47; Blanco-Ulate x Song GLRaV-3 rho=-0.21; other cells ≈ 0

Fig. S65 — Tier 1 log2FC concordance across Vitis stress datasets

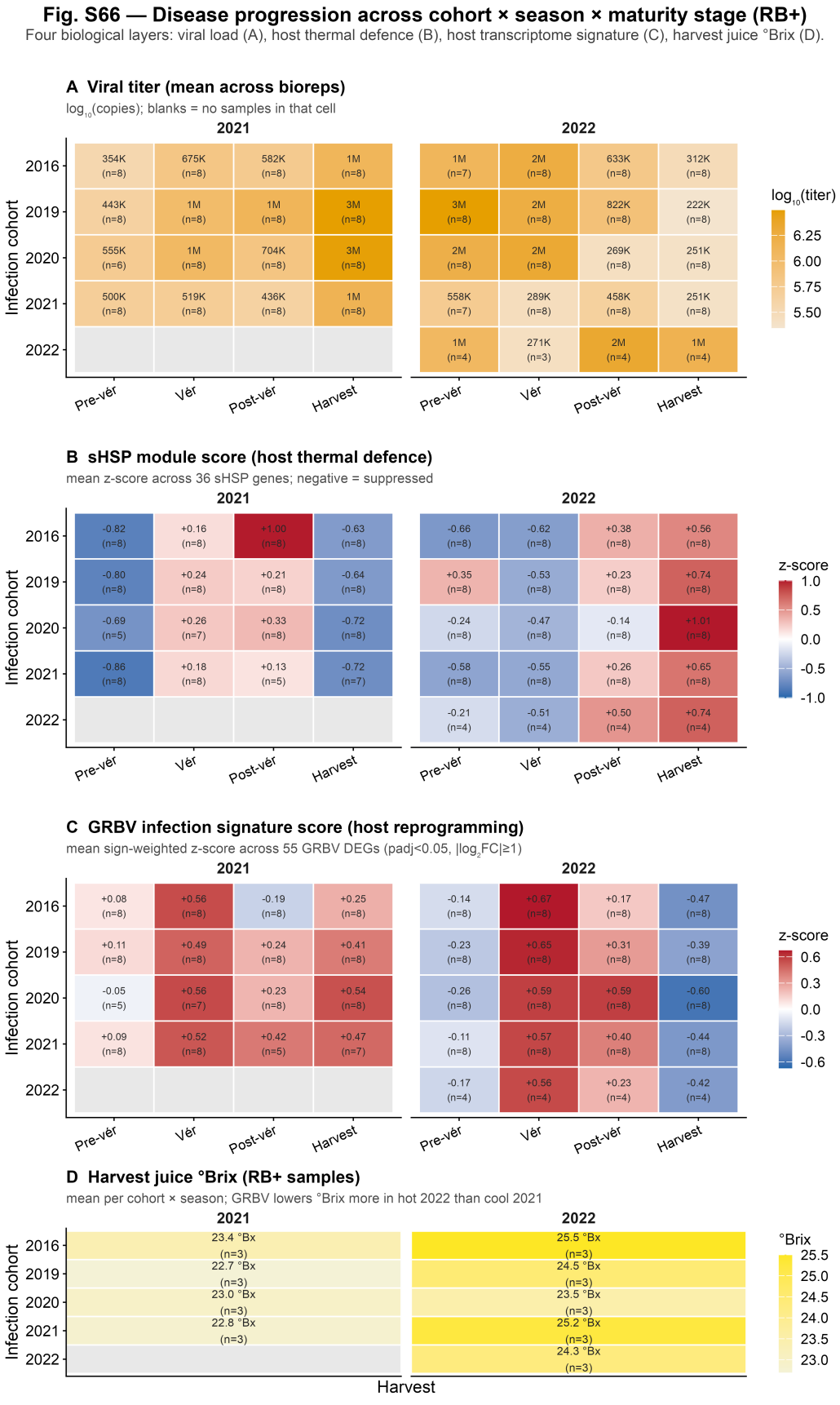
Spearman ρ on shared genes; 'no overlap' = no ID-space intersection. Side strip shows LOC-mapping coverage.



Supplementary Figure S66. Disease progression cube (RB+)

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Cohort x season x stage cube for titer / sHSP / GRBV signature / Brix



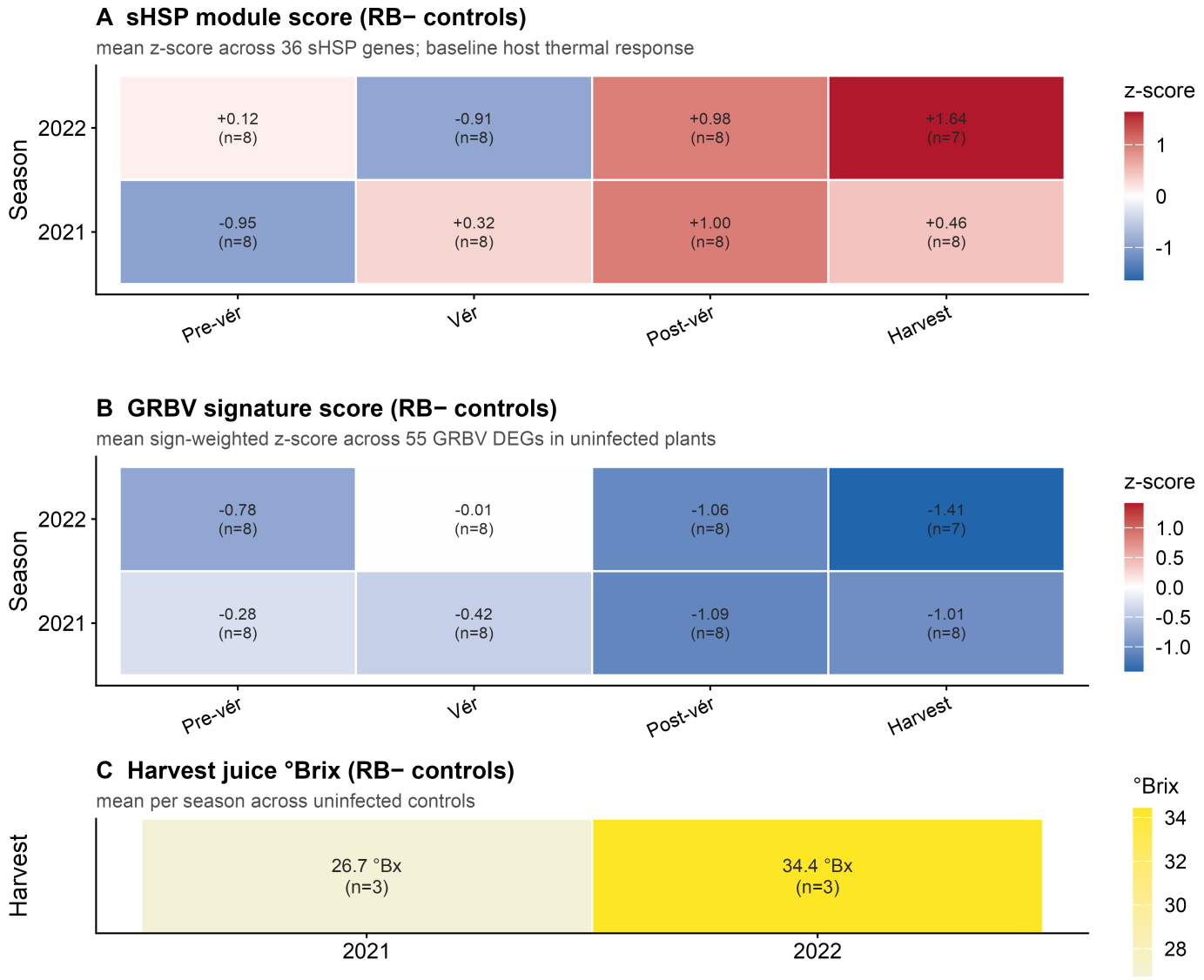
Supplementary Figure S67. RB(-) baseline cube

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Healthy comparator that makes the passive-failure reframe visible

Fig. S67 — RB(−) baseline: same grid, uninfected controls

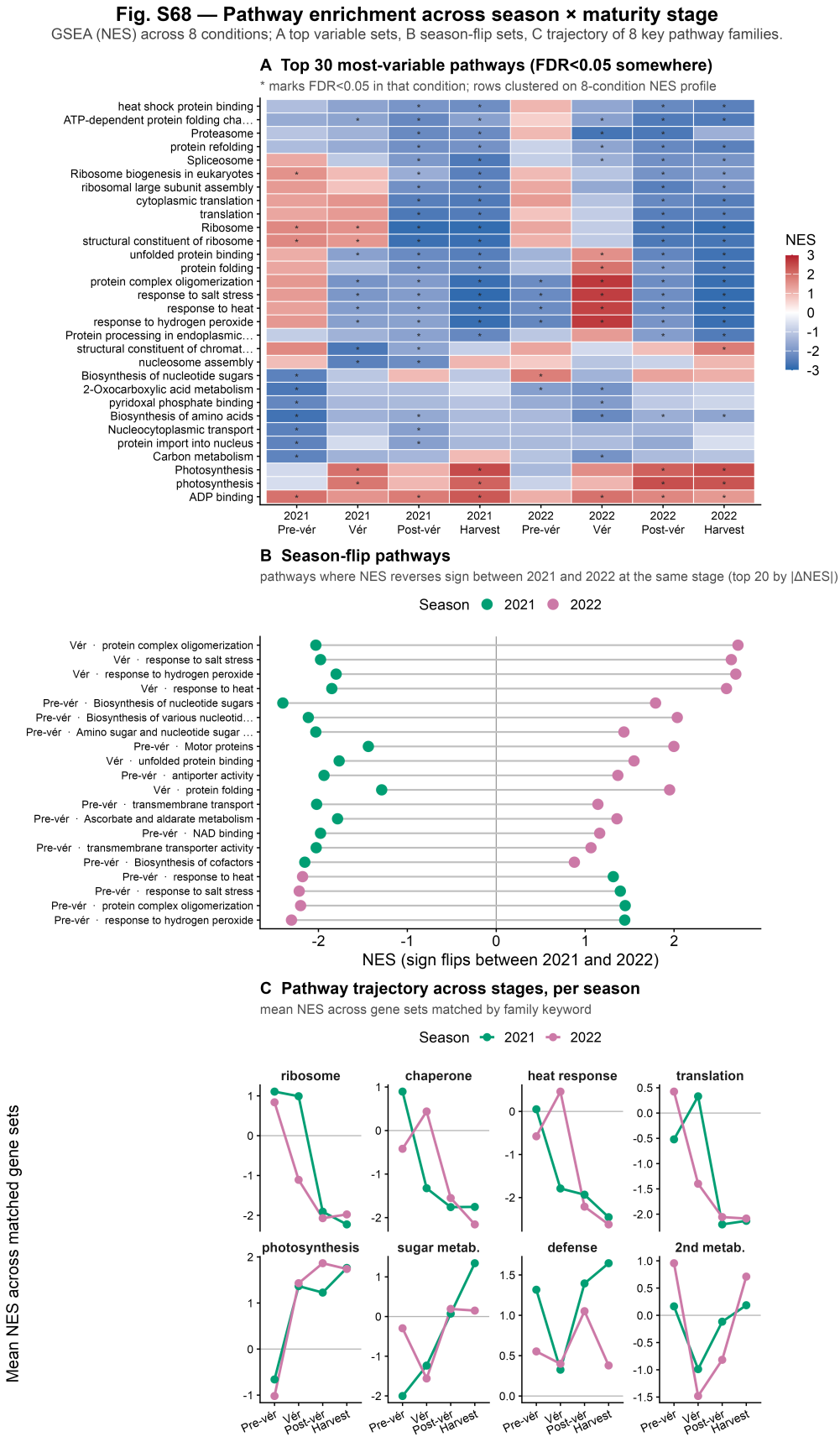
Null comparator for S66. No cohort axis: controls form a single group per season × stage. Compare directly to S66 B, C, D.



Supplementary Figure S68. Season x stage GSEA + season-flip catalog

Status: Regenerated under DESeq2, R/ggplot2, Arial

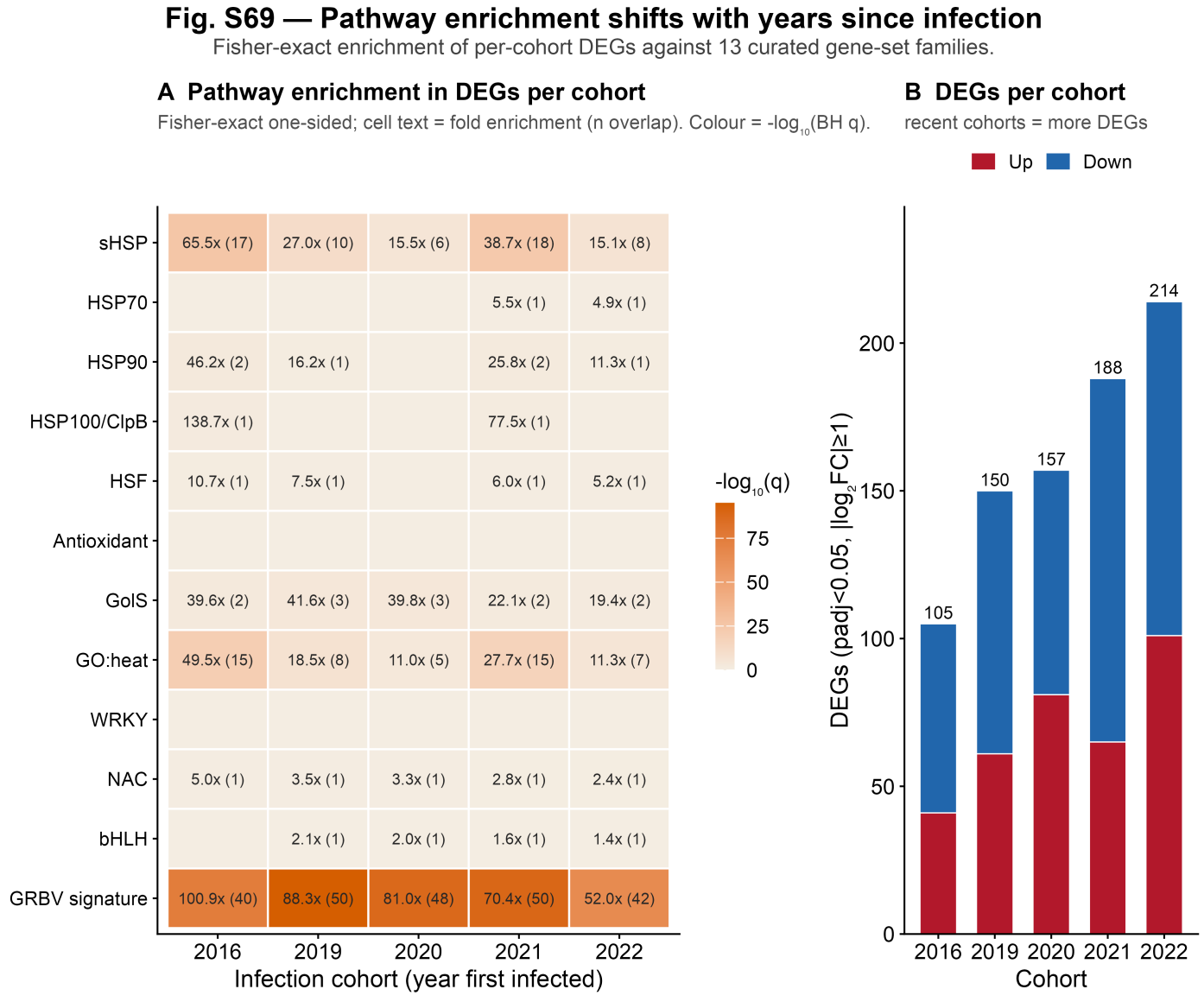
Note: Top variable pathways + 20-pathway sign-flip catalog + 8-family trajectory



Supplementary Figure S69. Per-cohort hyper-enrichment

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: 65x sHSP at 2016 (chronic) vs 5x sHSP at 2022 (acute); 7-year temporal architecture



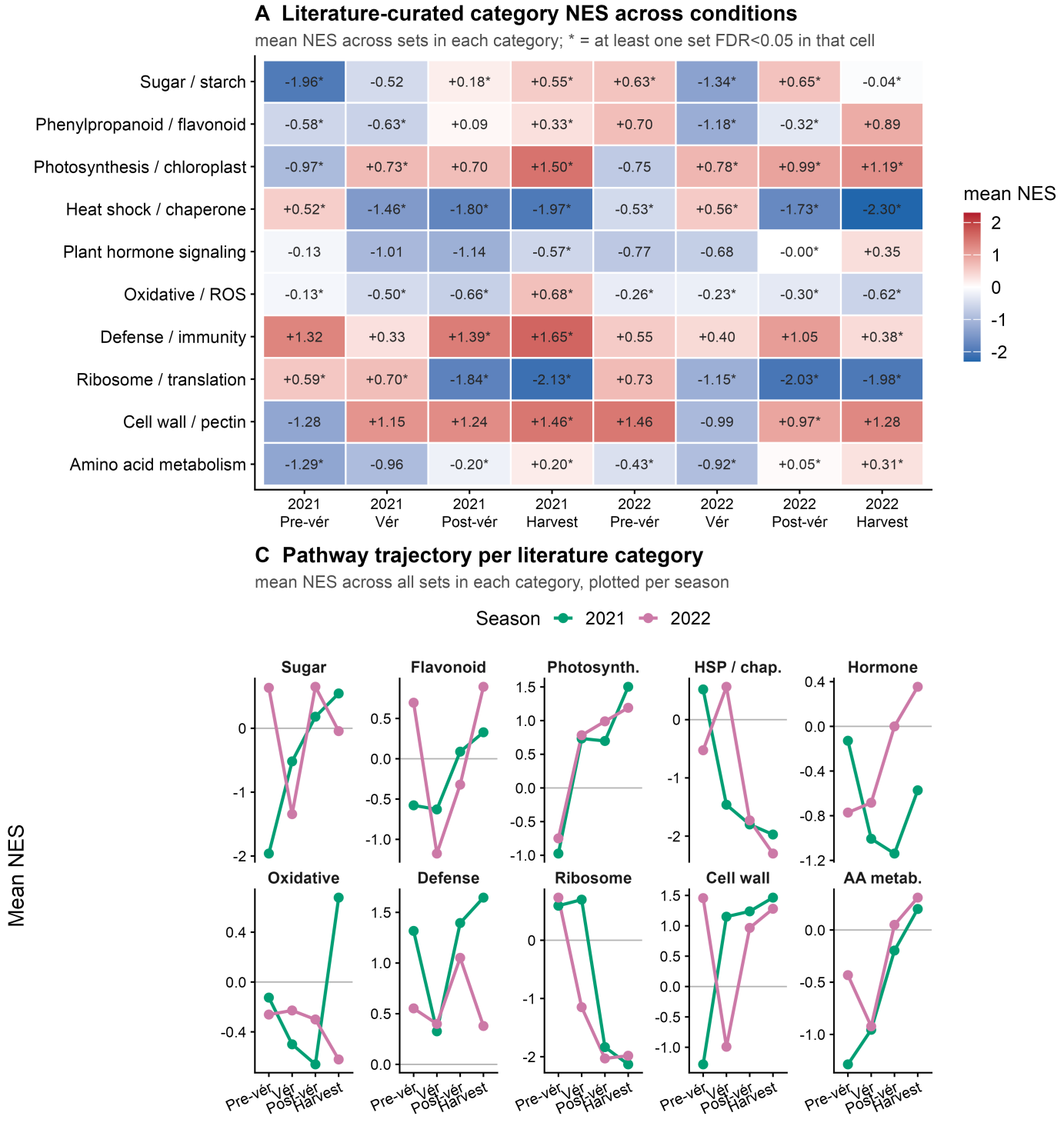
Supplementary Figure S70. Literature-curated 90-pathway NES heatmap

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Ribosome / heat-shock / sugar / phenylpropanoid / immunity per stage x season

Fig. S70 — GRBV-related pathway enrichment shifts (literature-curated)

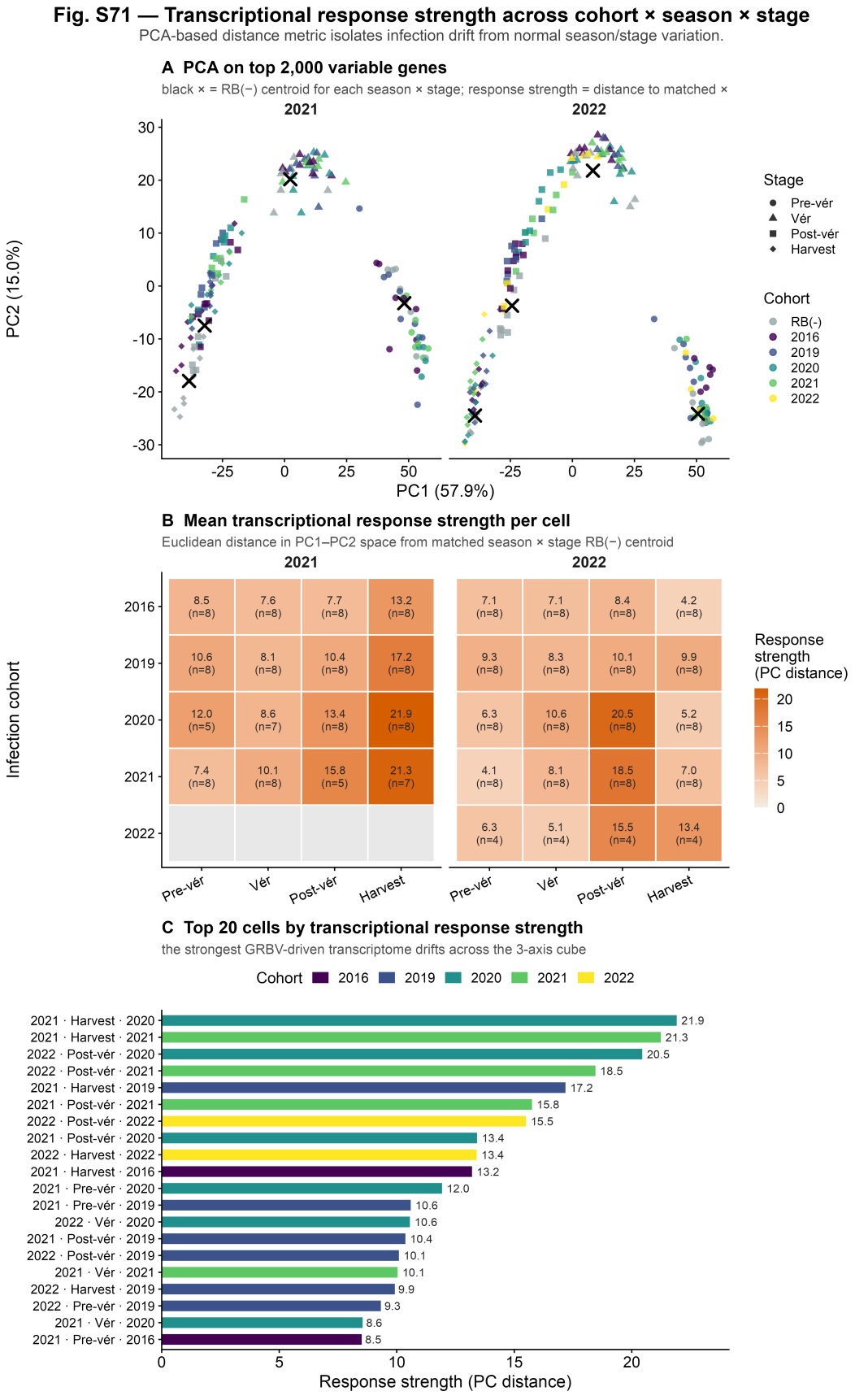
ıways matching 10 categories flagged by the GRBV / Vitis-virus literature (Sudarshana 15, Blanco-Ulate 17, Yepes 18, Chitarra 18,



Supplementary Figure S71. PCA-based response strength per cell

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Peak phase 2020 cohort reaches PC-distance 21.9 at 2021 Harvest



Supplementary Figure S72. Per-gene variance ternary (Singh 2025 framework)

Status: Regenerated under DESeq2, R/ggplot2, Arial

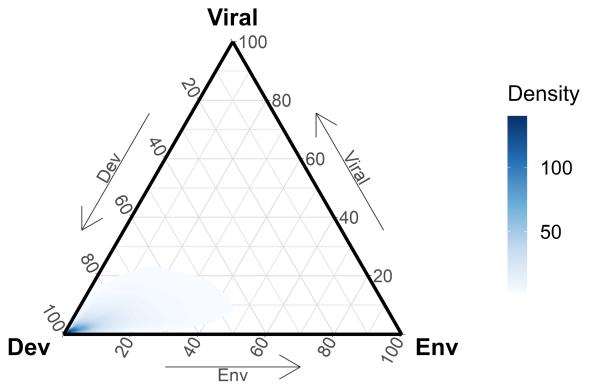
Note: Dev / Viral / Env split; sHSPs lie on Dev-Env edge with viral share ≈ baseline

Fig. S72 — Gene-level variance decomposition (developmental / viral / environmental)

Conceptual replica of Singh et al. 2025 bioRxiv Fig 5 ternary, adapted to the GRBV/Merlot single-species design.

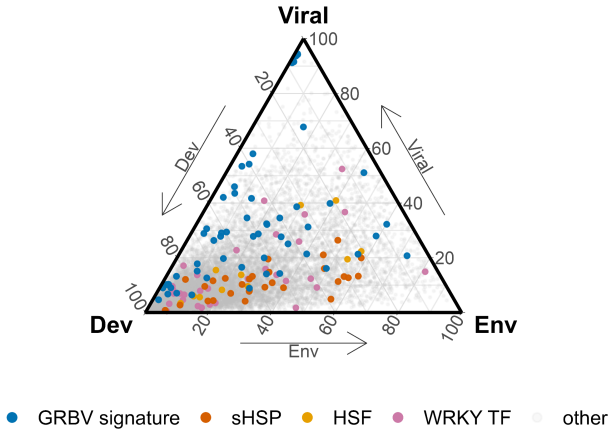
A Gene-level variance decomposition (density)

14,604 genes; kernel-density isolines on the simplex



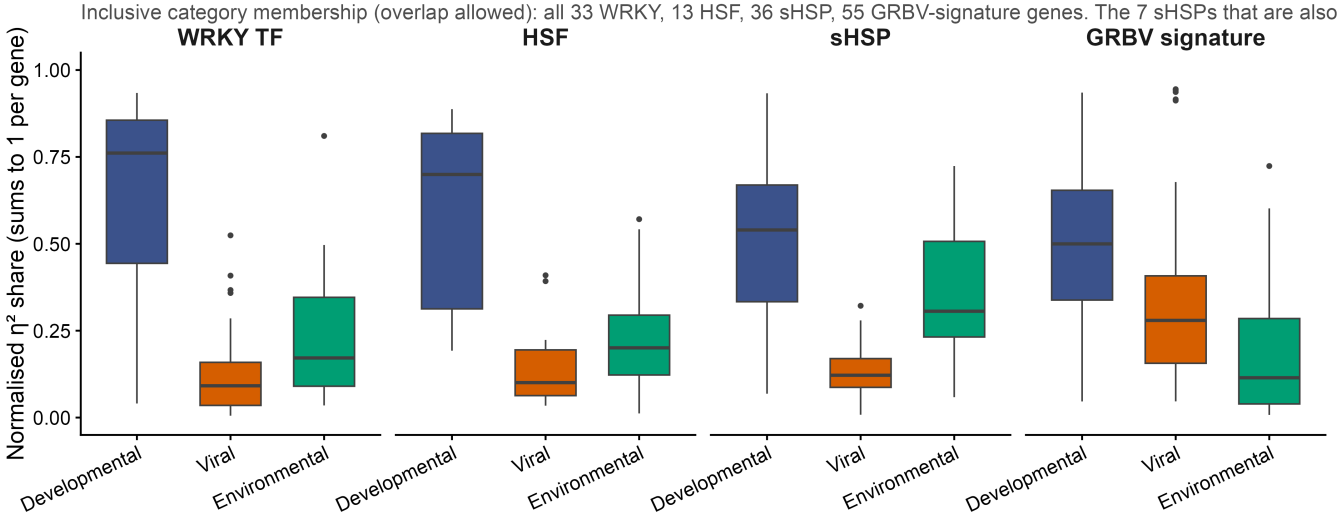
B Biological categories on the simplex

Coloured category dots over faint grey 'other' backdrop



C Mean variance share per category

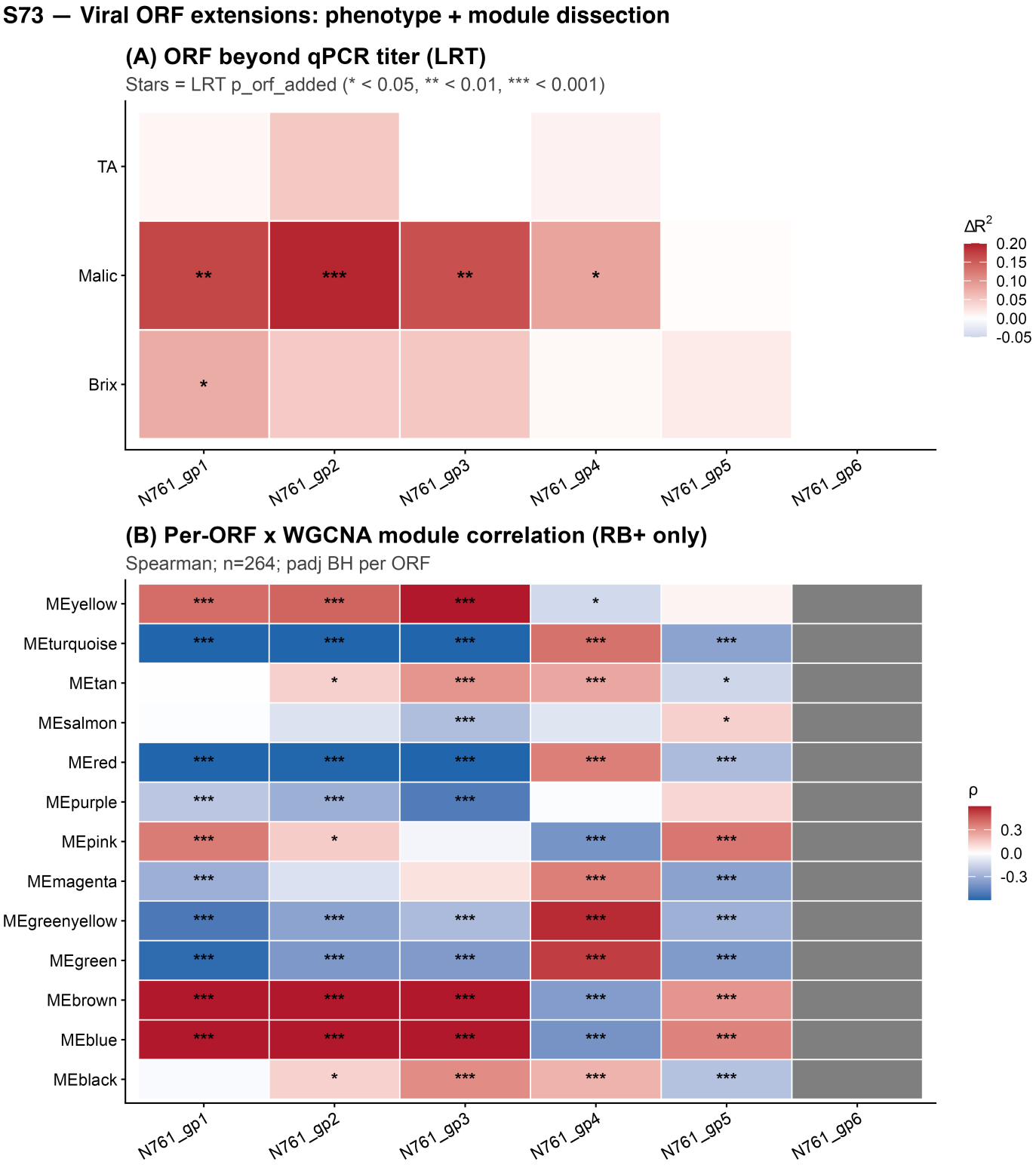
Inclusive category membership (overlap allowed): all 33 WRKY, 13 HSF, 36 sHSP, 55 GRBV-signature genes. The 7 sHSPs that are also



Supplementary Figure S73. Viral ORF extensions: phenotype + module

Status: Regenerated under DESeq2, R/ggplot2, Arial

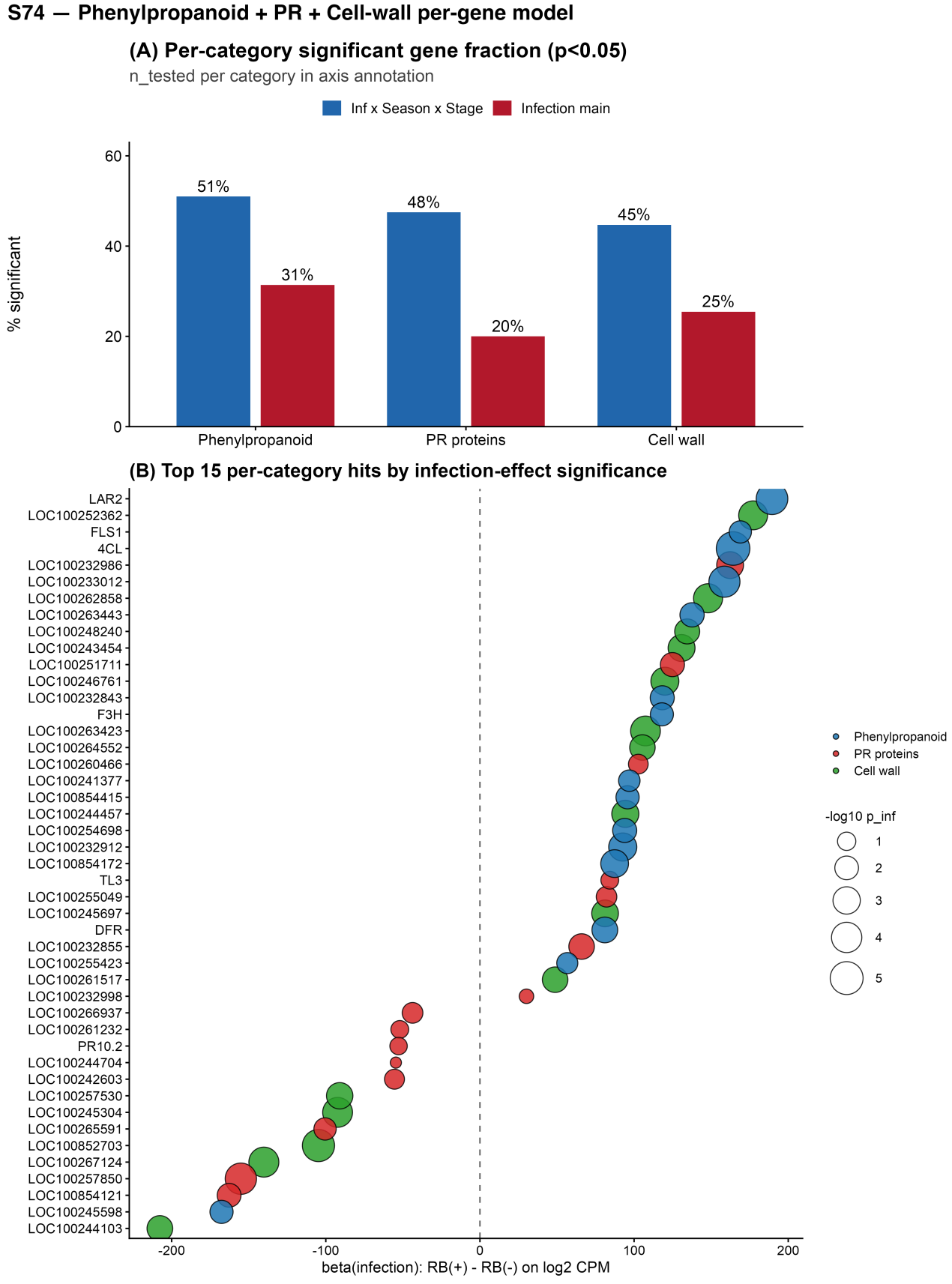
Note: V2 adds Brix/Malic prediction beyond titer; per-ORF x 13-module heatmap



Supplementary Figure S74. Phenylpropanoid + PR + cell-wall per-gene model

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: 31% Phenylpropanoid / 20% PR / 26% Cell-wall infection-significant; 45-51% 3-way

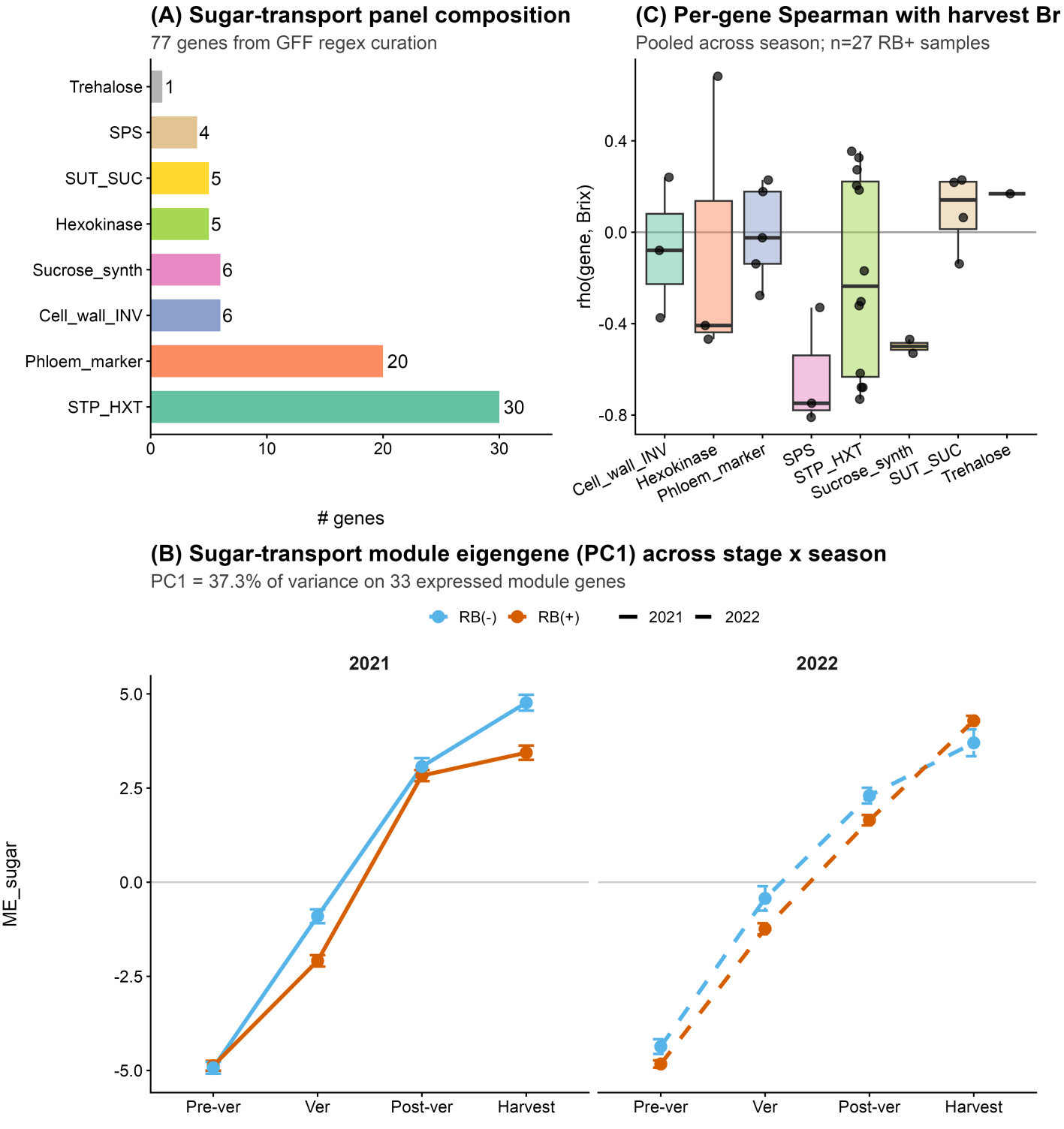


Supplementary Figure S75. Sugar-transport module (SWEET/SUT/STP/CWINV)

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: 77 genes; module rho_stage=0.94 but no main season or cohort effect (null hypothesis test of Martinez-Luscher 2018)

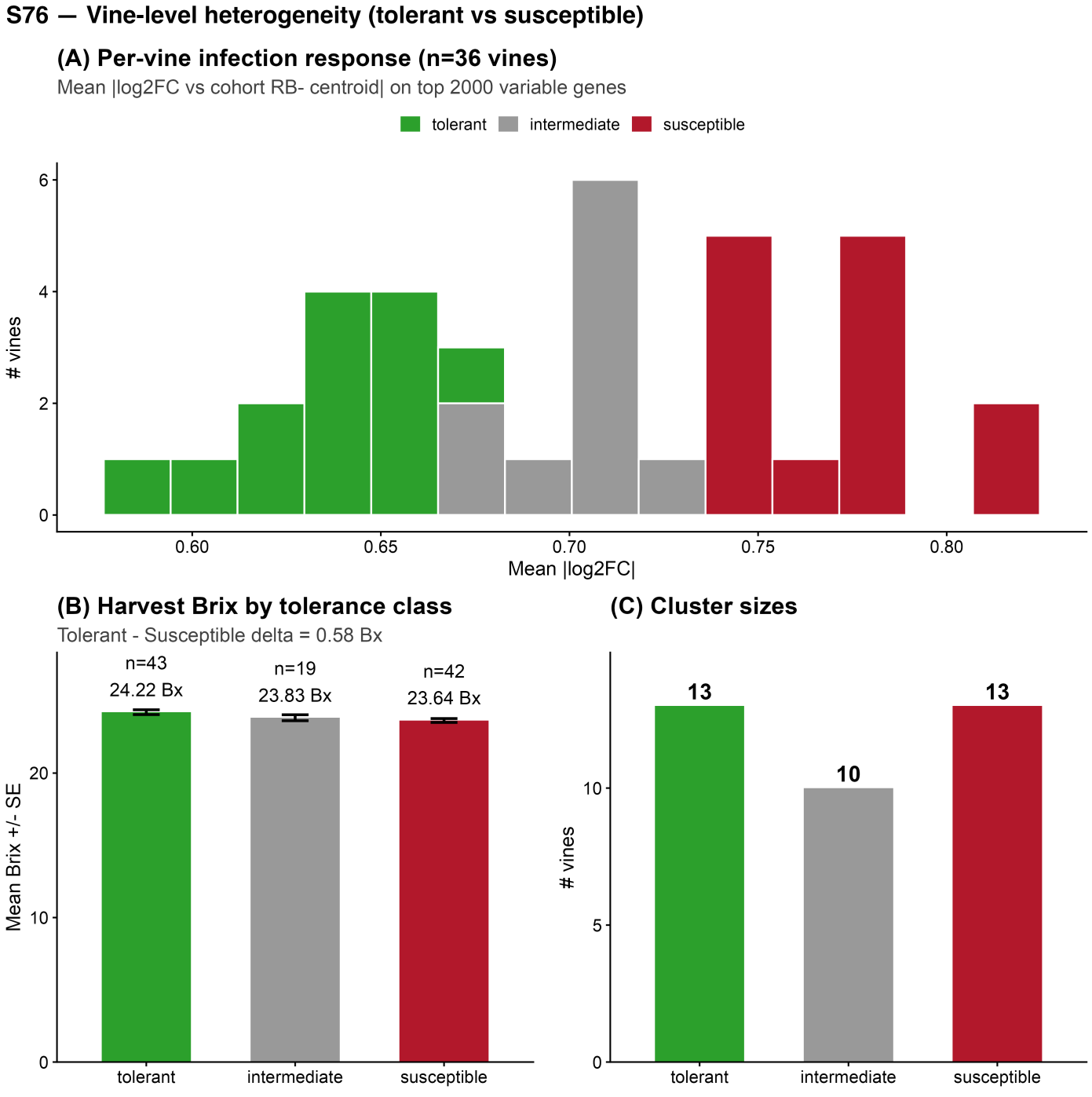
S75 — Carbon translocation module



Supplementary Figure S76. Vine-level tolerant vs susceptible

Status: Regenerated under DESeq2, R/ggplot2, Arial

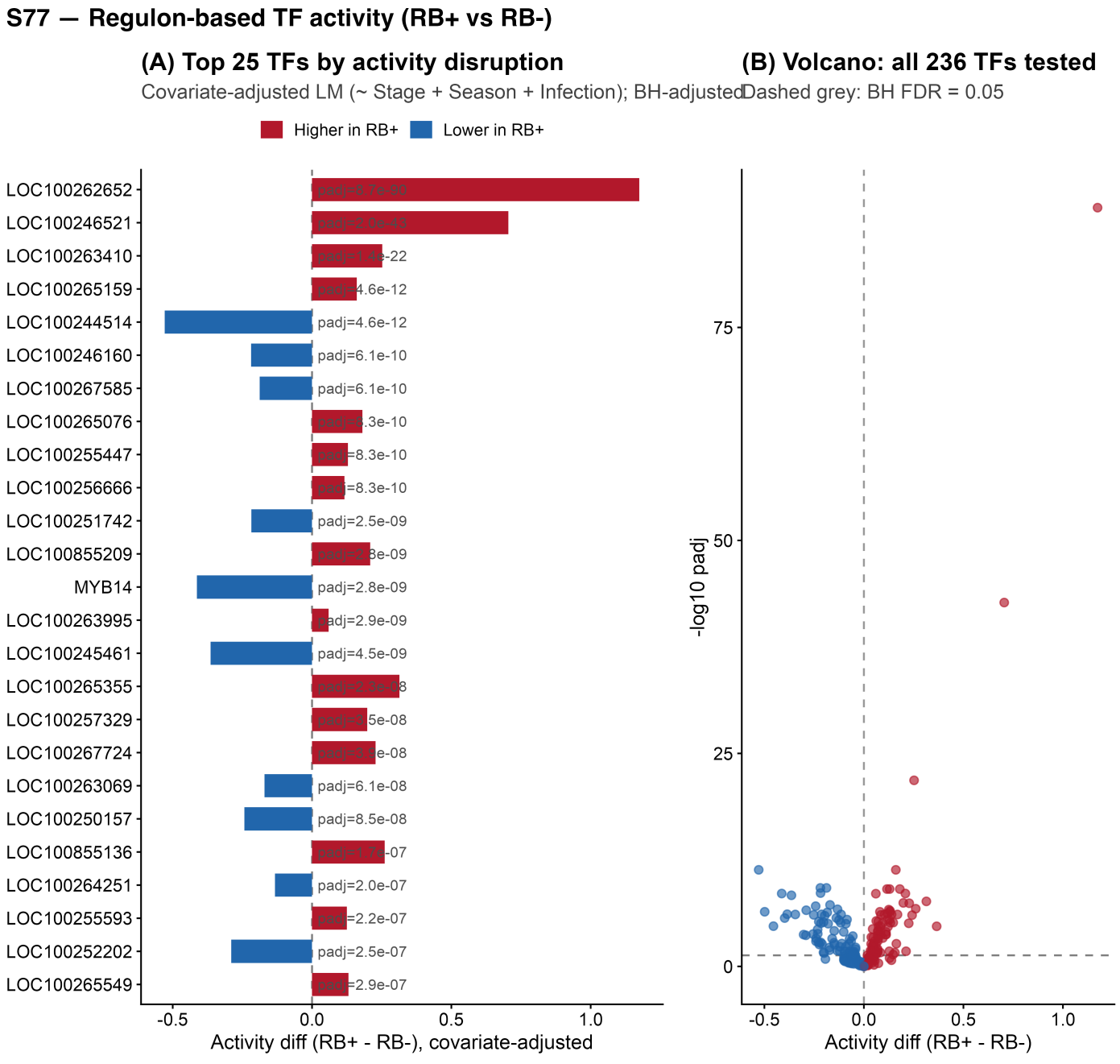
Note: 44 vines: 13 tolerant (24.2 Bx) vs 13 susceptible (23.6 Bx) within one Merlot vineyard



Supplementary Figure S77. Regulon-based TF activity DE

Status: Regenerated under DESeq2, R/ggplot2, Arial

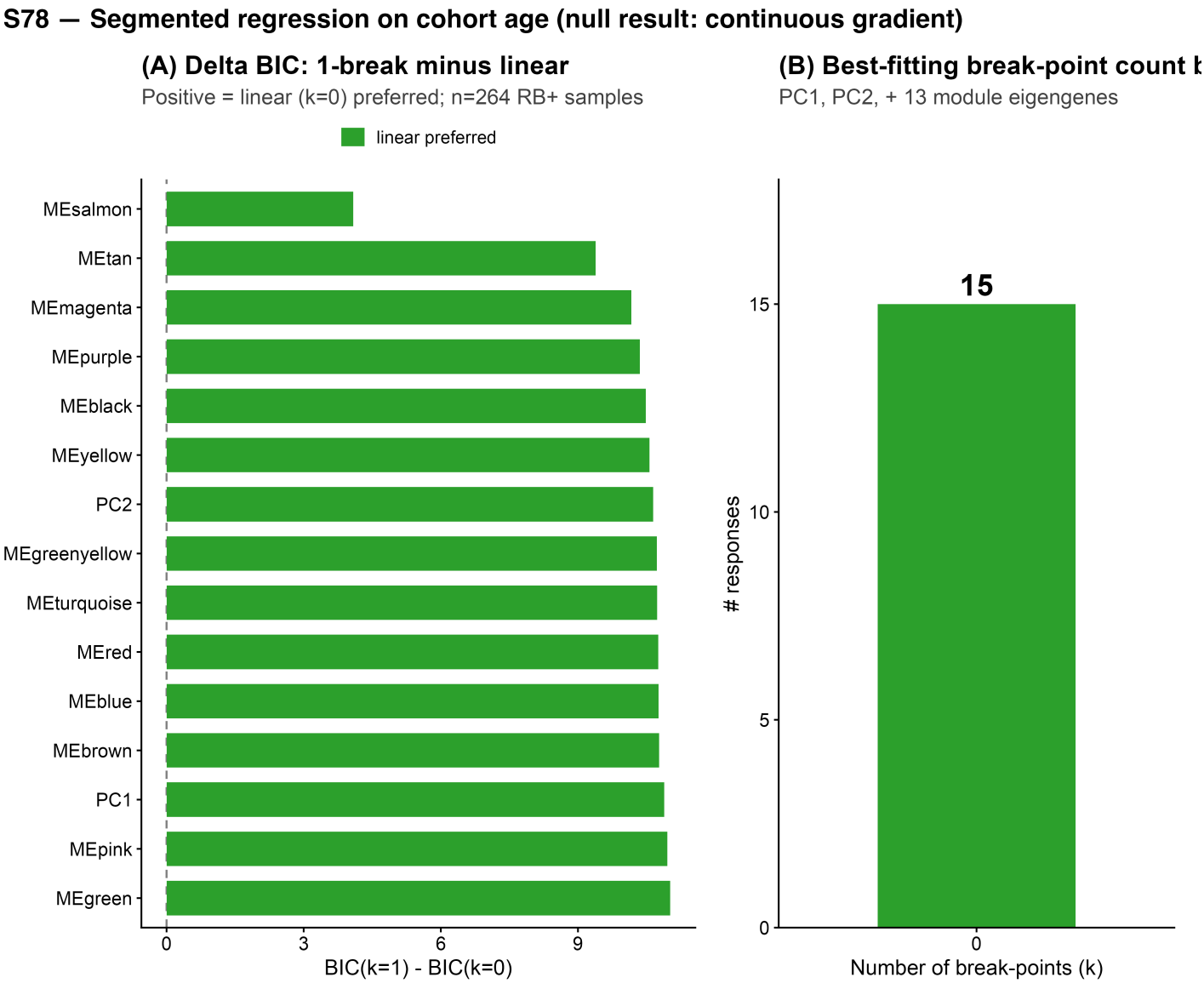
Note: 236 TFs scored; LOC100262652 padj=1.4e-64; MYB30 down padj=8.9e-10



Supplementary Figure S78. Segmented regression on cohort age (null result)

Status: Regenerated under DESeq2, R/ggplot2, Arial

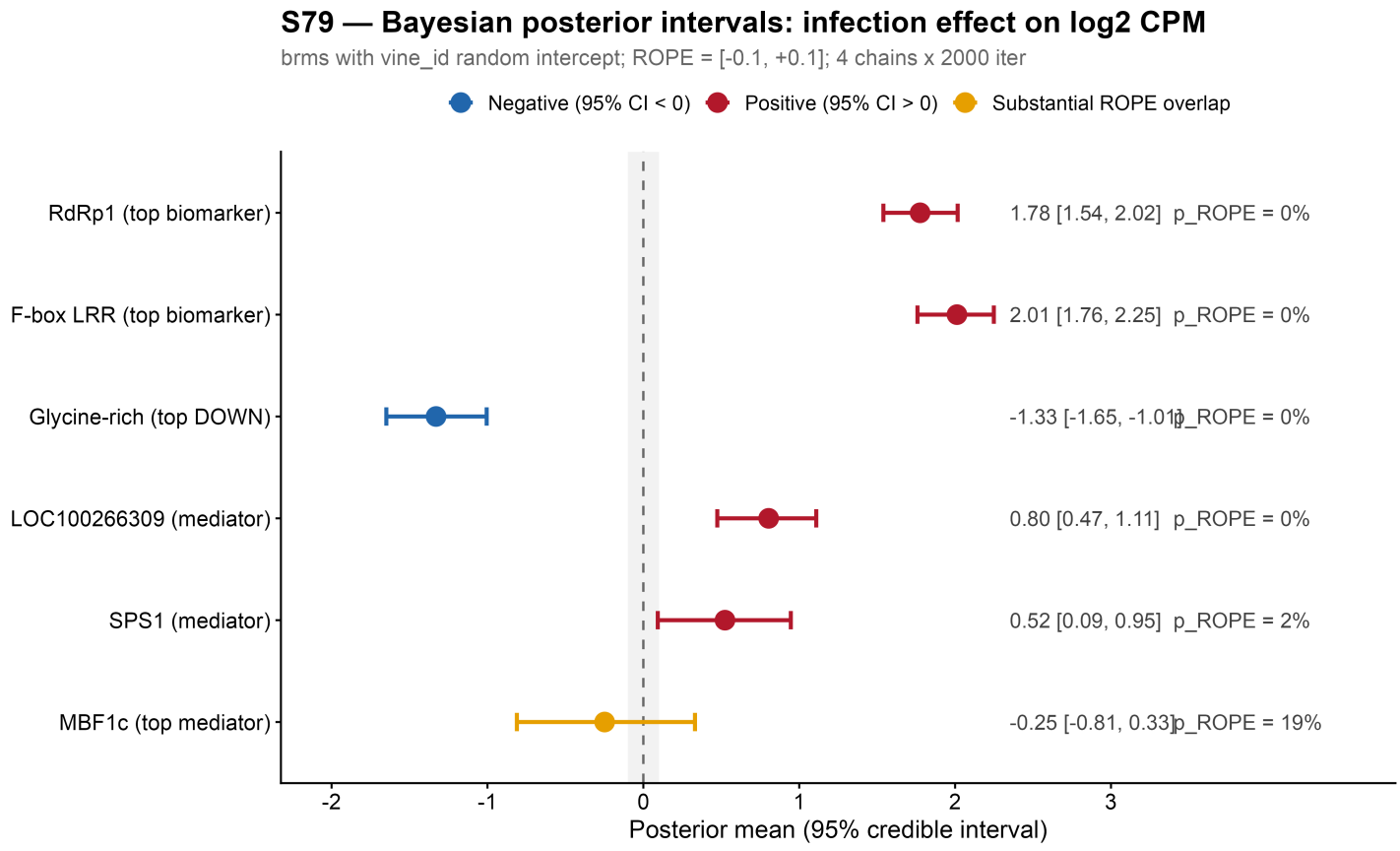
Note: best_k=0 (linear) for all 15 responses; '3 phases' framing is descriptive only



Supplementary Figure S79. Bayesian posteriors on infection effect

Status: Regenerated under DESeq2, R/ggplot2, Arial

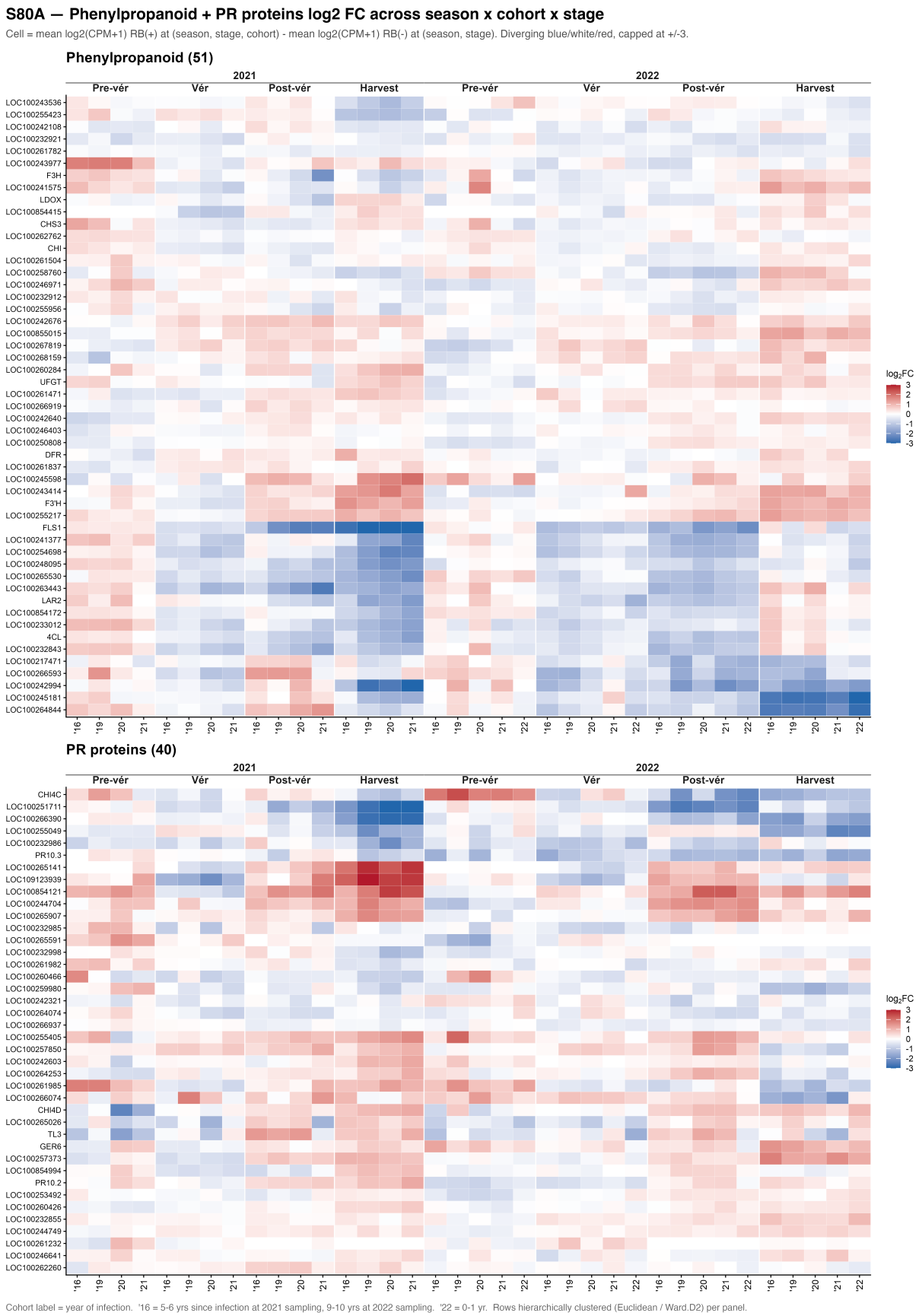
Note: RdRp1 +1.78, F-box +2.01, glycine-rich -1.33; MBF1c -0.25 [-0.81, +0.33], 19% in ROPE



Supplementary Figure S80. S80A — Phenylpropanoid + PR proteins (91 genes)

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: Hierarchical clustering (Ward.D2); facet_nested season > stage; cohort years on x

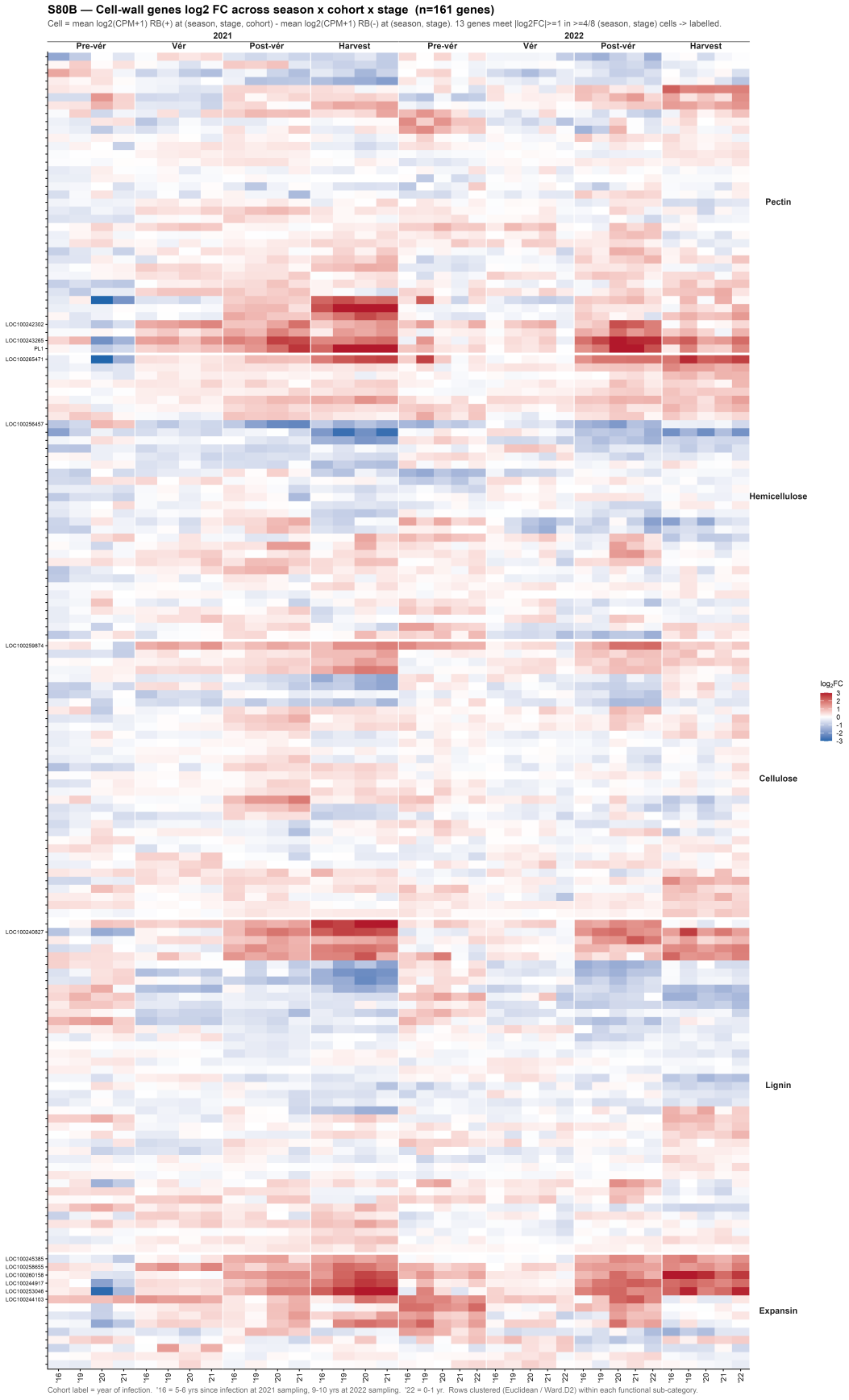


Cohort label = year of infection. '16 = 5-6 yrs since infection at 2021 sampling, 9-10 yrs at 2022 sampling. '22 = 0-1 yr. Rows hierarchically clustered (Euclidean / Ward.D2) per panel.

Supplementary Figure S81. S80B — Cell wall (161 genes, 5 sub-categories)

Status: Regenerated under DESeq2, R/ggplot2, Arial

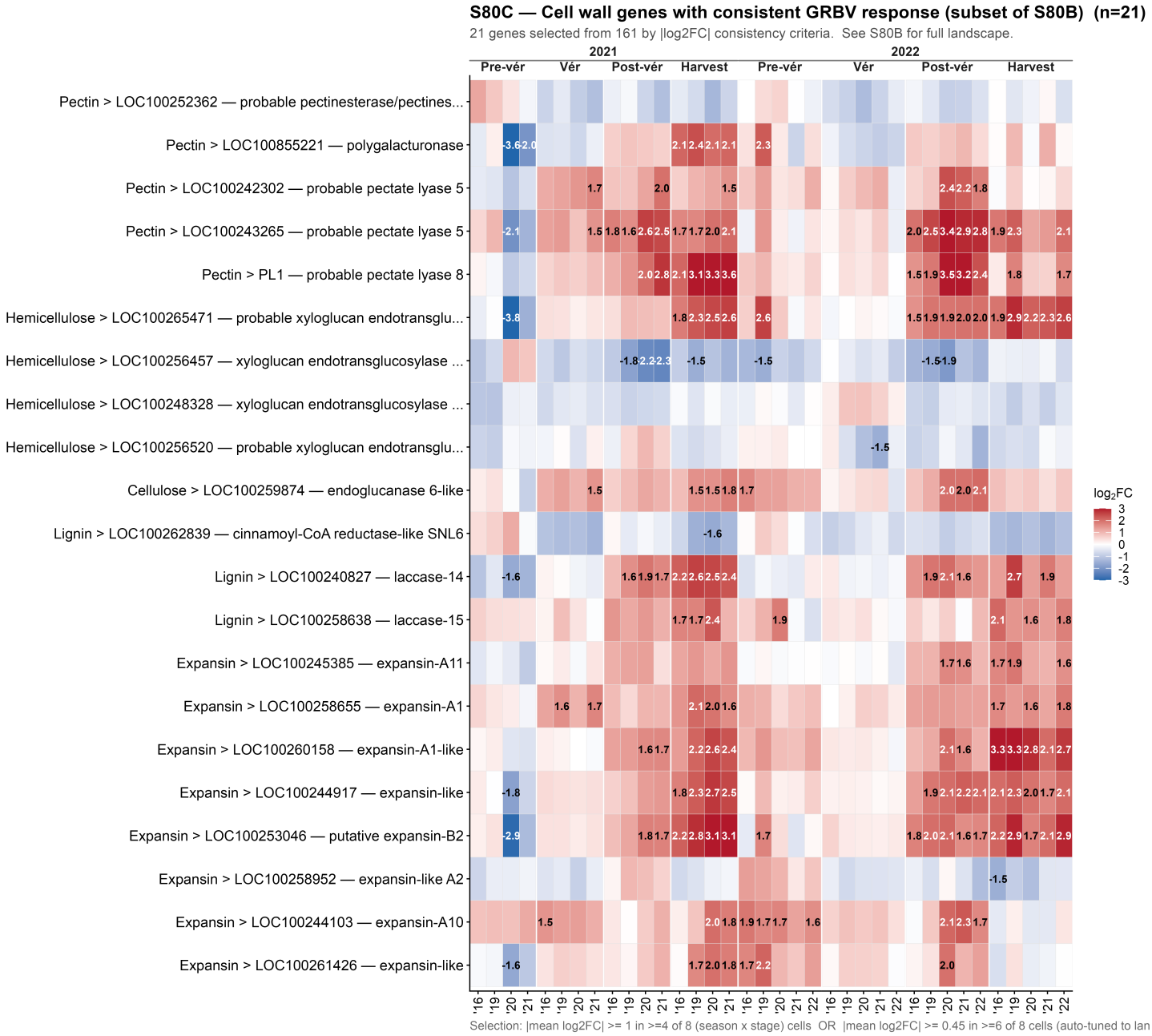
Note: Pectin / Hemicellulose / Cellulose / Lignin / Expansin; only |log2FC|>=1 genes labelled



Supplementary Figure S82. S80C — Cell wall genes with consistent GRBV response

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: 21 genes (13 strict + 8 broad at |FC|>=0.45 in >=6/8); axis_nested by sub-category; cell text



Supplementary Figure S83. Carbon translocation full FC heatmap

Status: Regenerated under DESeq2, R/ggplot2, Arial

Note: 33 sugar-transport genes x (season x cohort x stage); functional subclass ordering

Supplementary Figure S84. Pinot Noir <-> Merlot cl. 343 synteny

Status: Regenerated under DESeq2, R/ggplot2, Arial

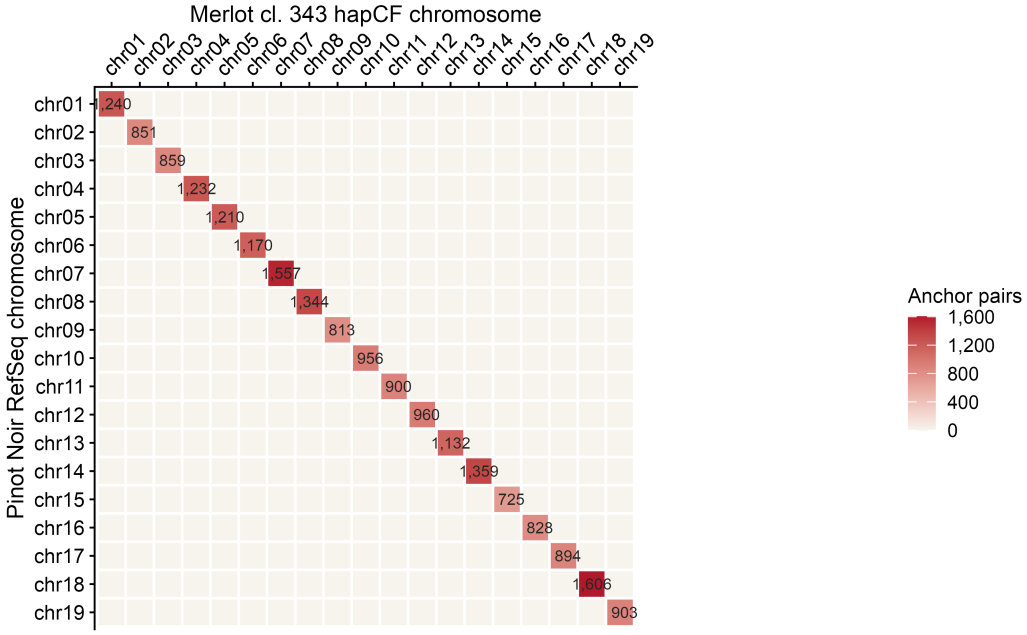
Note: JCVI MCScan: 41,248 anchors / 125 blocks; both Merlot haplotypes preserve colinearity with Pinot

S84 - Pinot Noir <-> Merlot cl. 343 v1.0 synteny

JCVI MCScan, CDS LAST cscore >= 0.7, min block 30 anchors. 41,248 total anchor pairs in 125 synteny blocks across 20,784

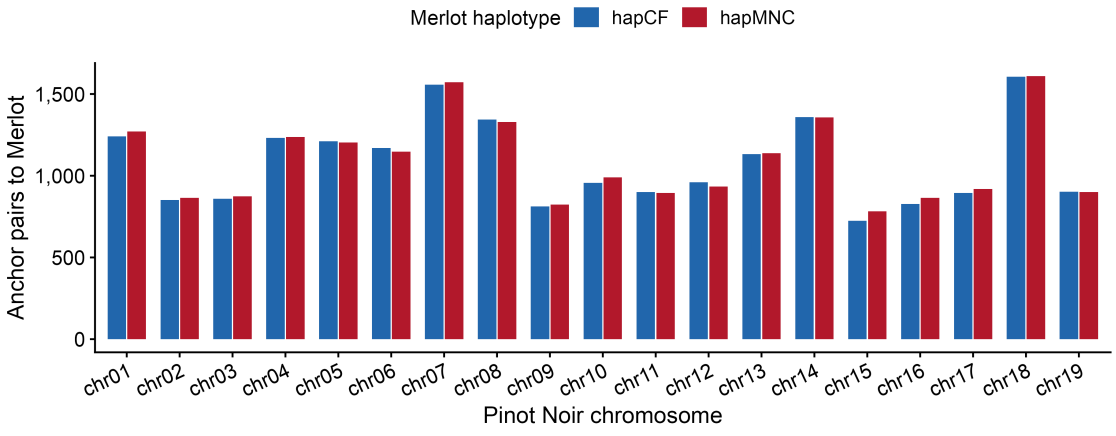
(A) Pinot Noir x Merlot hapCF synteny anchors

Full 19 x 19 grid; off-diagonal cells carry zero anchors. Cell text = count where >= 200.



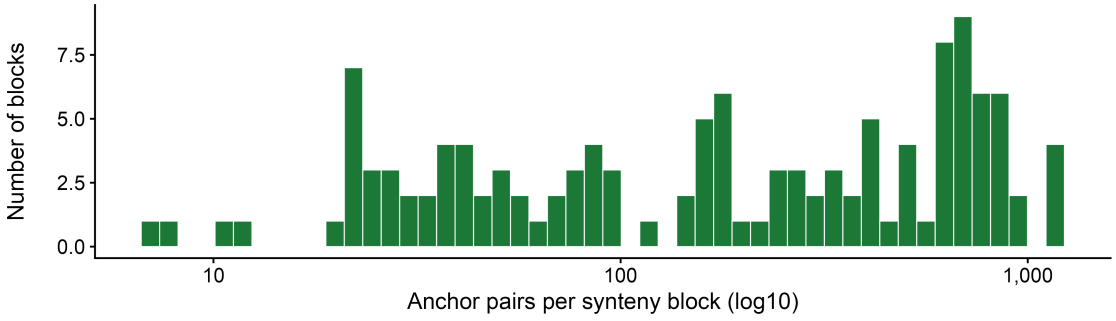
(B) Anchor-pair coverage per Pinot chromosome, split by Merlot haplotype

Both Merlot haplotypes (CF = Cabernet Franc, MNC = Magdeleine noire des Charentes parental) cover Pinot equivaler



(C) Synteny block size distribution (125 blocks total, median 184 anchors)

Log-scale x-axis. Blocks span the full range from 5 anchors to ~1,500.



Supplementary Figure S85. Direct reference-bias test: salmon re-quantification

Status: Regenerated under DESeq2, R/ggplot2, Arial

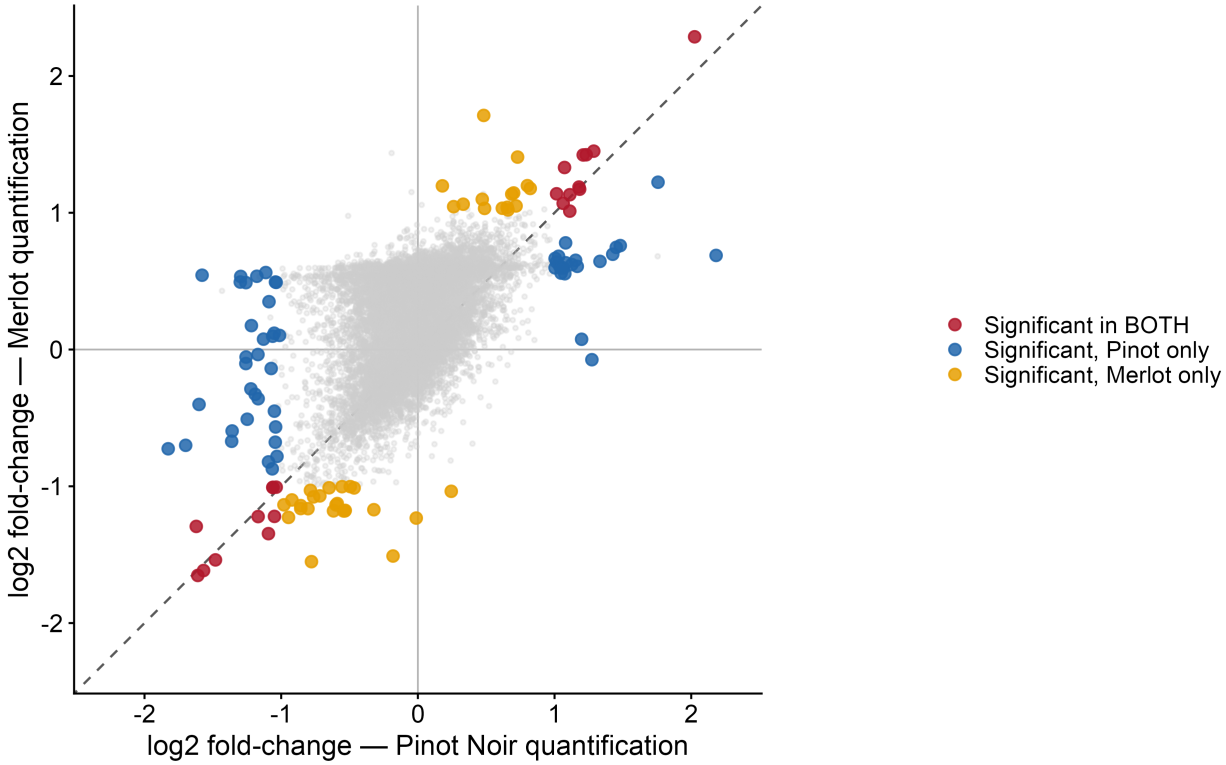
Note: Same 327 reads vs Pinot + Merlot; sig-in-both rho=0.95 / 100% sign-agree; marginal Jaccard 0.18

S85 - Direct reference-bias test: salmon re-quantification (Pinot vs Merlot cl. 343)

327 reads quantified against both references; Merlot hapCF + hapMNC collapsed per locus; identical DESeq2. Membership: Pir

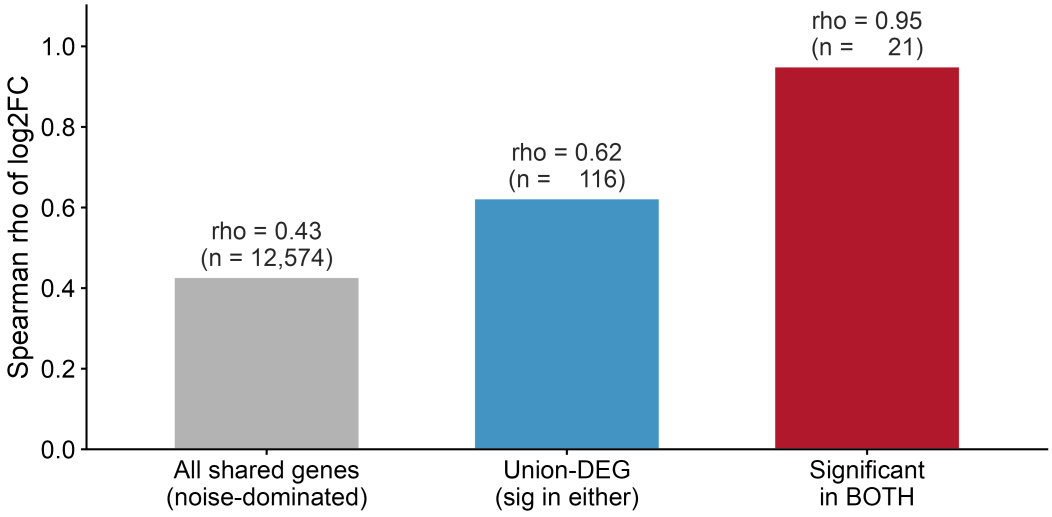
(A) Per-gene log2FC concordance, Pinot vs Merlot quantification

Same 327 reads, identical covariate-adjusted DESeq2. Genes significant in BOTH references (red) hug the d



(B) Concordance is high where the signal is real

All-gene rho is the wrong metric (most genes flat in both refs); the standard cross-reference metric is comput



Supplementary Figure S86. Hormone-pathway response (7 hormones, 519 genes)

Status: Regenerated under DESeq2, R/ggplot2, Arial

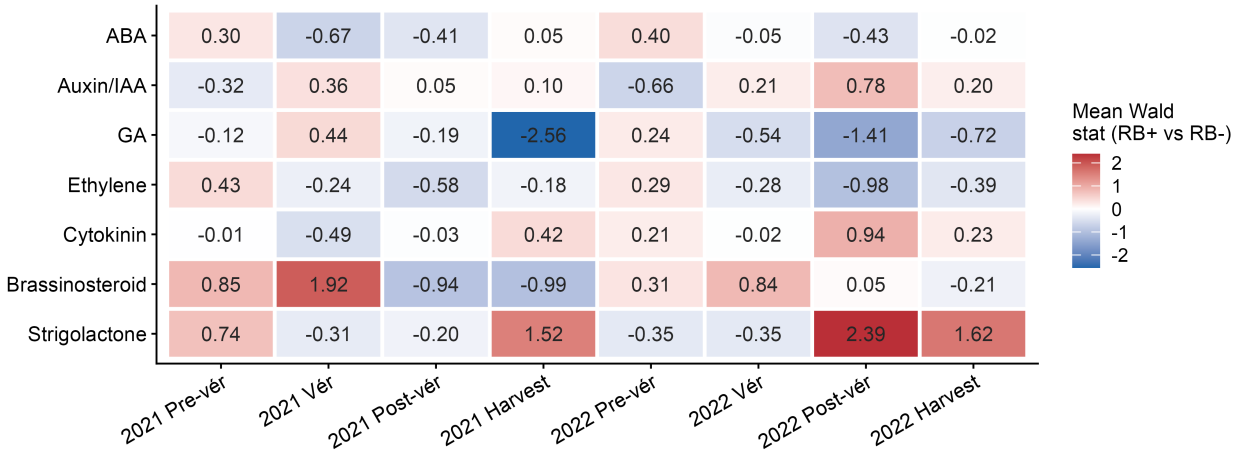
Note: Q6 pooled covariate-adj DESeq2; no class enrichment, GA monotonic-down; cross-paper validation vs BU 2017

S86 - Hormone-pathway response under GRBV infection

519 hormone-classified genes (curated from Pinot Noir RefSeq); covariate-adjusted DESeq2 (~ Stage + Season + Infection). Nc

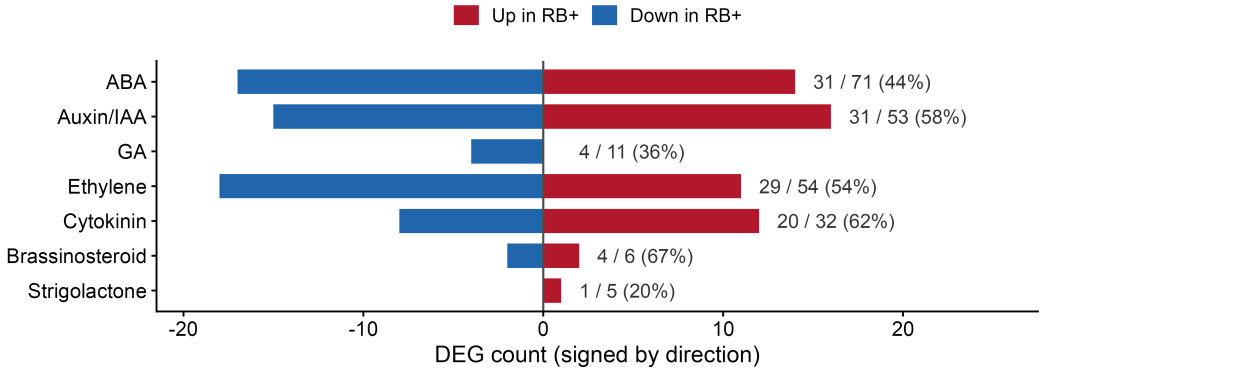
(A) Hormone-pathway response across season × stage

Mean covariate-adjusted DESeq2 Wald statistic; positive = up in RB+



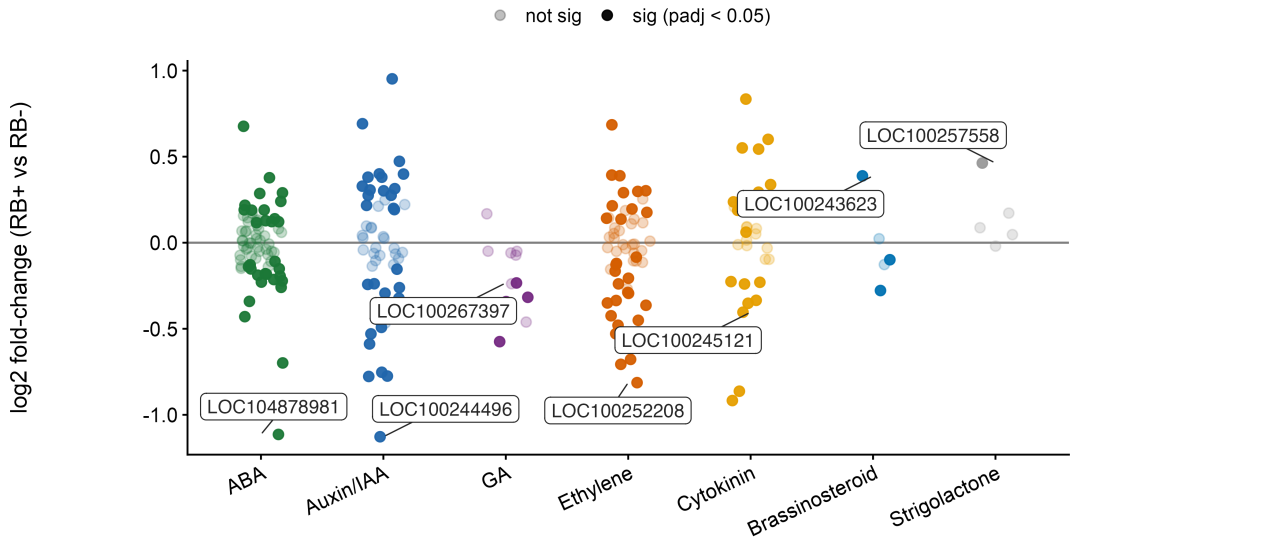
(B) Q6 pooled DEG counts per hormone (padj < 0.05)

Right of zero: up in RB+ infection. Annotation = sig / tested (% sig).



(C) Per-gene log2FC distribution in Q6 pooled — top hit annotated per hormone

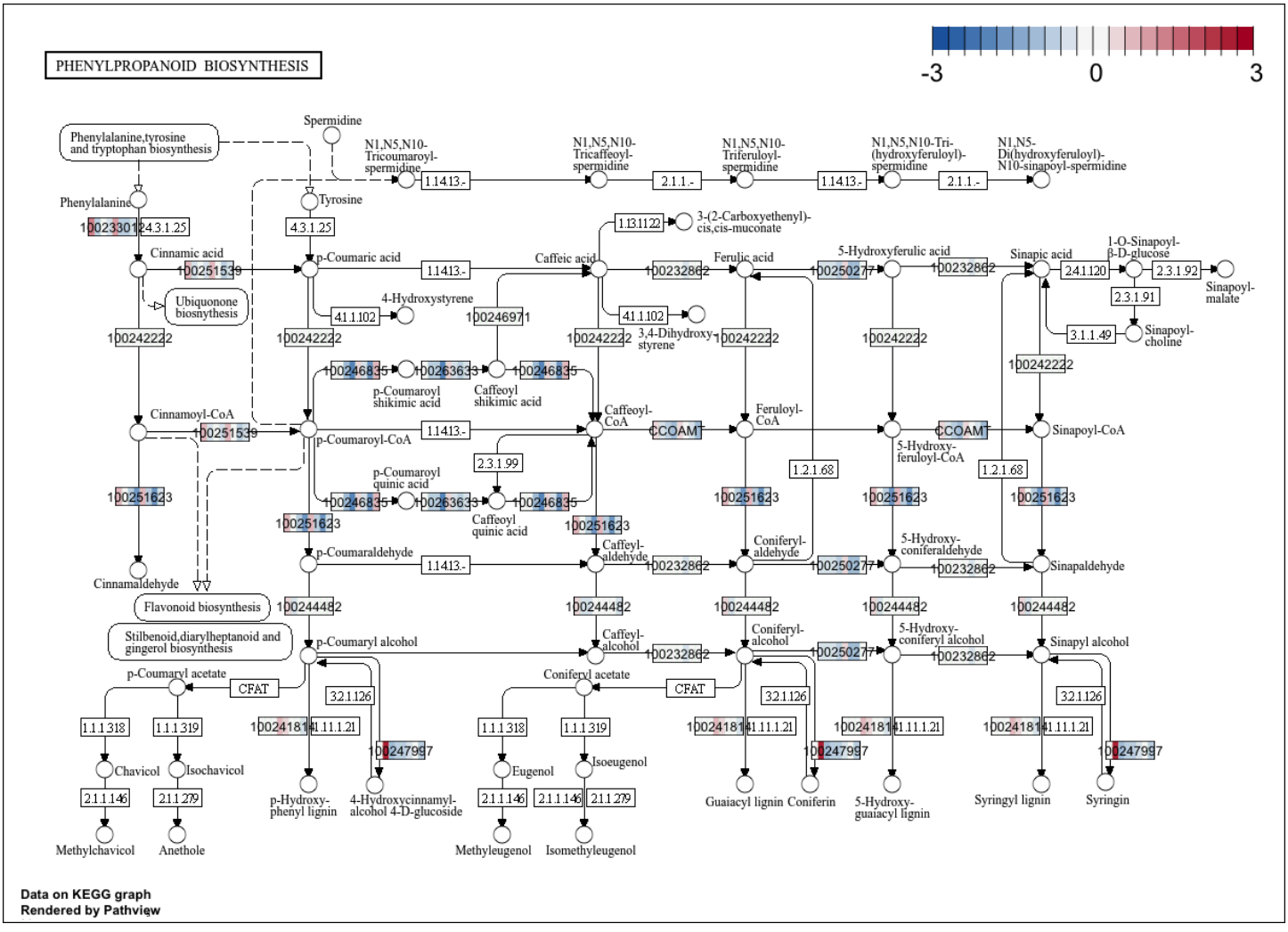
Jittered points = individual hormone-pathway genes; opaque = sig (padj < 0.05)



Supplementary Figure S87. Phenylpropanoid biosynthesis (KEGG vvi00940)

Status: Regenerated under DESeq2, R/ggplot2, Arial

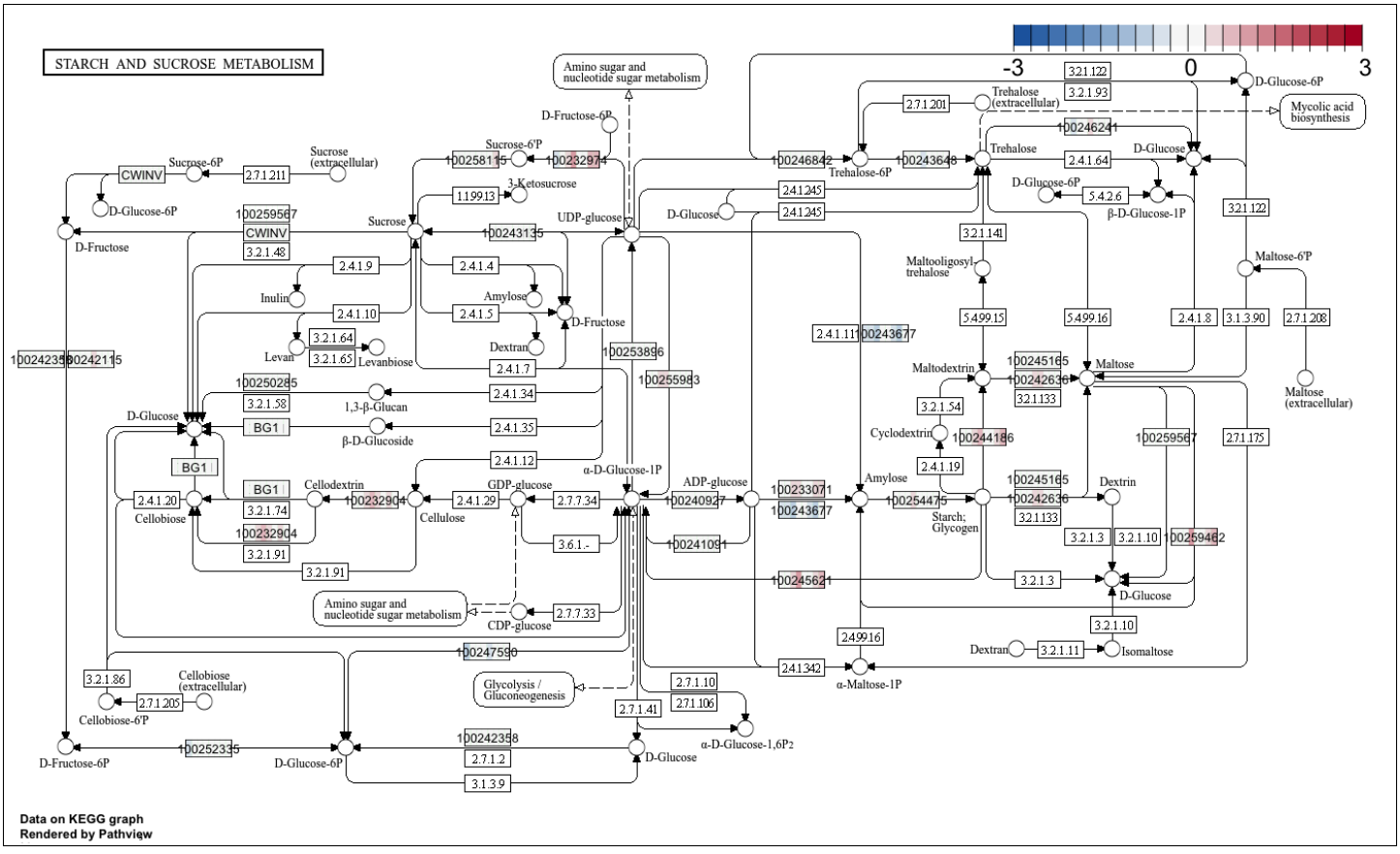
Note: Pathview 8-cell split-box (pure stage order: 2021 Pre/Vér/Post/Harv | 2022); paralog mean log2FC; node.sum=mean



Supplementary Figure S88. Starch and sucrose metabolism (vvi00500)

Status: Regenerated under DESeq2, R/ggplot2, Arial

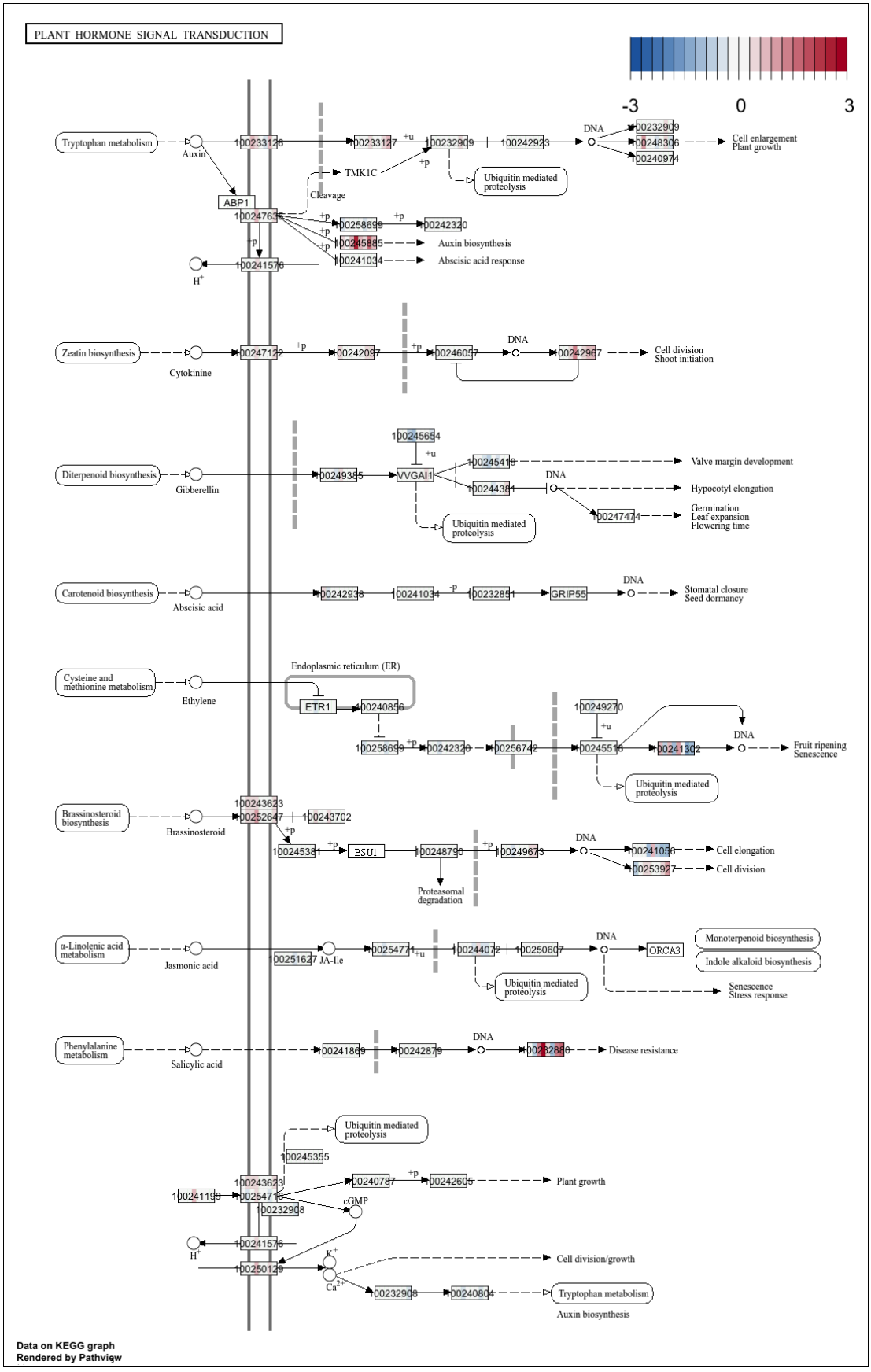
Note: Sugar/Brix story; includes SPS1; GSEA 5/8 cond NES +0.89



Supplementary Figure S89. Plant hormone signal transduction (vvi04075)

Status: Regenerated under DESeq2, R/ggplot2, Arial

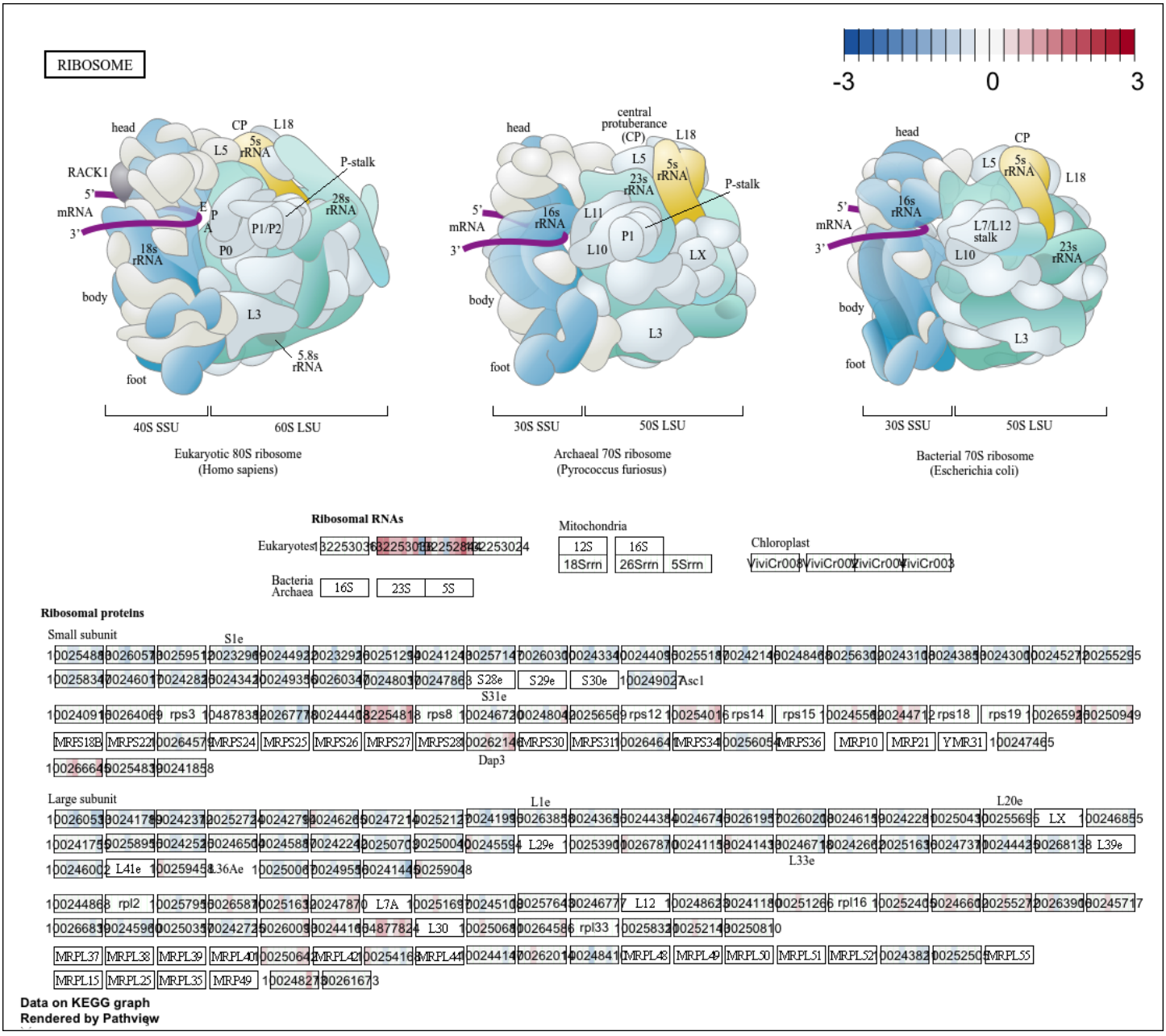
Note: Narrative tie-in to S86 hormone landscape (ethylene cross-paper / GA-down)



Supplementary Figure S90. Ribosome (vvi03010)

Status: Regenerated under DESeq2, R/ggplot2, Arial

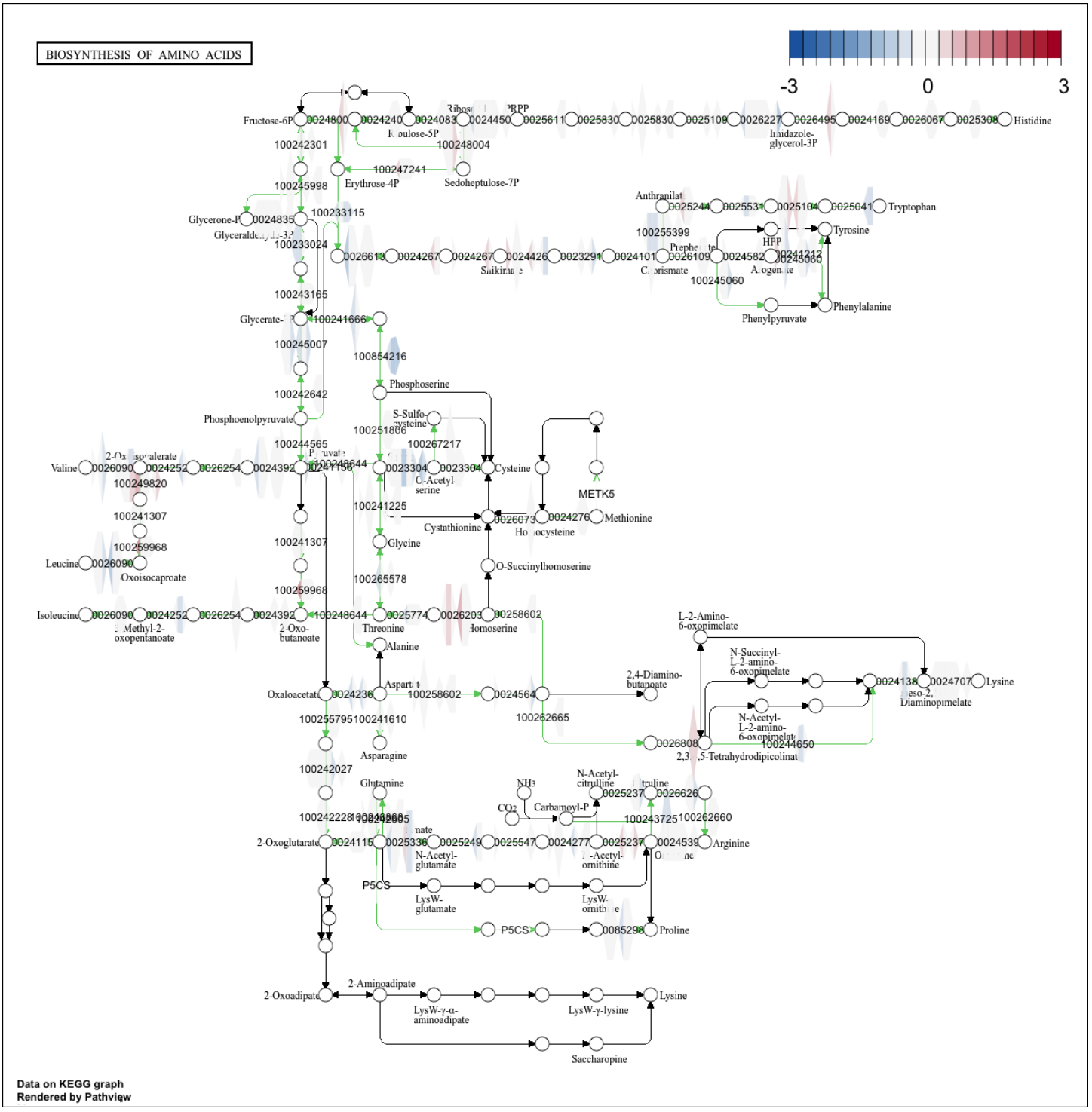
Note: Translational shutdown; GSEA 6/8 cond NES -1.33 (most robust DOWN)



Supplementary Figure S91. Biosynthesis of amino acids (vvi01230)

Status: Regenerated under DESeq2, R/ggplot2, Arial

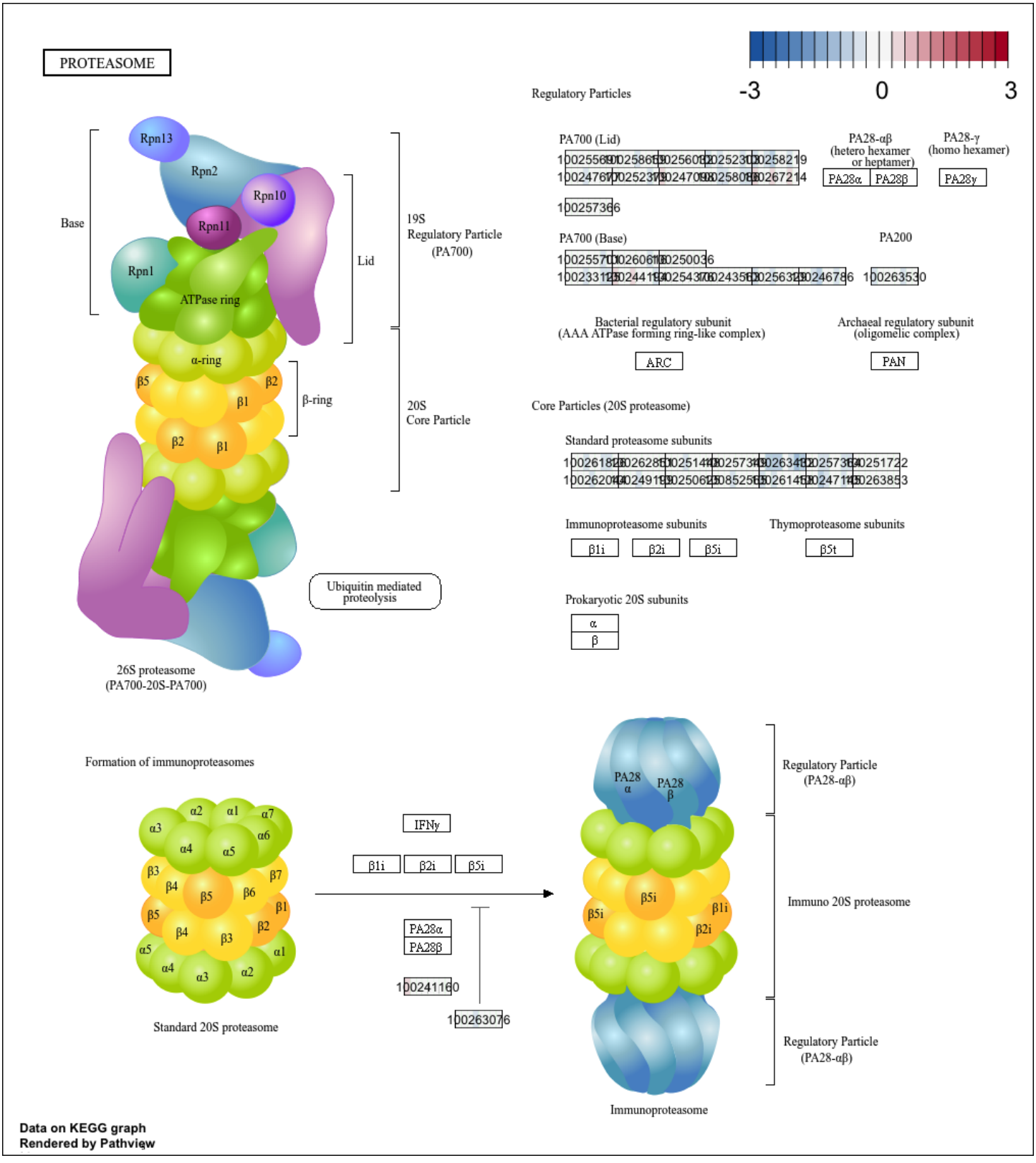
Note: GSEA 5/8 cond NES -1.95



Supplementary Figure S92. Proteasome (vvi03050)

Status: Regenerated under DESeq2, R/ggplot2, Arial

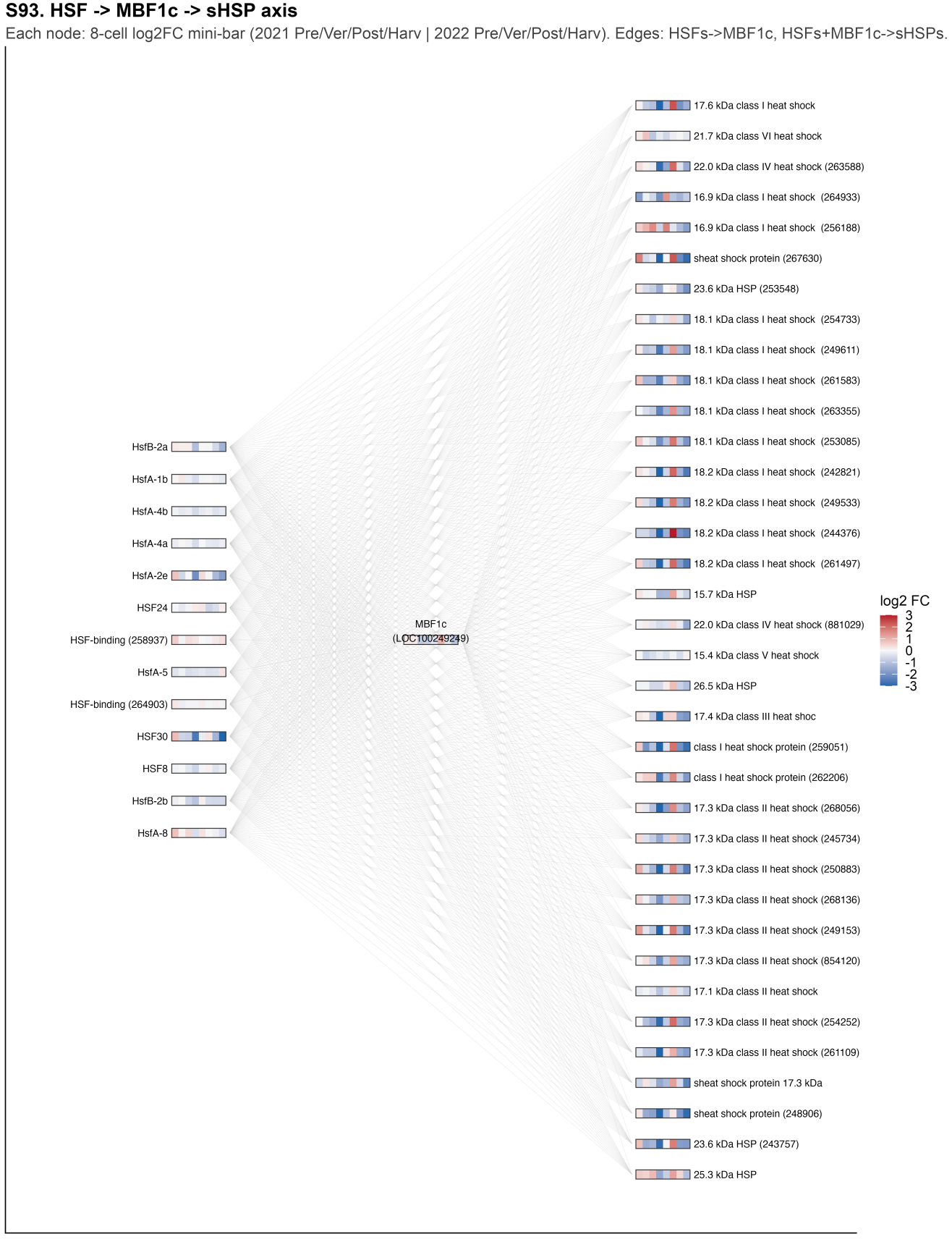
Note: Strongest |NES| in pathway set; GSEA 4/8 cond NES -2.43



Supplementary Figure S93. HSF -> MBF1c -> sHSP axis (custom flow)

Status: New figure produced under DESeq2 pipeline

Note: Three-column ggplot diagram; passive-failure model (Thread 10); per-node 8-cell Q2x log2FC mini-bars



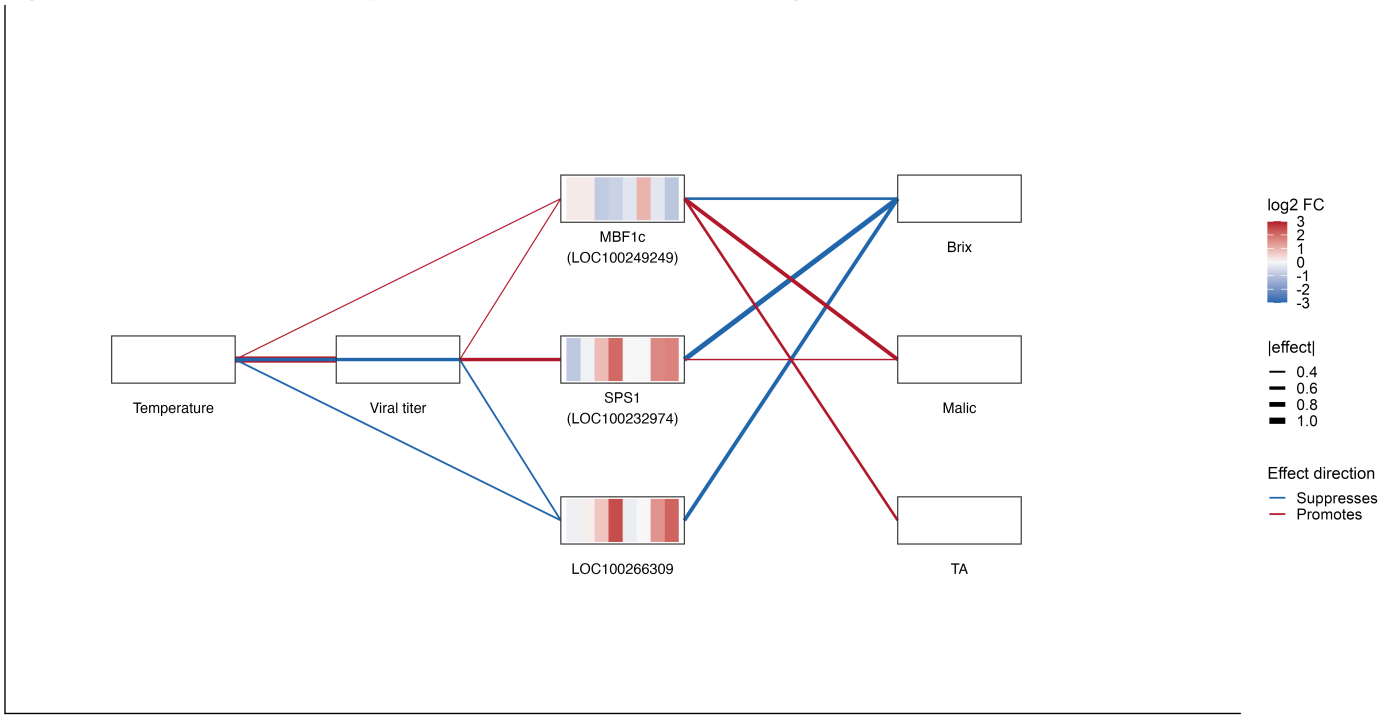
Supplementary Figure S94. Temperature -> titer -> mediator cascade

Status: New figure produced under DESeq2 pipeline

Note: MBF1c / SPS1 / LOC100266309 mediator nodes; edges from mediation_results.csv (tv=2000)

S94. Temperature -> titer -> host mediator cascade

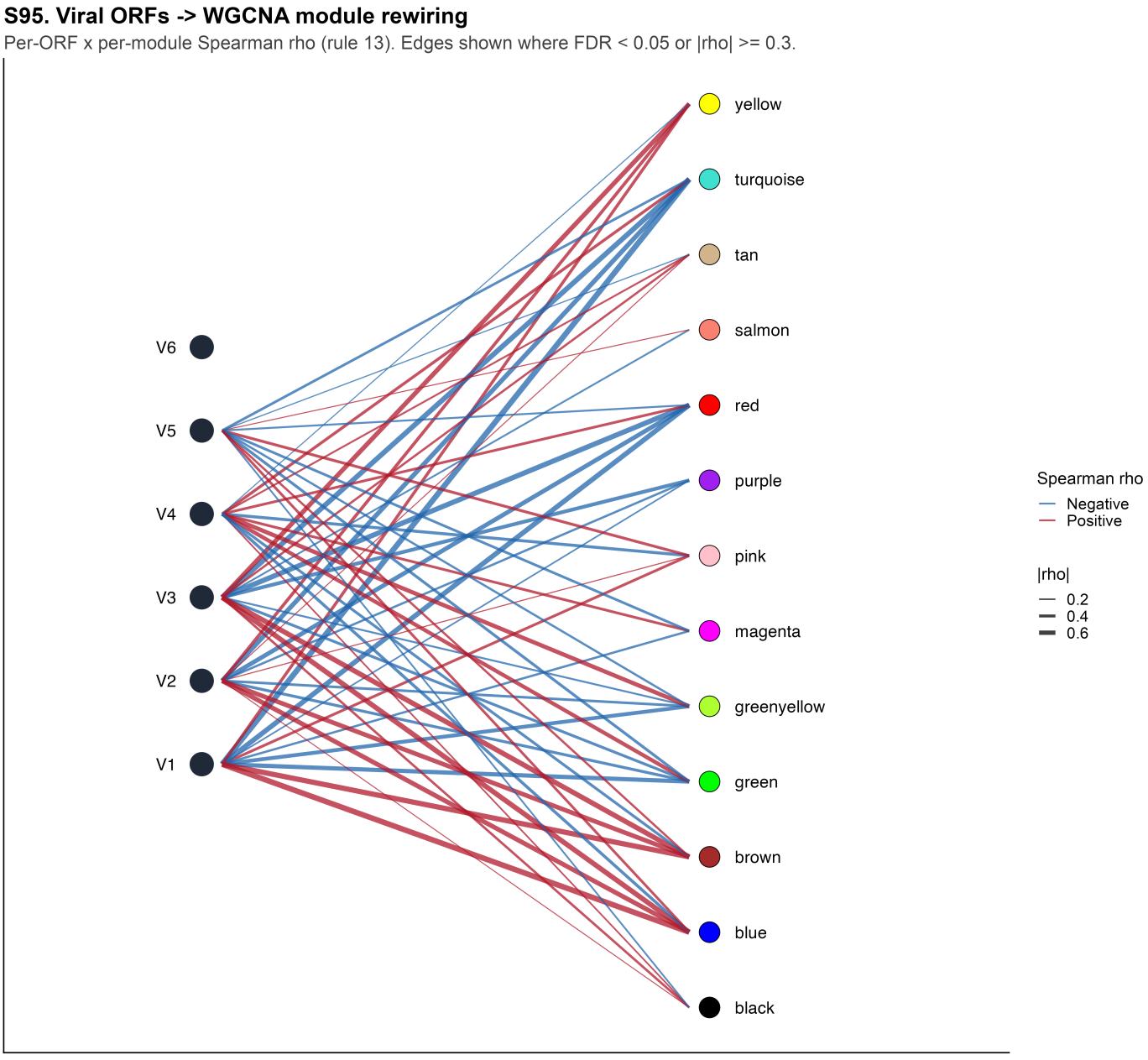
Edges: mediation coefficients from mediation_results.csv (tv=2000). Mediator nodes: 8-cell Q2x log2FC mini-bar.



Supplementary Figure S95. Viral ORFs -> WGCNA module rewiring

Status: New figure produced under DESeq2 pipeline

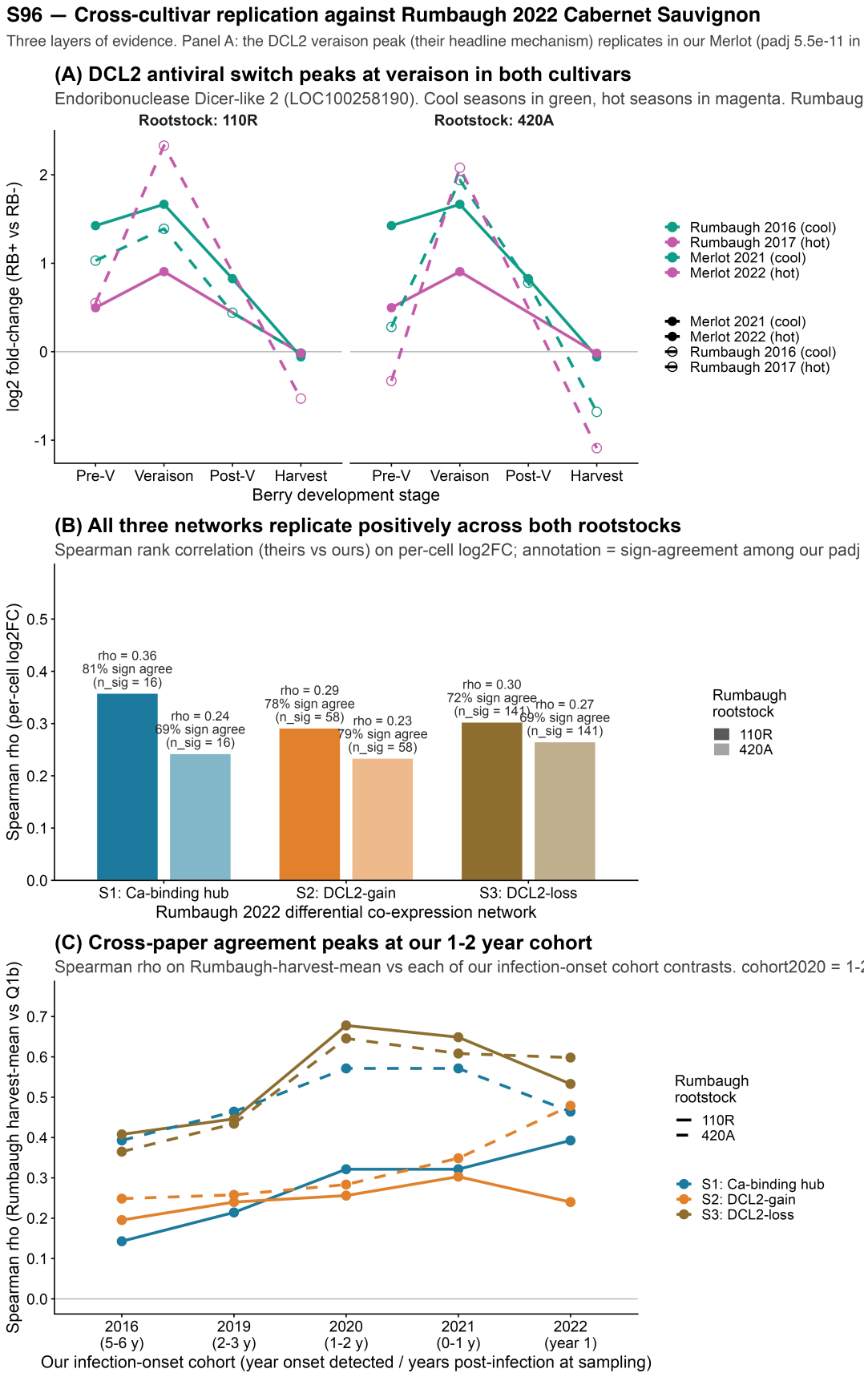
Note: Bipartite: 6 ORFs x 13 modules; rule 13 Spearman; FDR<0.05 or |rho|>=0.3



Supplementary Figure S96. Cross-cultivar replication vs Rumbaugh 2022 CS

Status: New figure produced under DESeq2 pipeline

Note: 3 panels: DCL2 veraison-peak trajectory (their headline mechanism, padj 5.5e-11 in our 2021); per-network Spearman concordance (S1 Ca



Supplementary Figure S97. Within-cultivar replication vs Rumbaugh 2023 Merlot

Status: New figure produced under DESeq2 pipeline

Note: 4 panels: per-gene harvest scatter (rho = 0.79 across 37 mapped genes); per-category Spearman bars (cell wall rho = 0.69 with 100% sign